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Manual for the

Category B licence in

Easy Read

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Definitions related to vehicles

Vehicle

Device prepared to travel on roads, streets, roads, and land of all kinds.

Types of vehicles

1. Vehicles without an engine
2. Vehicles with an engine
3. Other vehicles with an engine

Watch video
Vehicles without an engine
Animal-drawn vehicle
Cycle and bicycle
Trailer
Semi-trailer

Animal-drawn vehicles

For example, a horse-drawn carriage.

Cycle

Vehicle with two or more wheels that has pedals. It moves by the energy and effort of the person who rides it. A bicycle is a two-wheeled cycle.

Watch video
Trailer

Platform with wheels that is attached to a motor vehicle, by means of an axle, to carry loads.

Axle. Bar that goes through an object and helps it move and turn. In a trailer, the axle makes the load spread out and makes it easier to drive.

Light trailer
Heavy trailer

With load it weighs a maximum of 750 kilos.

With load it can weigh more than 750 kilos.

Semi-trailer

Trailer that is attached directly to a motor vehicle, without an axle.

Light semi-trailer

Non-light semi-trailer

With load it weighs a maximum of 750 kilos.

With load it can weigh more than 750 kilos.

Motor vehicles

Vehicles that need an engine to work. Motor vehicles are classified into:

- Motor vehicles They are used to carry people and things. They are also used to move or tow other vehicles.
- Special vehicles They are used to do certain jobs or services. Some have their own engine and others are towed. For example, excavators to move earth and tractors to work in the countryside. There are different types of motor vehicles and special vehicles:

Motor vehicles

Two-wheeled motorcycles

Motorcycles with sidecar
Three-wheeled vehicles
Heavy quadricycle (or quadricycle)
Car
Pick-up
Car-derived van
Adaptable mixed-use vehicle
Bus / Coach
Trolleybus
Lorry

Van
Articulated tractor unit
Articulated vehicle
Road train
Special vehicles
With an engine
Quad - ATV
Tractor
Motor cultivator
Self-propelled machine
Towed
Agricultural trailer
Towed machine

Motor vehicles
Motorcycles
Watch video

Two-wheeled vehicles that must meet at least one of these two features related to engine capacity and speed:

- If it is petrol, have an engine capacity greater than 50 cubic centimetres or, what is the same, 0.05 litres.
- Reach a speed greater than 45 kilometres per hour.

Engine capacity. Measure used to calculate the power of an engine. The more engine capacity an engine has, the greater its power will be.

Engine capacity is measured in litres or in cubic centimetres.

Motorcycles with sidecar

They are three-wheeled motorcycles that have a seat at the side for another person to sit. They must meet the same features of engine capacity and speed as two-wheeled motorcycles.

Three-wheeled vehicle

Vehicle with three symmetrical wheels that must meet the same features of engine capacity and speed as two-wheeled motorcycles and sidecars.

Heavy quadricycle

Four-wheeled motor vehicle that meets these features:

- With load and passengers it weighs a maximum of 450 kilos if it carries people or 600 kilos if it carries goods.
- The maximum power the engine can have is 15 kilowatts.

Kilowatt. Unit used to measure the maximum power a device can have.

Car

Motor vehicle used to carry people. It has at least four wheels. It can have a maximum of nine seats, counting the driver's seat.

Car-derived van

Motor vehicle with the body of a car, but used to carry goods instead of people. It only has one row of seats to leave more space for goods.

Body. Structure that gives shape to the motor vehicle.

Pick up

Vehicle in which the seats for people and the load area are separated into two different spaces. With load and passengers it weighs a maximum of 3,500 kilos.

Adaptable mixed-use vehicle

Vehicle used to carry people or goods. The seats can be put in or taken out depending on the use you want to give the vehicle. A maximum of nine people can travel in it.

Bus or coach

Vehicle with more than nine seats used to carry people and their luggage.

Lorry

Vehicle with four wheels or more used to carry goods.

Watch video

It has two parts:

- The cab. Space with seats where passengers and the driver travel. It can have a maximum of nine seats.
- Rest of the lorry. Space that is at the back. It is the place where the goods travel.

Van

Vehicle with four wheels or more used to carry goods. Unlike the lorry, the cab and the rest of the lorry are together, not separate.

Articulated tractor unit

Vehicle used to tow the semi-trailer.

Combination of vehicles

Motor vehicle made up of a motor vehicle and a trailer or semi-trailer attached or joined to it.

Watch video

There are three types of combinations of vehicles:

1. Road train. The trailer is attached to the motor vehicle by means of an axle.
2. Articulated vehicle. The semi-trailer is attached to the motor vehicle, without an axle.
3. Euromodular configuration. Attached vehicles that have at least 6 axles in their wheels. The axles

connect the wheels to each other and help them turn.

Special vehicles
Quad ATV

Vehicle with four wheels or more used more for driving on hills or tracks than on roads. It is driven with handlebars, like a motorcycle.

Tractor

Vehicle with four wheels or more that has a very large engine. It is used to carry or tow materials and heavy loads.

Watch video

The tractor is used mainly in field work (agricultural tractor) and in construction (works tractor).

Motor cultivator

Small vehicle that has an engine and wheels. The person who drives it walks and moves it by means of a long part, like handlebars. This long part is part of the motor cultivator. It is used mainly in gardening and agriculture to turn over the soil in vegetable gardens or small plots of land.

Self-propelled machine

Vehicle with four wheels or more used to do work in the countryside and in construction.

Towed vehicles

There are two types:

1. Agricultural trailer Vehicle to do work in the countryside. It must be towed by a tractor or by other special vehicles.
2. Towed machine Vehicle used to do work on building sites, in the countryside, or in other services. It must be towed by another special vehicle.

Other types of motor vehicles

Some vehicles that have an engine are not considered motor vehicles because they have special features.

Vehicles not considered motor vehicles
Tram
Moped

Vehicles for people with reduced mobility

Pedal bicycle with an engine

Personal mobility vehicle

Tram

It works through rails placed in the road or in the street.

Moped

Vehicles with two, three, or four wheels. They have low-power engines. The maximum speed they reach is 45 kilometres per hour. A maximum of two people can travel on mopeds.

Vehicles for people with reduced mobility

They are made for use by people who have a physical disability. The maximum speed they reach is 45 kilometres per hour. It weighs a maximum of 350 kilos.

Pedal bicycles with an engine
Personal mobility vehicles

They have an electric motor that reaches a speed of between 6 and 25 kilometres per hour. For example, the electric scooter. In this type of vehicle only one person can travel.

Definitions related to people
Driver

Person who drives a vehicle. It is also the person who controls the extra controls in driving school vehicles to learn to drive.

Pedestrian

Person who walks on the pavement and the road. Pedestrians are also people who:

- Pull or push a pram, a wheelchair, or any small vehicle without an engine.
- Walk and take with them a cycle or a two-wheeled moped.
- Use a wheelchair.

Vehicle owner

Person who has the vehicle registered in their name in the relevant official register. This person will always carry in the vehicle the documents that say they are the owner and will show them to the authorities when asked. The vehicle owner is responsible for making sure that people who do not have a driving licence do not drive that vehicle.

Usual driver

Person who usually drives a vehicle, even if they are not the owner. The usual driver must have the driving licence needed to drive that vehicle. The vehicle owner is the person who authorises that person to be the usual driver.

Vehicle categories depending on their use

Category	Use
M	Motor vehicles made to carry people and their luggage. For example, cars and buses.
N	Motor vehicles made to carry goods. For example, vans and lorries.
O	Trailers made to carry people and goods.

Topic 2. Documents

Contents

Documents needed to drive

To drive a vehicle you need the following documents:

Documents

Driving licence

Vehicle registration document

Roadworthiness test certificate

Third-party liability insurance

Watch video

You must always carry these documents in the vehicle when you drive to show them to the authorities who are responsible for monitoring traffic. These documents must be the originals. You can also present a photocopy of the documents that is certified by an official public body. For example, the town hall, the police or a notary.

To certify. An official body confirms that the copy of the document is genuine

When you need to adapt the vehicle to drive, you must ask for a special permit and make the necessary changes to the vehicle. For example, fitting controls on the steering wheel for people who have a disability in their legs and find it hard to use the car pedals. We are going to learn about the 4 documents needed that a driver must have, what they are for and how they are used.

Driving licence

Document that authorises a person to drive and makes sure they meet the requirements to do so.

Types of driving licence

You must get one licence or another depending on the vehicle you are going to drive:

Watch video

Type of licence	**Vehicles you can drive**	**Minimum age to drive**
AM	Mopeds with two or three wheels and light quadricycles. Vehicles for people with reduced mobility.	15 years
A1	Motorcycles with a maximum engine size of 125 cubic centimetres and a maximum power of 11 kilowatts. Motor tricycles with a maximum power of 15 kilowatts.	16 years
A2	Motorcycles with a maximum power of 35 kilowatts.	18 years
A	Motorcycles and tricycles with any type of engine and power.	20 years

Type of licence	**Vehicles you can drive**	**Minimum age to drive**
B	Mopeds. Vehicles for people with reduced mobility. Cars that can carry up to nine passengers. Special agricultural vehicles. Special non-agricultural vehicles that reach a maximum speed of 40 kilometres per hour and do not weigh more than 3,500 kilos. Combination of vehicles when the trailer does not carry a weight over 750 kilos.	18 years
B+E	Combination of vehicles. The trailer can carry a load of up to 3,500 kilos.	18 years

Points-based driving licence

When you get your driving licence you are given 8 points. You will lose some of these points or all of them if you commit a serious or very serious offence. For example, using your mobile while driving. You can get the points back after some time by doing a re-education course for driving. This course lasts 12 hours.

New drivers

A person who gets a driving licence is called a new driver during the first year.

New drivers must display in the car a green rectangular plate with a white letter L. This plate is placed on the rear window at the back of the car. It must be clearly visible so that other drivers know that car is being driven by a new driver.

When must you renew your category B driving licence?

- Every 10 years if you are under 65.
- Every 5 years if you are over 65. You must apply to renew the licence before it expires. You must also apply again for the driving licence if you lose it, it is stolen, or any of your personal details change. For example, your address.

Vehicle registration document

Document that confirms that a vehicle has a number plate and is authorised to be on the road.

Which vehicles must have a vehicle registration document?

- All motor vehicles.
- Mopeds.
- Trailers and semi-trailers that can carry more than 750 kilos.

Number plate. A set of letters and numbers that identify vehicles.

Each vehicle has a different number plate that is stamped on a metal plate and placed on the vehicle.

What information appears on the vehicle registration document?

- First name, surname, and address of the vehicle owner.
- Vehicle number plate and the date it was registered.
- Number of seats in the vehicle.
- Use of the vehicle. That is, whether it is used to carry people or goods.
- Maximum weight the vehicle can carry. The owner of a vehicle must report any change in their details to the Traffic Headquarters.

They must do it no later than 15 days after the change. They must also report if they sell or hand over the vehicle to another person. In this case, they have 10 days to report it. The person who buys or receives the vehicle must apply for the vehicle registration document to be renewed in their name.

Watch video
Roadworthiness test certificate (ITV)

Document that shows that a vehicle has no faults and is in good condition to work.

Which vehicles must have the roadworthiness test certificate?

- All motor vehicles.
- All mopeds.
- All trailers and semi-trailers.

What information appears on the roadworthiness test certificate?

- Vehicle characteristics.
- Record of the inspections the vehicle has passed.

When must you take your vehicle for the roadworthiness test (ITV)? To know when each vehicle must be checked, you must take into account what type of vehicle it is, how old it is and what it is for: whether it is a car, a special vehicle...

Types of vehicle

When to take the roadworthiness test (ITV)

Two-wheeled mopeds First test at 3 years. After 3 years, a test every 2 years.

Three-wheeled mopeds and light quadricycles

Motorcycles, three-wheeled vehicles and heavy quadricycles

Quads

First test at 4 years. After 4 years, a test every 2 years.

Motor vehicles that can carry up to nine passengers First test at 4 years. After 4 years, a test every 2 years. After 10 years, a test every year.

Motor vehicles that can carry goods with a weight of up to 3.5 tonnes First test at 2 years. After 2 years, a test every 2 years. After 6 years, a test every year. After 10 years, a test every 6 months.

Trailers to carry goods or people or to house people

Except the towed caravan First test at 1 year.

Up to 10 years, a test every year. After 10 years, once every 6 months.

Towed caravan

First test at 6 years. After 6 years, every 2 years. You must also take a roadworthiness test when the vehicle is modified, or if it has had an accident and the structure has been damaged. The vehicle owner is the person responsible for making sure the roadworthiness tests are done.

Roadworthiness test results

When the vehicle is in good condition, the ITV technicians:

- Write on the roadworthiness test certificate that the vehicle is OK and the date of the next test.
- Give you a sticker. You must place it inside the vehicle, on the right side of the windscreen.
- Give you a report. You must always carry it in the vehicle. When the vehicle has a fault or a serious problem:
- You cannot continue driving.
- You must take it to a garage to be repaired.
- Once repaired, you must take it for a new test to check that it can be driven again.

Compulsory third-party liability insurance

People who have a motor vehicle must take out and pay for insurance. This insurance is to protect other people, objects or vehicles if you have an accident. This insurance is called third-party liability insurance.

Which vehicles must have compulsory third-party liability insurance?

All motor vehicles if the owner lives in Spain, except:

- Trains and trams.
- Wheelchairs.
- Motorised toys.

Other vehicles that do not need third-party liability insurance are trailers, semi-trailers and towed machines that cannot carry more than 750 kilos. What does compulsory insurance cover?

- Damage that your vehicle causes to an object. For example, repairing a street lamp if you crash into it.
- Damage to another person. For example, if you run over a pedestrian. However, the insurance does not cover damage to another person if it is them who behaves badly. For example, if a pedestrian throws themselves at your car so that you run them over. It will also not cover damage to a person for reasons that have nothing to do with driving. For example, a traffic accident caused by a hurricane or a fire.

What does compulsory insurance not cover?

- Injuries that in an accident are suffered by the driver of the insured vehicle. For example, it does not pay medical costs for a driver who loses a leg in an accident.
- Damage suffered by objects that are inside the car of the person driving.
- Damage that the accident causes to the objects of the insured person, the person driving, or their family members. For example, it does not cover repairing the door of a garage that a driver breaks when entering the home of their brother.
- Damage to people who are not wearing a helmet if it is compulsory to wear one.
- Accidents and damage that happen in a stolen vehicle. When there is an accident, the drivers of all vehicles must inform their insurers. They must do it no later than 7 days after the accident.

What are the consequences of not insuring the vehicle?

- The vehicle cannot be driven. If a traffic officer stops you, they will immobilise the vehicle at that moment.
- The vehicle owner must pay the costs of the place where the vehicle is kept while it has no insurance.
- The vehicle owner must pay a fine. You must always carry in the vehicle the proof of payment for the insurance. This way you can show that the vehicle is insured. This proof will show the following information:
- Name of the insurance company.
- Vehicle number plate.
- Insurance certificate.
- Date when the insurance must be renewed.

- Indication of what damage the insurance covers.

Responsible people

As a general rule, the person who commits an offence or infringement as a driver or pedestrian is the one who must pay the fine or penalty.

However, there are cases in which drivers are responsible for a fault or offence, even if they did not cause the accident. These cases are:

- Drivers and passengers who do not wear a helmet in vehicles where it is compulsory to wear one.
- Drivers who carry under-age people in vehicles where it is not allowed for children of that age to travel. When traffic faults or offences are committed by a person under 18 years old, the fine must be paid by their father, mother, or guardian. The person who is the registered keeper of a vehicle is responsible for faults or offences related to:
- The vehicle documents.
- Not keeping up with the vehicle inspections.
- The condition the vehicle is in.

Topic 3. The condition of the driver

Contents
Physical and psychological condition of the driver

Factors that affect the condition of the driver

- Fatigue
- Sleepiness
- Alcohol
- Other drugs
- Illnesses and medicines
- Food
- Suitable clothing

Accidents due to distractions

- Smoking in the vehicle
- Use of the mobile phone
- GPS sat navs

Physical and psychological condition of the driver

You must be in good physical and psychological condition to drive. There are many physical and psychological factors that affect your safety and the safety of other people when you drive a vehicle. For example, your ability to see well, your mood, and the time you take to react to unexpected events that happen on the road.

Also, you must have enough training to handle the vehicle and understand all traffic rules.

Eyesight

Through our eyes we receive most of the information to drive well. Some of this information is: road condition, traffic signs, distance to objects and the speed of other vehicles. To drive well it is necessary to see things in front of you, to the sides, and behind.

But some factors make you see less:

- Drinking alcohol.
- Taking drugs.
- Being tired.
- Driving too fast.

Mood

Changes in mood can make you lose concentration and cause a road traffic accident. For example, having a big argument while you are driving or receiving very sad news before driving can change your mood.

Reaction time

It is the time between hearing or seeing something and reacting. For example, the time between seeing a red traffic light and stopping the car. In normal conditions, the reaction time of a person who is driving is between half a second and one second.

Factors that make a person take longer to react:

- Being tired (fatigue).
- Being sleepy.
- Being very old.
- Hearing or seeing badly.
- Illnesses.
- Some medicines.
- Alcohol and drugs.
- Eating a lot before driving.
- Too much heat in the car.
- Altered mood.
- Paying little attention to driving. !

Factors that affect the condition of the driver

Being tired (fatigue)

It is one of the main risks that can cause accidents on the road. Fatigue can be physical or psychological.

Physical fatigue produces a feeling of tiredness and psychological fatigue makes it harder to concentrate.

Watch video

What can cause fatigue?

- What the road is like.
- What the vehicle is like.
- What the driver is like. What the road is like:
- Road with heavy traffic.
- The road surface is in poor condition.
- You do not know the road.
- Weather difficulties: rain, fog, snow or too much heat. What the vehicle is like:
- Too much heat inside the vehicle. The right temperature is 23 degrees more or less.
- You drive at night with poor lighting.
- You drive in a vehicle in poor condition or that makes too much noise.
- You are uncomfortable in the seat.

What the driver is like:

- You drive for many hours without resting or you take very short breaks.
- You drive fast for a long time.
- You drive sleepy, after drinking alcohol, or you feel unwell.
- You do long journeys and at night when you are not used to doing them.
- You have had your driving licence for a short time.
- You drive with your body in a bad posture. What are the symptoms of fatigue? There are some signs that warn you that you are suffering from fatigue and you must stop driving. These signs are:
- Difficulty concentrating on the road.
- Your eyes feel heavy and you start to see badly.
- You hear badly.
- Feeling that your arms are numb.
- Feeling of pressure in your head.
- You move a lot in the seat and change posture.
- You make slower movements.
- Feeling more nervous and irritable.
- Difficulty making decisions about driving.
- You take longer to react.

Being sleepy (sleepiness)

Many road traffic accidents are related to driving while sleepy. You do not need to fall completely asleep to have an accident for this reason. The symptoms of sleepiness appear before falling fully asleep. Sleepiness. A state in which you feel tired, heavy in the body, and sleepy.

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What can cause sleepiness?

- Sleeping fewer hours than usual.
- Changing the hours when you usually sleep.
- Sleeping badly, even if you sleep many hours.
- Driving on roads with little traffic.
- Drinking alcohol or taking medicines before driving.
- Having illnesses related to sleep.
- Driving at dawn or at midday, after eating. What are the symptoms of sleepiness? There are some signs that warn you that you are suffering from sleepiness and you must stop driving. These signs are:
- Difficulty keeping your head upright, or keeping your eyes open.
- Blurred vision.
- Yawning many times.
- Losing concentration or having meaningless thoughts.
- Being distracted by anything.
- Being restless or irritable.
- Difficulty remembering the last kilometres you have travelled.
- Drifting out of your lane.
- Not noticing traffic signs or the place where you must leave the road.
- Driving very close to the vehicle in front of yours. Effects of sleepiness:
- Taking longer than normal to react.
- Difficulty making decisions about driving.
- Feeling that it is hard to make movements with your body, or you make them more slowly.
- Falling asleep for a few seconds without noticing.

How can you avoid fatigue and sleepiness?

- Stop the vehicle in a safe place when you feel the first symptoms and sleep for 20 or 30 minutes.
- Rest for 20 or 30 minutes every two hours or every 200 kilometres, even if you do not feel fatigue or sleepiness.
- If you have a fatigue detector follow what it tells you. It is a system that works through sensors and recognises if the person is very tired or about to fall asleep. In those cases it warns you by a light, sound, or vibration in the steering wheel so that you stop the vehicle and do not have an accident.

Sensor. Devices that capture information about things that happen outside the vehicle.

Watch video

Alcohol

It is very dangerous to drink alcohol when you are going to drive, even if you drink a small amount.

Alcohol spreads through your whole body through the blood and affects, above all, the brain and eyesight. Alcohol is the cause of many road traffic accidents.

Blood alcohol level

Blood alcohol is the total amount of alcohol in the blood after drinking. The blood alcohol level is the amount of alcohol in each litre of blood. It can be calculated in two ways:

- Grams of alcohol in each litre of blood.
- Milligrams of alcohol in each litre of air that we breathe out from the lungs.

The permitted blood alcohol level depends on the type of vehicle and the driving licence.

1. For people who got their driving licence less than two years ago. And for vehicles that carry:

- Goods with a weight over 3,500 kilos.
- More than nine people.
- Minors.
- People in emergency services.
- Dangerous loads.

The permitted alcohol level is:

- 0.15 milligrams of alcohol per litre of air.
- 0.3 grams of alcohol per litre of blood.

2. For any other vehicle and driver, the permitted alcohol level is:

- 0.25 milligrams of alcohol per litre of air.
- 0.5 grams of alcohol per litre of blood. In any case, the only blood alcohol level that is safe for driving is 0.0, that is, not drinking alcohol.

What factors affect the blood alcohol level?

Alcohol does not affect all people in the same way. The factors that affect the blood alcohol level are:

Amount of alcohol

Sex of the person

Type of drink

Speed at which you drink

Age

Personal circumstances

Drinking alcohol without eating

Weight of the person

Sleeping after drinking

Time since you drink alcohol

- The amount of alcohol you drink. The more alcohol you drink, the higher the blood alcohol level will be.
- The weight of the person. Alcohol affects thin people more.
- The sex of the person. Alcohol usually affects women more.
- The time that passes since you drink alcohol. The moment when the blood alcohol level is highest is one hour after drinking alcohol. After that, the effects of alcohol go down very slowly.
- The type of drink and the way you drink it.

Alcohol reaches the blood faster after drinking some drinks such as gin or whisky than after drinking other drinks such as wine or beer.

Also, alcohol reaches the blood faster when it is mixed with tonic or some soft drinks.

- Sleeping after drinking.

Alcohol is removed more slowly when we sleep. So, it is not safe to drive after drinking a lot of alcohol and sleeping a few hours.

- The speed at which you drink. The body removes alcohol better when you drink slowly than when you drink fast.
- Drinking alcohol without eating.

Food helps alcohol reach the blood more slowly.

- Age.

Alcohol usually affects people more under 18 years old and people over 65 years old.

- Personal circumstances. There are circumstances that can make alcohol affect a person more. For example, being pregnant, stress, or having an illness.

Effects of alcohol while driving

On behaviour

- False confidence in yourself.
- You take more risks.
- You commit more faults that cause accidents.
- You may treat other drivers in a more aggressive or impulsive way.

On the way you see the surroundings

- You see traffic signs and traffic lights worse.
- You judge the distance to other vehicles worse.
- Less ability to see what happens to one side and the other.
- You are more dazzled by vehicle lights.
- Possibility of being distracted by elements in the surroundings. On movements
- Difficulty coordinating your body movements. On decision making

- You need more time to react.
- Higher chance of making bad decisions or not knowing how to carry them out.

Other drugs

Taking drugs before driving is very dangerous.

One in every 10 people who die in a road traffic accident had taken drugs before driving. It is forbidden to drive any type of vehicle when a person has taken drugs and they are still in their body. You can only drive after taking substances prescribed by the doctor and that do not affect driving. How do drugs affect driving?

All drugs are dangerous and it is forbidden to drive when you have taken them. But each one produces effects that are different, and these are the cause of the danger.

Type of drug Possible dangers while the person is driving

Cannabis

- Seeing the surroundings in a different way. For example, colours look different.
- Judging distances worse
- Losing concentration.
- Needing more time to react.
- Falling asleep while driving.

Cocaine

- Becoming more impulsive or aggressive.
- Losing the sense of danger.
- Driving in a more dangerous way.
- Seeing the surroundings in a different way.
- Losing concentration.

Watch video
Ecstasy

- Having hallucinations.
- Being more sensitive to light or seeing blurred.
- Losing concentration.
- Suffering depression or anxiety.
- Feeling tired when the effects of the drug wear off.

LSD

- Having hallucinations.
- Reacting in an aggressive way.
- Having anxiety and even panic.

Amphetamines

- Losing patience.
- Having impulsive and violent behaviour.
- Having little sense of danger.
- Delayed tiredness and sleep. This can cause the person to suddenly feel very tired and fall asleep without meaning to when the effect of the drug wears off.

Tests to detect alcohol and drugs

Driving after drinking or taking drugs is forbidden and punished by law. The person who does it will have to pay a fine and may even go to prison. To know if a person who is driving has taken alcohol or drugs, they are given a test. What does the test involve? To detect if a person has drunk alcohol the traffic officer asks them to blow into a device that measures the amount of alcohol in their blood. If the test is positive or the person shows signs of having drunk the test is repeated to confirm it. To detect if a person has taken drugs a saliva sample is taken and the sample is put into a device.

If the person does not agree with the results they can ask for a blood test.

Test results

If the alcohol or drug tests are positive, the person who is driving will have committed a very serious offence. The consequences are:

- Paying a fine.
- Losing between 4 and 6 points from the driving licence.
- Possibility of losing the driving licence for a period of time.
- Possibility of going to prison if they have put at risk the lives of other people. The traffic police officer can forbid the person to continue driving. They will leave the car immobilised until the effects of the alcohol or drugs wear off.

Who must take the alcohol and drug detection tests? People who are driving a vehicle and:

- Have a road traffic accident.
- Show symptoms of driving under the effects of alcohol or drugs.
- Are reported for breaking a traffic rule or committing some other offence.
- Go through a checkpoint to prevent the use of alcohol and drugs.

Pedestrians must also take the test if their behaviour could cause a road traffic accident. The law imposes fines and penalties on people who refuse to take these tests when security officers ask them to.

Illnesses and medicines

Some illnesses and medicines make the person lose abilities to drive safely. When you have an illness or take medicine you must always ask the doctors if you can drive. It is also important to go to the Medical Centre for Driver Medical Checks. There they will tell you if it is safe to drive with the medicine you are taking.

If you have a chronic illness you can follow these tips to avoid accidents while you drive:

- Know the illness well.

- Know the effects of the medicines you take.
- Know how to act in case of a crisis.
- Avoid driving when you feel unwell.
- Do not stop the treatment until the doctor authorises it.
- Do not drink alcohol while you take the medicine.

What illnesses can affect driving?

Illness
Tips for driving

Respiratory allergy

- Drive with the windows closed.
- Set the air conditioning low.
- Keep the vehicle and the ventilation ducts clean.
- Do not mix alcohol with medicines.
- Do not do long journeys driving.
- Wear sunglasses. The sun, many times, makes you sneeze.
- Do not drive at dawn or in damp areas.
- Do not take medicine without a doctor's prescription.

Stress

- Do not drive in the phases of most stress.
- Do not drive if you take medicine against stress.
- Seek help to reduce stress.

Depression

- Do not take drugs or alcohol to improve depression.
- Take only the medicine that the doctor has prescribed for you.
- Put yourself in the hands of specialists and follow the treatment.
- Do not drive in the periods of most depression.

What medicines can be dangerous for driving?

Medicine

What is it for?

Possible effects
Painkillers

Helps pain to reduce or go away. Sleepiness. Vertigo.

Lack of concentration.

Cough suppressants

Calm a cough.

Mood changes.

Loss of reflexes.

Antihistamines Treat allergies. Sleepiness. Depression.

Loss of reflexes.

Psychotropic medicines

Treat depression, anxiety and sleep disorders. Sleepiness.

Loss of reflexes. Dizziness. Blurred vision. Confusion.

Food

Eating a large amount and foods that are hard to digest before driving can cause sleepiness and tiredness. On the other hand, eating too little can cause dizziness. The best thing is to have several light meals instead of one very heavy one. It is also recommended to wait a while (between 15 and 20 minutes) after finishing eating before driving. It is important to drink water or juices so that the body is hydrated and to have less chance of feeling sleepy or tired.

Suitable clothing

You must wear comfortable clothes when you drive. Very tight clothing will not let you move well. In winter, you must take off your coat before you start driving.

Footwear must be comfortable and light to use the pedals better. It is not advisable to drive with high-heeled shoes, flip-flops, or shoes with very thick soles.

Watch video
Motorcycle

You must pay special attention to the clothes you wear when you ride a motorcycle. In this vehicle there is no bodywork to protect you in an accident.

Bodywork. Metal part that covers most parts of a vehicle. The most suitable clothing to ride a motorcycle is:

- Leather suit or a similar material. This suit must fit your body well.
- Leather gloves with protection.
- Strong boots that support the foot and ankle.
- Wear bright and striking colours.

Accidents due to distractions

Distractions while driving happen when the person who is driving looks at something that happens inside or outside the vehicle and that has nothing to do with driving. For example, picking up the mobile phone or looking at a shop window. When are accidents due to distractions more common?

- In young drivers between 18 and 25 years old and in those over 70 years old.
- When there are several people inside the vehicle.

- In the summer months.
- On weekend trips and during the day.
- On roads more than in the city.
- On motorways and dual carriageways.

Causes of accidents due to distraction

Distractions related to the road

Distractions of the person who is driving

Knowing the road well and being too confident. The signs are hidden or cannot be seen well for some reason.

Complicated traffic situation with many vehicles, pedestrians and signs to pay attention to.

Not seeing well because it is night or because the lights of other cars dazzle you. Suffering tiredness or sleepiness.

Suffering stress, anxiety or depression. Being very old.

Drinking alcohol, taking drugs or taking medicines. Some behaviours such as: using the mobile phone, lighting a cigarette or using the GPS sat nav. Looking at a map, throwing an insect out of the vehicle, eating or drinking while driving.

Smoking in the vehicle

People who smoke while driving have twice as many accidents as people who do not smoke.

Watch video

The reasons are:

- Lack of attention on the road while you look for the cigarette and light it.
- One hand is busy with the cigarette.
- You see the road worse because of the smoke that comes from the cigarette.
- The air inside the car is worse quality and can affect the abilities to drive.

Using the mobile phone

It can be very useful to have a mobile phone in the car in case there is a breakdown or emergency. But using it badly can cause accidents. The risk of an accident when using the mobile in the vehicle is four times higher than when it is not used.

The reasons are as follows:

- You pay more attention to the conversation than to the road.
- You judge distances worse and you stop seeing some signs because of lack of attention to driving.
- Leaving the road and going into the opposite lane because you pay attention to the conversation.
- Difficulty handling the steering wheel when you hold the mobile in your hand or on your shoulder.
- Disorientation and loss of sense of time. In Spain it is forbidden to use the mobile phone while you

drive. But you can use it through a hands-free device that lets you talk on the mobile without touching the screen.

However, hands-free devices are also dangerous because the person who is driving pays part of their attention to the conversation and not to the road.

Recommendations for using the mobile and the hands-free device

- Stop the vehicle in a safe place when you need to make a call. Drive again only when the call ends.
- When you talk through the hands-free tell the other person that you are driving.
- Do not have conversations that last more than one minute.
- Never call a person if you know they are driving and do not have hands-free.
- When you drive, be careful with pedestrians who are talking on the mobile. They may be distracted.

GPS sat navs

It is a system that calculates the route from one place to another in real time. It helps you to find your way and to know where you have to go. If you make a mistake, the GPS sat nav calculates the route again and tells you again where you must go.

Tips to use the GPS sat nav well:

- Set up the device before you start driving. Do not do it while you drive.
- Keep it fixed in one place. So it does not move or roll around the vehicle.
- Put it in a place where you can see it without taking your eyes off the road.
- Put it in a place that allows the airbags to open if necessary.

Watch video

Airbag. Safety device that is placed in the front of a car and on the sides to protect passengers in case of an accident.

Topic 4. Duties of drivers and pedestrians

Contents

Road safety

Road safety depends on drivers and also on pedestrians.

Watch video

That is why it is necessary that all people work together by respecting others and travelling in an orderly way when we are walking or in any vehicle. To achieve this, it is important that we follow all the instructions and duties set by the law. For example, it is forbidden to throw objects and materials onto the road or leave them abandoned. Especially objects that can damage the road, make it hard for vehicles to pass or cause an accident.

Duties of drivers
General duties

When you drive a vehicle you must not get distracted to prevent harm or accidents. In this way, you will avoid dangers for you, for the people travelling with you, those travelling in other vehicles and pedestrians. In particular, you must pay attention to pedestrians, especially children, older people, people who cannot see or who use a wheelchair.

Watch video

It is forbidden for all drivers:

- To drive a vehicle that gives off more noise, gases or fumes than allowed.
- To keep the vehicle doors open or open them before it has fully stopped.
- To open the doors and get out of the vehicle before making sure there is no danger for you or for other vehicles and pedestrians. For example, it is dangerous to open the car door without looking because at that moment a bicycle may pass in front and you can cause an accident.
- To use headphones connected to the mobile phone or another device to listen to music or any kind of sound. Only the use of some headphones is allowed, specific ones in the helmet of motorcycles and mopeds when the headphones are used as GPS navigators.

Watch video

- To use the mobile phone. You can use it through hands-free devices.
- To put fuel into the vehicle when the engine is running, the lights are on or the radio is on.

Everything must be switched off or disconnected to put fuel into the vehicle.

Watch video

Fuel. Material that when mixed with oxygen gives off heat. Many vehicles need fuel to work. For example, petrol.

Terms related to safe driving

Defensive driving A way of driving in which the person is alert to predict what other road users (drivers and pedestrians) are going to do and react in the right way. For example, taking into account that another driver may need to change lane and being ready if this happens.

Unsafe zone

Space where unexpected things can happen because of other drivers and pedestrians. For example, a child runs out into the road after a ball or a bicycle crosses in front of your vehicle suddenly. These unexpected things can happen in front, behind and at the sides of your vehicle.

Watch video
Two- or three-wheeled vehicles

These vehicles are seen less, are less stable and are also more fragile than other cars. That is why their passengers are more likely to suffer injuries in accidents. Which vehicles are these?

- Motorcycles and motorcycles with a sidecar.
- Three-wheeled vehicles, quads and heavy quadricycles.
- Mopeds.

Whenever you travel in these vehicles as a driver or passenger you must wear the safety helmet properly fastened and fitted to your head. When you ride a bicycle at night or if it is hard to see for another reason, you have to switch on the vehicle lights and wear a piece of clothing or an object that shines and can be seen from a distance of 150 metres. You must wear this bright item if you are the driver and also if you travel as a passenger. Some two- or three-wheeled vehicles must have a seat belt. In those cases, you have to fasten it. In vehicles with a seat belt you do not need to wear a helmet.

Special vehicles

Vehicles that enter the roads to do works or provide a special service must have a yellow light switched on while they do improvement work or road maintenance. When they work on motorways or dual carriageways they must switch on the yellow light from when they enter the motorway or dual carriageway until they reach their destination. This light is so that other drivers know the vehicle is there. They do not have priority to go through or overtake. The maximum speed these vehicles can travel at is 40 kilometres per hour.

Duties of pedestrians
General duties

When walking in towns and cities When walking on the road When crossing the road When walking in towns and cities

- Walk on the pavement and not on the road whenever you can. This rule must also be followed by pedestrians who use skates, skateboards or other similar devices that are not electric.
- Always keep to the right, in the city and on the road, when you pull or push a bicycle or a two-wheeled moped, handcarts or a similar device.

People who use a wheelchair must also keep to the right.

- People who carry a very large object or a small vehicle without an engine can walk on the carriageway if the hard shoulder or the pavement are narrow and get in the way of pedestrians.

Hard shoulder. Sides of the road that can be on the right and on the left. On the right one only some vehicles can travel, such as, for example, mopeds.

Watch video

When walking on the road

- On roads that are outside towns or cities you must always walk on the left side of the road. Unless it is safer to walk on the right.
- At night you must wear a light-up or bright item that can be seen from a distance of 150 metres. For example, a high-visibility vest, a light-up wristband...
- Groups of people must walk in single file, one behind the other. The people at the front must carry a white or yellow light that shines and the people at the back must carry a red light that also shines. This way drivers will be able to know where the line of pedestrians starts and ends.

Watch video

When crossing the road or the street

- Cross in a straight line without stopping until you reach the other pavement.
- At pedestrian crossings, make sure that cars have stopped or are coming slowly before you start to cross.
- Go around squares or roundabouts. Do not cross them on the road.

Personal mobility vehicles

Electric scooters, electric unicycles other personal mobility vehicles are forbidden to travel on:

Unicycle. Vehicle that has only one wheel joined to the seat by means of a metal bar.

- Pavements and pedestrian areas.
- Tunnels.
- Through roads.
- Motorways and dual carriageways.
- Roads that are outside the town or city.

What duties do you have if you drive these vehicles?

- You cannot use headphones while you drive.
- You cannot use the mobile phone while you drive.
- You must take alcohol and drug tests when an officer asks you to, like the rest of drivers.
- You must wear bright and light-up clothing if you drive at night or in places where it is hard to see.
- You must wear a helmet when the law requires it.
- You cannot carry passengers.

Animals on the road

Livestock, animals that carry materials or those that transport people, may only travel on the road when there are no other paths to go from one place to another. They must always be guided by a person over 18 years old.

Livestock and pack animals will always go on the right hard shoulder. When there is no hard shoulder they will go close to the right edge of the road. Animals that go in a herd or flock will also go on the right side of the road, taking up as little space as possible.

No animal may travel on motorways or dual carriageways.

Livestock. Group of animals raised by people to produce meat, milk, wool and other products. For example, cows, sheep, goats and pigs.

Topic 5. Safety devices in the vehicle

Contents

The importance of the vehicle in road traffic accidents For a vehicle to be safe and have less risk of having accidents it must meet two requirements:

1. Follow the inspection programme and the compulsory maintenance for each vehicle.
2. Have a good safety system.

Most new vehicles have safety systems that help make the vehicle safer and have fewer accidents. But the reality is that new vehicles have the same number of accidents as older vehicles. This can happen for two reasons:

1. Few drivers know what the safety systems in their vehicle are for and how they work.
2. Drivers take more risks because they trust too much in the safety systems of their vehicle. They do not realise that these safety systems have limits.

The safety systems of a vehicle are divided into two types:

Active safety

Passive safety

Watch video
Active safety

Vehicle elements designed to prevent accidents. The driver has to use them for them to work. For example, brakes, wheels and the lighting system.

Passive safety

Elements that help passengers and other users suffer less harm in an accident. They work automatically. For example, seat belt, airbag and head restraint.

Active safety elements
Lighting systems

They are the different lights in the vehicle. They let you see what is around you and let pedestrians and drivers of other vehicles see you when it is night or it is hard to see. Thanks to the lighting system you can light up roads and streets to see nearby dangers and act in time. It is very important to use the lights well by switching on the right ones and switching off the ones you do not need. Some lights can dazzle other drivers when they are used wrongly.

Today's vehicles have new lighting systems to protect more. These new lighting systems are:

Xenon and bi-xenon lamps

Automatic lighting activation

Adaptive lights

Xenon and bi-xenon lamps

Headlights that light up the road with a strong bluish-white light, more like natural light than the headlights used before. They light up the road better, help the driver see further ahead and reduce eye strain.

Also, they dazzle other drivers less than other lights.

Xenon. Gas used in some lighting systems.

Automatic lighting activation

System that measures the light outside and switches the lights on or off automatically so that the vehicle can be seen better.

Adaptive lights

System that adjusts the strength of the lights and where they shine depending on what is needed at each moment. It does this taking into account: the distance to the vehicle coming in the opposite lane, the type of road you are driving on and the speed your vehicle is travelling at. Adaptive lights are divided into two types of lights:

1. Cornering lights. They switch on when you turn the steering wheel to show the side your vehicle is going to turn to. This light helps you see better on bends.

At roundabouts it lets other drivers know where you are going to turn.

Roundabout. Circular square that is a junction between streets.

2. Automatic main beam lights.

System to switch on vehicle lights that light up more of the road when there is little light outside. For these lights to switch on your vehicle has to be travelling at a speed higher than 45 kilometres per hour and it must not detect another vehicle nearby. These lights switch off when they detect another vehicle on the road coming towards you in the opposite lane.

Brakes

Brakes are responsible for reducing the speed of the vehicle until it stops completely. A vehicle's brakes do not usually fail. But sometimes it does happen and this can cause a serious accident.

Watch video

What safety systems are there in brakes?

Engine brake

Anti-lock braking system (ABS)

Autonomous emergency braking (AES)

Emergency braking warning (EBD)

Engine brake

System used to reduce the speed at which a vehicle is travelling without using the wheel brake. It works when you stop accelerating and, in this way, the engine is the one that holds the vehicle back. If you drive in low gears (1st or 2nd) the engine slows the vehicle more. It is recommended to use engine braking when going down long hills.

Anti-lock braking system (ABS)

Device that prevents the wheels from locking when braking. This system is very important because it helps you keep control of the vehicle and to brake in less space. The more weight there is in the vehicle the harder it is for the wheels to lock when braking. For example, it will be harder for the wheel of your motorcycle to lock when you have a passenger with you.

Autonomous emergency braking (AEB)

System that calculates, through a radar, the distance there is and should be between your vehicle and the one in front of you. If there is very little distance between the vehicles, this braking system will warn you by means of a voice and a sound. If you do not pay attention to these signals and there is a risk of an accident the system will do an emergency brake.

Watch video

Radar. System that can detect where an object is, for example another car, and how fast it is going.

Emergency braking warning (EBD)

System that warns the driver of a vehicle that the vehicle in front has to do an emergency brake. When the driver in front presses the brake firmly and quickly the brake lights of their vehicle flash. In this way, the system warns the driver behind so they have more time to brake and try not to crash into them. How are brakes used? To brake safely and in a controlled way it is necessary to:

1. Brake gently, little by little and with enough time.
2. Take into account the condition of the road. When it is not in good condition, you must brake more gently and with more time.
3. Do not use the brake too much because it will get hot and will brake worse.

To do an emergency stop in a vehicle without ABS, you must press the brake pedal hard and fully until you notice that the wheel starts to lock. Then lift your foot off the brake a little to prevent the wheels from locking. But do not stop braking. To do an emergency stop in a vehicle with ABS, you must press the brake pedal hard and fully until the vehicle stops completely. What to do when the brakes fail? You can act in different ways depending on what the situation is in which the brakes fail.

Problem
Possible solution

The brakes get very hot because they have been used too much in a short time.

Release the brake pedal so they can cool down. The brakes fail when the vehicle goes down a long and very steep hill. Do not accelerate.

Change down gears so the engine acts as a brake. If you cannot change down gears pull the handbrake lever gently.

Other times when the brakes fail.

Press and release the pedal several times. This will reduce the speed of the vehicle.

Wheels

The wheels of a vehicle are made up of two parts:

1. Rim. Metal part in the shape of a circle that is the inner part of the wheel.
2. Tyre. Elastic rubber part that is placed around the rim. A piece of rubber is elastic when it can be stretched. When you stop stretching it, it returns to its original size.

Watch video

The rim and the tyre of a wheel must be compatible with each other so that the tyre fits well on the rim.

Tyres

Tyre faults are a common cause of road accidents. That is why it is very important that they meet quality characteristics.

On new tyres that are in good condition you can read on the side what their characteristics are.

Watch video

They must not have blisters, deformations, or tears. Nor should any of their layers be coming loose or have exposed cords or cracks. The part of the tyre that is in contact with the road is called the tread. The tread of cars must have grooves with at least 1.6 millimetres of depth.

These grooves are used so that the tyre:

- Does not get hot.
- Is more flexible and rolls better on the road.
- Grips the road better.
- Removes the water it picks up from the road when it rains.

Watch video
Inflation pressure

Tyres are inflated with air so that they are more resistant and support the load and the weight of the vehicle better. The manufacturer of each vehicle is the one who sets how much air the tyres must have. This amount of air will be the inflation pressure.

What happens when the inflation pressure is not correct?

Inflation pressure lower than recommended

Inflation pressure higher than recommended The tyre gets hot, deforms and wears out sooner. More fuel is used. The vehicle is less stable. The vehicle has more risk of skidding when the ground is wet. There is more chance that the tyre will burst. The tyre has less contact with the ground.

Therefore, it grips the road less. The centre of the tyre wears more than the sides. The vehicle vibrates more because the tyres cannot absorb small stones and other elements. For example, potholes in the road. The shock absorbers get damaged more. You must check the tyre pressure of your vehicle at least once a month. You must also always carry a spare wheel with the correct inflation pressure.

Tyre wear

The rubber of tyres can wear out due to rubbing against the road. For that reason, you must change the tyres of your vehicle every five years, even if they are in good condition.

Tyre rubber can wear out earlier for these reasons:

- Driving in a rough or aggressive way.
- Driving very fast.
- The climate.
- Tyres wear more in summer.
- Carrying a lot of load in the vehicle.
- Not having the correct inflation pressure.
- Driving on roads in poor condition.
- Problems with the brakes or with other parts of the vehicle that affect the wheels.

What should you do when a tyre gets a puncture? The steps you must take if a tyre gets a puncture while you are driving are:

- Reduce speed slowly until you stop the vehicle. Do not stop suddenly.
- Secure the vehicle in a safe place. Off the road and off the hard shoulder if possible.
- Change the punctured wheel for the spare one. Some vehicles have other systems to be able to keep driving and there are vehicles that have a wheel that only serves to get you to the nearest garage. It is called a space-saver wheel. You can travel about 200 kilometres with this wheel at a maximum speed of 80 kilometres per hour.

Watch video

Wheel balancing

Wheel balancing consists of putting weight on the rims so that the wheels support the weight and the load of the vehicle. This balance is very important so that the wheels turn well.

Wheels can lose balance for these reasons:

- Tyre change.
- The rims lose the counterweights.
- The rims get dented.
- The wheel wears out or gets cuts.

Watch video

How to know that a wheel is losing balance? You may notice one of these things:

- The vehicle makes noises and vibrates.
- The tyres wear more than normal.
- The wheel bolts are loose.

Other active safety elements

Traction control

Electronic stability control

Speed limiters

Cruise control

Adaptive cruise control

Traction control

Safety system that identifies if one wheel is turning faster than the others and brakes it so it does not skid. This system helps to keep the stability of the vehicle on bends, when going up a hill or when it is raining.

Electronic stability control

Safety system that helps the vehicle to follow the path set by the steering wheel and not to drift when it does not turn as much as the driver asks or turns too much. This system helps you keep control of the vehicle in dangerous situations. For example, when avoiding an obstacle or when taking a bend.

Speed limiters

System that allows you to choose the maximum speed that your vehicle can reach. When you activate the speed limiter, the vehicle will not go above the speed chosen by the driver.

Cruise control

System that helps you keep the speed you choose to drive without needing to press the pedals to speed up or slow down. For example, driving all the time at 100 kilometres per hour on the motorway. This system is activated from the steering wheel. It switches off when you accelerate or brake. Then you have to choose the speed again and activate it.

Adaptive cruise control

As well as helping to control speed, this system helps you keep the distance from the vehicle in front.

Your vehicle will reduce speed or brake if you do not keep that distance and you get too close to the vehicle in front.

Passive safety elements

The chassis and the bodywork The chassis is the internal structure of the vehicle on which all the parts that make it up are placed. It is the skeleton of the vehicle and cannot be seen at first glance. The bodywork is the external metal structure that covers the vehicle. In the event of an accident, the chassis and the bodywork deform and protect the vehicle so that the passengers inside suffer less damage.

Rigid passenger compartment
Deformable parts

The seat belt What is it for? To protect all the people travelling in a vehicle in the event of a sudden impact or if the vehicle rolls over. In these cases, the belt helps people not to move from their seats and not to be thrown out of the vehicle.

Wearing the seat belt correctly means that a person has double the chance of surviving an accident. In fact, there have been fewer deaths from road accidents since the seat belt exists. For a seat belt to be useful and safe it must meet quality requirements and be well attached to the vehicle bodywork. It must be checked from time to time and taken to the garage to be repaired or replaced if it has any damage. Who must wear a seat belt?

All people travelling in a vehicle that has seat belts installed.

Drivers must do it and also passengers on all roads and streets.

There are some exceptions where drivers are not required to wear a seat belt, although it is recommended that they always wear it. These exceptions can only be applied when the vehicle is travelling within a town or city. Never when travelling on roads, dual carriageways, or motorways. Exceptions:

- Drivers who are parking moving the vehicle backwards.
- Taxi drivers who are on duty.
- Children who are less than 135 centimetres tall and travel in a taxi that does not have child safety systems. In those cases, the children will travel in the back seats and will fasten the seat belt that is on the seat.
- Delivery drivers who have to get out continuously of the vehicle to collect and deliver orders.
- Drivers and passengers of vehicles in emergency services. For example, ambulances.
- Driving school instructors who take a learner and are in charge of the additional controls of the vehicle.

Only people who cannot wear it for medical reasons or disability can travel without a seat belt in any situation or on any road. These people must carry a medical certificate that explains the reasons.

Seat belts for children

Children who are less than 135 centimetres tall must use a different safety system, more suitable for them. They are called child restraint systems and they are adapted to each child's height and weight. It is best for the child to try the child restraint system before starting to use it to check that it fits their size and is comfortable.

To fit a child restraint system in a vehicle you must take into account the following recommendations:

- Do not install the child seat in a seat that has an airbag in front of it.
- For small children, place the child seat facing the opposite direction to the vehicle's travel. That is, facing backwards. It is safer.
- Place the child seat in the centre rear seat. This way they will be more protected in case there is an accident from either side.

Watch video

It is important that child restraint systems have passed all quality tests so that they are safe. In vehicles with nine seats or fewer children must travel in the back seat and have the child restraint system properly secured.

Children may only travel in the front seat when:

- The vehicle has no rear seats.
- All the rear seats are occupied by other children.
- The child restraint systems cannot be installed in the rear seats. On buses, children who are less than 135 centimetres tall and are three years old or more must also use child restraint systems.

If there are none, they will have to fasten the seat belt on the seat, as long as it is suitable for their height and weight. How do you fasten the seat belt? You must wear the seat belt properly fastened and fitted to the body. It must not be too loose or too tight so that you can move easily. The seat belt has two straps that you must wear in the right place so that they protect you.

Watch video

Chest strap It must pass over the collarbone, between the shoulder and the neck, and go down the centre of the chest.

Placing the chest strap on the neck or on one breast can cause serious injuries in an accident.

Placing it on the shoulder can make the belt slip and protect less.

WRONG

RIGHT

Lap strap It must be placed over the hip bones. Always below the stomach. If it is placed above, it can cause serious injuries inside the body in an accident.

WRONG

RIGHT

After fastening your seat belt pull it slightly upwards to check that it fits well. Check that it is not caught or twisted in any part.

The submarining effect This effect happens when in an accident the body slides down, under the belt. This happens because the seat is tilted backwards and the belt is badly positioned. To prevent this you must:

- Fasten your seat belt properly.
- Check that the seat belt is well fitted to the body.
- Do not place towels, cushions, or covers on the seat that could make you slip.
- Have a correct driving posture, without tilting the seat too far back.

Submarining effect

Airbags

What are they? A safety device that is a bag of air that inflates inside the vehicle in case of an accident to protect passengers.

Airbags are placed at the front of the vehicle and on the sides. What are they for? The main functions of the airbag in an accident are:

- To slow down the sudden movement of the body.
- To prevent people from hitting violently against any part of the vehicle.
- To protect people's face and eyes from broken glass and other items that come loose because of the accident. Precautions when using airbags:
- Always use the seat belt so that the impact against the airbag is smaller.
- Place your chest at a distance of at least 25 centimetres from the steering wheel so that the front airbag does not hit you when it inflates.
- Switch off the front passenger airbag if you are going to place a child seat in that seat.

The head restraint A device that protects the neck and the cervical vertebrae in case of a sudden impact. It is very important to place the head restraint at the height of the passengers' heads, both in the front seats and in the back seats.

Cervical vertebrae. Bones that are in the upper part of the spine. The top of the head restraint must be at the same height as the top of your head. You should try to keep the distance between your head and the head restraint to four centimetres or less.

The helmet

Head injuries are the main cause of death in accidents in two-wheeled vehicles. Three out of ten people who have an accident in a two-wheeled vehicle save their life because they wear a helmet. What does the helmet do in an accident?

- It protects the head from impacts against the ground, other vehicles, or road elements.
- It prevents stones, metal, or other sharp objects from entering the head
- It spreads the force of the impact over the whole helmet so that it does not concentrate on just one point of the head. This prevents serious injuries.
- It helps prevent the face and head from being burned when sliding along the ground after the fall.

Type of helmet

Full-face model. That is, it also protects the lower part of the face and the jaw.

Watch video
Material

Some helmets expire after a few years and can lose properties if they are painted or if stickers are put on them.

Size

It must fit the head well.

Strap fastening

Check that the helmet is properly fastened and does not come off even if you pull hard.

Watch video
Colour

Light and bright helmets are safer.

Ventilation

It must have enough holes so that air can go in and out.

Visor

It is recommended to wear visors over the face that do not mist up and let you see well through them. These visors are called anti-fog.

Approval

The helmet must have a label that explains that it has passed the quality tests and is a safe helmet.

Topic 6. Vehicle parts and rules for using them

Contents

Vehicle controls
Controls of vehicles in general

Some vehicle controls are used with the feet and others with the hands.

Controls that are used with the feet

Accelerator pedal

Brake pedal

Clutch pedal

Controls that are used with the hands

Steering wheel
Handbrake
Accelerator pedal

It is used to control the amount of fuel that goes into the vehicle's engine. The harder you press the accelerator, the more fuel will go into the engine and the faster the vehicle will go. This pedal is pressed with the right foot. If you do not press it, the vehicle receives the right amount of fuel to keep running and not stop.

Watch video
Brake pedal

It is used to reduce speed or stop the vehicle. The brake pedal acts on all the wheels of the car. This pedal is pressed with the right foot and you must press it gently.

Clutch pedal

It is a mechanism that allows the gearbox to choose the gear you want to use and connect it to the engine. It is used to change gear while driving and to allow the vehicle to go forwards or backwards. When you press the clutch pedal fully the engine's movement is not transmitted to the gearbox or the wheels. Then you can change gear. The gearbox is used to change gear.

Watch video

When you release the clutch pedal the engine power works again by itself so that you can keep driving.

Steering wheel

Part of the vehicle that allows you to control which direction you go and where the wheels move.

Watch video

How should you hold the steering wheel?

- With both hands. You should only let go with one hand for just long enough to operate other car controls.
- On the outside. You must never hold the steering wheel on the inside.
- Firmly, but without using too much force.
- Without crossing your hands when turning the steering wheel.

Handbrake

A system that brakes the rear wheels of the vehicle and makes the vehicle stay stopped. The handbrake is a lever that is used with the right hand.

Controls of motorcycles

Motorcycles have the same controls as other motor vehicles. But they are in a different place and are used differently.

Watch video

With the left hand Clutch. It works by squeezing a lever. Control to make sounds (horn). Switch to turn on the lights.

With the right hand Front brake. Lever to brake the front wheel.

Accelerator. It is on the handlebar and is activated by twisting the grip.

With the feet

Brake pedal. Activates the brake of the rear wheel. It is usually on the right side of the motorcycle. Gear pedal. It is used to change gear. It is usually on the left side of the motorcycle.

Visibility when driving

It is necessary to see well everything that is around the vehicle so that driving is safe. To achieve this, all the windows must be clean. It is forbidden:

- To stick films or stickers on the windows that make an area harder to see.
- To fit coloured windows that are not approved or allowed by law.

Which vehicle parts help you see better?

Windscreen wipers
Windscreen washers
Heated rear window
Rear-view mirror
Windscreen wipers

They keep the front windscreen of the vehicle clean. Some cars also have a rear window wiper to clean the rear window. They move from side to side to clean the windows well. They have different speeds. Try to use the slowest one. The windscreen wiper and the rear window wiper have a wiper blade that you must change when you notice it leaves marks on the glass or it does not get properly clean.

Do not use the windscreen wiper when the glass is dry because it can scratch it. There are automatic windscreen wipers that switch on by themselves when they detect water. They work faster or slower depending on how much rain falls on the vehicle. The driver can switch on or switch off the automatic windscreen wiper.

Windscreen washers

A device that sprays a jet of liquid onto the glass so that the windscreen wiper can clean better. There are special liquids that work as windscreen washer fluid, although you can use water mixed with a little detergent. It is important to check that the tank where the windscreen washer is has liquid and refill it when it is running low.

Watch video
Heated rear window

Lines that you can see on the rear window of the vehicle. They remove the ice and the mist that form on this glass so that you can see well.

Rear-view mirrors

Mirrors that allow the driver to see better what happens at the sides and behind the vehicle.

Rear-view mirrors in cars, vans and lorries

Cars, vans and lorries that can carry up to 3,500 kilos in weight must have the following rear-view mirrors:

- A mirror outside the vehicle, on the left side.
- A mirror inside the vehicle, in the centre, above the front windscreen.
- A mirror outside the vehicle, on the right side. This mirror is recommended, but it is not compulsory.

Watch video

You can fit sun blinds on the rear side windows of the vehicle only when both exterior mirrors are fitted, on the right and on the left.

Rear-view mirrors on motorcycles

Motorcycles that travel at less than 100 kilometres per hour must have one mirror on the left side of the vehicle.

Motorcycles that travel at more than 100 kilometres per hour must have two mirrors. One on the left side of the vehicle and one on the right side.

Rear-view mirrors on two-wheeled mopeds

These vehicles must have one mirror compulsory on the left side of the moped. The mirror on the right side is optional.

Rules for driving in a comfortable and safe way

Having a good driving position helps to avoid tiredness and makes driving safer. Sitting in a good position in front of the steering wheel helps you respond better and faster to unexpected situations. Driving too close to the steering wheel causes tiredness because you have to strain your body more to make movements.

Driving too far from the steering wheel forces you to lean forwards to reach the controls and to lift your back off the seat backrest. This can be dangerous.

Your head must be above the steering wheel. If this is not possible because you are not tall enough, you can use a suitable support firmly fixed to the seat, bearing in mind that it cannot be a cushion or similar.

Rules for driving cars, vans and lorries Before you start driving you must check that: The seat and the backrest are well adjusted The mirrors are well positioned The seat belt is well fastened The seat and the backrest are well adjusted when:

- You can reach the pedals well and you can press them fully without straining your ankles.
- Your legs are not fully stretched. They are slightly bent and do not rub against any part of the vehicle.
- Your head is above the steering wheel so that you can see over it and not through it.
- The position of the seat backrest lets you reach all the controls. To check this, rest your back on the backrest, stretch your arms and check that your wrists can rest on the top part of the steering wheel.
- The head restraint is at the height of your head. The rear-view mirrors are well positioned when:
- Looking at the mirror inside the vehicle, you can see the four edges of the rear window.
- Looking at the mirrors on the sides, you can see the road, the vehicles coming from behind and the ones at your sides. By turning your neck a little you can see a part of the left rear of your vehicle. If you find that the mirrors are badly set while you are driving, you must stop the vehicle and set them correctly. To do this, try to stop in a flat and straight place. The seat belt is well fastened

Once you have done these checks fasten your seat belt properly before you start driving.

Rules for riding motorcycles

- Keep a natural position with your body. Do not force your posture.
- Lean your body just enough to reach the handlebars.
- Keep your arms and hands relaxed. This will help you avoid tiredness.
- When the motorcycle starts moving put your feet on the footrests. Do not leave them hanging.
- Lean your body a little to take bends. Do not lean too much, like professional riders in races.

Watch video

Which postures are not suitable for riding a motorcycle?

- Leaning your body too much.
- Keeping your elbows tucked inwards.
- Having your arms completely straight. These postures cause tiredness and can make it harder for you to react.

Systems that help you drive safely

Traffic sign recognition system

(TSR) It detects speed limit signs on the road so that the person driving can reduce the vehicle speed if needed. In some vehicles the system shows the number that matches the maximum speed the vehicle can travel at. This number appears on the instrument panel. In other vehicles, as well as warning, the system reduces the speed by itself when the driver is going faster than they should.

Instrument panel. A set of indicators in front of the driver that give information about the condition and operation of the vehicle.

Lane departure warning (LDW)

This system warns the driver when they make an unexpected lane change because they are distracted or because they have fallen asleep. The system warns by a light on the instrument panel, a sound, or by making the driver's seat or the steering wheel vibrate.

Watch video
Lane keeping assist (LKA)

A system that detects the white lines on the road to help the driver guide their car and not leave their lane. It warns the driver when they leave the lane.

Watch video
Reversing camera and 360-degree camera

The reversing camera is located at the rear of the vehicle. Its function is to show the driver the obstacles behind the car when reversing. For example, when they have to reverse to park the vehicle. What this camera sees appears on the screen at the front of the vehicle. The 360-degree camera captures everything around the vehicle and shows it on the screen at the front. These types of cameras are very useful for parking the vehicle.

Parking assistant

A system to help the driver park or leave a parking space. There are different parking assistants with different functions such as:

Finding a place to park, showing what is around the vehicle, or turning the steering wheel without help from the driver while the vehicle parks. The driver must pay close attention to make sure these movements can be done safely and can stop the manoeuvre at any time.

Hill start assist system

This system helps the vehicle not roll backwards when the driver moves off on a very steep slope. Or when the driver moves their foot from the brake to the accelerator to continue driving on a slope.

Rear cross traffic alert system (RCTA)

This system warns the driver if there are other vehicles behind when reversing. For example, when leaving a parking space. The system will make a sound when it sees that there is a vehicle behind that is too close and there may be a risk of a crash.

Topic 7. Vehicle lighting system

Contents

General rules for using the vehicle lights

Vehicle lights

- Position lights
- Parking light
- Clearance lights
- Main beam or full beam

- Dipped beam or passing beam
- Fog lights
- Reflector
- Direction indicator
- Hazard warning lights
- Brake light
- Third brake light
- Reversing light

Audible warnings

General rules for using the vehicle lights When must you switch on the vehicle lights? At night you must have the lights on that apply at each moment whenever it is night. During the day, you must switch them on in these cases:

- It is a dark day with little natural light.
- You go through a tunnel.
- You drive in a reversible lane Reversible lane. A lane that is sometimes open in one direction and other times in the opposite direction.
- You drive in an additional lane or one that goes in the opposite direction to the one it normally has. For example, an additional lane is added or the direction of a lane is changed for a time because of roadworks or because of an accident in that area. Additional lane. A new lane that is opened temporarily because of roadworks, queues, an accident...

If some lights break while you are driving, you must reduce the car's speed and switch on the lights that work, even if they are less bright. When you reach a lit area, you must park the vehicle and not continue driving until the lighting system is repaired. A road is considered poorly lit when:

- You cannot read the number plate of the vehicle in front from 10 metres away.
- You cannot see a vehicle painted a dark colour from 50 metres away.

Vehicle lights

1. Front position light
2. Dipped beam or passing beam
3. Main beam or full beam
4. Direction indicators
5. Front fog light
6. Daytime running lights
7. Rear position light
8. Brake light
9. Rear fog light
10. Direction indicators

11. Reversing light
12. Third brake light

Position lights

They are used so that the vehicle can be seen well and so that people know how wide it is. There are position lights at the front, at the rear and on the sides of the vehicle.

Watch video

What colour are the position lights? The front lights are white. The rear lights are red. The side lights are amber. When are they used?

- When you drive at night.
- When you go through a tunnel.
- On dark days when there is little natural light for driving. When the vehicle is moving, the position lights always switch on with another type of lights.

Watch video

The position lights are only switched on by themselves, without another type of lights, when the car is stopped because of some circumstance.

For example, a car that has to stop because of a breakdown on a poorly lit road.

Parking light

It is used when a vehicle has to stop in an area that is poorly lit. It can be used instead of the sidelights. They are placed at the front and at the back of the vehicle. They are the same colours as the sidelights.

Outline marker lights

They are used to show clearly the total width of the vehicle. They are two front lights and two rear lights that are placed on the outer edges of the vehicle, in the highest area, so that they can be seen well. They are used in the same cases as the sidelights.

Outline. Width of a vehicle

Watch video

What colour are they? The front lights are white. The rear lights are red.

Which vehicles must have them? It is compulsory for cars, trailers and semi-trailers that are more than 2.10 metres wide. It is optional for cars that are between 1.80 metres and 2.10 metres wide.

Main beam or full beam

It is used to light a long distance in front of the vehicle that is driving on the road. This light is compulsory for all cars and optional for mopeds. It is a powerful white light that can dazzle other vehicles.

Watch video

When is it used? On roads that are outside towns or cities. It is compulsory to have it on:

- When you drive on a road at night, there is little natural light and you are going faster than 40 kilometres per hour.

- At any time of day in tunnels outside a built-up area when the tunnel is not well lit. It is optional to switch it on when you drive at less than 40 kilometres per hour. When is it forbidden to switch on the main beam?
- Inside towns and cities.
- When the vehicle is stopped. You can use the main beam to warn other drivers of a danger. You do this by switching the lights on and off in an intermittent way so as not to dazzle.

Dipped beam or passing beam

It is used to light the road in front of the vehicle without dazzling or bothering the drivers of other vehicles. It is white.

It is compulsory for:

- Cars: they must have two lights.
- Two- and three-wheel mopeds: they must have one or two lights.

Motorcycles, three-wheel vehicles and heavy quadricycles can have one or two. When is it compulsory to use it? At night, on all types of roads, inside and outside towns and cities. During the day, it is used when driving through tunnels or through temporary lanes that are put in place because of roadworks, accidents or other circumstances.

These lights must also be used when:

- The vehicle does not have main beam.
- You drive at less than 40 kilometres per hour.
- There is a chance that you could dazzle the drivers of other vehicles when you switch on the main beam. Motorcycles must always have the dipped beam on during the day when they travel on any street or road.

Watch video
Fog lights

They are used to see the road better in fog, snow, heavy rain and smoke. There are front and rear fog lights. The front lights are white or yellow. The rear lights are red

How are they used?

Front fog lights are used with the sidelights. You can also use them at the same time as the main beam and the dipped beam.

Rear fog lights you must only use in cases of great need, when the weather and visibility conditions are very bad. They can be switched on at the same time as the main beam and dipped beam and the front fog lights.

Watch video
Reflector

It is a device that reflects external light from other vehicles. It is used so that the vehicle can be seen. This type of lights are on the front, rear and side of the vehicle. Reflectors must be:

- White at the front.
- Red at the rear.

- Yellow at the sides.

Watch video

Trailers and semi-trailers must have them in a triangular shape at the rear.

Direction indicator

It is used to warn that a vehicle is going to turn and move to the right or to the left. You must switch it on whenever you are going to change direction, direction of travel, or change lane. You have to switch it off when you finish that movement. Its light is yellow and flashes. It is compulsory for all motor vehicles, trailers and semi-trailers.

Watch video
Hazard warning lights

They warn that the vehicle has a problem and can be a danger to other vehicles that are on the road. To show the hazard warning lights all the vehicle's direction indicators are switched on. These lights are compulsory for cars, their trailers and semi-trailers.

Watch video

When are they used? They are used in the following cases both by day and by night.

- When a vehicle has a breakdown and cannot reach the minimum speed that you must drive at on that road.
- When a vehicle is making an emergency journey. For example, taking to hospital a woman who is about to give birth.
- To warn that another vehicle is stopped because of a problem or an emergency on any road or in a tunnel.
- When picking up and dropping off passengers in school transport vehicles.

Brake light

It shows that the vehicle is braking. The driver must warn with these lights whenever they are going to brake the vehicle, even if it is for a short time. For example, stopping at a red traffic light. Its light is a strong red and they are placed at the back of the vehicle. It is compulsory for all cars, mopeds, trailers and semi-trailers.

Watch video
Third brake light

It is a single light that is placed at the back of the vehicle, above the brake lights. It switches on at the same time as the brake lights and is the same colour. This light is optional for cars. Motorcycles cannot have it.

Reversing light

It lights the road from the back of the vehicle to warn that the vehicle is going to move backwards. They are one or two white lights that switch on automatically when the vehicle starts to go backwards. This light is compulsory for all cars, except motorcycles, which are not allowed to have it. It is optional for three-wheel vehicles and heavy quadricycles.

Watch video
Audible warnings

They are sounds used to warn other drivers that a vehicle is there. They can only be used in the following cases.

- To avoid a possible accident.
- On roads that are outside a town or city, to warn a driver that you are going to overtake them.
- To warn other vehicles that you need to pass or park there because you are doing a priority service. For example, when you take a seriously injured person to hospital.

Topic 8. Traffic signs

Contents

Types of signs
General rules

- It is forbidden to change road signs without a justified reason.
- The instruction of each sign must be followed across the full width of the street or road. In cases where there is a different instruction for each lane there will be road markings that show it.
- All drivers and pedestrians must obey the instructions of traffic signs.

Watch video

Which traffic signs must you obey first?

1

Signs and orders from traffic officers.

2

Signs that show that the road has been changed temporarily for some reason and signs that warn of bends and obstacles.

3

Traffic lights. 4 Vertical traffic signs. 5 Road markings and lines.

Watch video
Signs on the right and left

- You must obey vertical signs and traffic lights that are on your right on the road.
- When you want to turn left or go straight on and there are no vertical signs or traffic lights on the right you must obey the signs that are on the left.
- You must obey the signs on your left when you want to turn left or go straight on and there are vertical signs or traffic lights with different instructions on each side.

Contradictory signs

Sometimes there are signs placed together, but they give different instructions.

Different signs

Which one must you obey?

Stop sign and traffic light on green

Traffic light on green

Stop sign and give way sign

Stop sign

Traffic light on green and no left turn sign You have to obey both. You can go straight on or turn right.

Traffic officers
Signs and orders from traffic officers

The signs given by traffic officers must always be followed. To give instructions they will use objects and clothing that can be seen from 150 metres away.

Traffic officers will give instructions by the following means:

Arm signals

Sound signals with a whistle

Signals from a vehicle

Arm signals

Arm raised vertically

All drivers approaching the officer must stop. When this signal is made at a junction drivers who were already inside the junction can continue driving.

Arm or arms extended horizontally

All drivers approaching the officer must stop. The officer’s arm or arms act as a barrier for vehicles that are approaching. This order must be obeyed until the officer gives another instruction, even if they lower their arms.

Swings a red or yellow light with one arm

Drivers towards whom the officer directs the light must stop.

Arm extended moving up and down

All drivers approaching the officer from the side from which the officer makes the arm signal must reduce the vehicle’s speed.

Sound signals with a whistle

Several short whistle blasts repeated Stop the vehicle.

One long whistle blast Start driving again and continue driving.

Signals from a vehicle
Red flag

Vehicles cannot drive on that road.

Green flag

Vehicles can drive again on that road.

Yellow flag

Drivers and pedestrians must travel with great care because there is a possible danger on the road.

Arm extended downwards and fixed You must stop on the right-hand side drivers at whom the arm points.

Police vehicle with a red or yellow flashing light and making sounds You must stop the vehicle on the right-hand side, in front of the police vehicle and stay inside the vehicle following all the officer’s instructions.

Temporary signs

Panels with changing messages

Roadworks signs

Panels with changing messages

They are panels placed on roads that change the information depending on traffic conditions. They are used to:

- Give information to drivers.
- Warn of possible dangers.
- Give recommendations and instructions that must be followed.

Roadworks signs

They are lights, signs and devices to highlight roadworks. They show the direction you must follow on a street or road and the obstacles you may find on them.

Temporary direction panel It forbids entry and tells you which way you must go.

Small flags and cones

They forbid passing through them and through the space between each flag or cone.

Steady red light

The road is closed to traffic.

Steady or flashing yellow lights They forbid passing through the space between the lights.

Permanent direction panels

Devices that show which way you must go in a place where there is always a possible danger. The number of panels warns how much danger there is in that area:

- One panel means moderate danger.
- Two panels mean quite a lot of danger.
- Three panels mean a lot of danger.

Traffic lights

There are different types of traffic lights: For pedestrians

Circular for vehicles

Lane

For some vehicles

Traffic lights for pedestrians

Their indications are for pedestrians.

Not for drivers

Watch video
Circular traffic lights for vehicles
Steady red light

No entry.

One or two flashing red lights It forbids entry for a time. They are placed before a level crossing or a bridge.

Steady yellow light It forbids entry like the steady red light. But it does allow vehicles to go through if they cannot stop for safety reasons.

One or two flashing yellow lights They make you give way to vehicles coming from the right and from the left. Also to pedestrians. Steady green light It allows vehicles to go through and it also gives them priority at the junction.

Black arrow

The vehicle can only go towards the side shown by the arrow.

Also, you must obey the colour behind the arrow. For example, do not go through if it is red.

Green arrow

It allows vehicles to go forward in the direction shown by the arrow. You must do it carefully, watching for pedestrians crossing the road and vehicles entering that lane.

Watch video

Square traffic lights for vehicles or lane traffic lights

Only vehicles travelling in the lane where the traffic light is placed must obey them.

Meaning of their lights

Red light in the shape of a cross You cannot use that lane.

All drivers must leave it as quickly as possible.

Green light in the shape of a downward arrow You are allowed to drive in that lane.

Drivers must obey the rest of the signs in that lane.

White or yellow light in the shape of a downward arrow It tells drivers that they must move to the lane shown by the arrow because the lane they are in now is going to close.

Traffic lights for some vehicles
Traffic lights for cycles and mopeds

A cycle is shown on the traffic light. Their indications do not apply to other vehicles.

Traffic lights for trams, buses and other vehicles They have a white line on a black circular background. Their indications do not apply to cars.

Vertical signs

They are plates with information that stay fixed on posts or other structures on the road and in the streets.

Vertical signs depending on what their function is:

1. They warn of dangers
2. They order or forbid
3. They inform and guide Remember... A route is the place where you travel, it can be a street, a road, or a track. The carriageway is the part of the route where vehicles travel.

Signs that warn of danger

The name of these signs starts with a P for danger.

P-1. Priority to go through.

Vehicles on that road have priority to go through before vehicles coming from side roads or streets.

P-1a Priority to go through before vehicles coming from the road on the right.

P-1b Priority to go through before vehicles coming from the road on the left.

P-1c Priority to go through before vehicles that want to enter that road from the right.

P-1d Priority to go through before vehicles that want to enter that road from the left. P-2 Junction.

Vehicles coming from the right have priority to go through. P-3 Traffic lights.

Nearby there is a junction or a section of road with traffic lights to control traffic. You must drive carefully because there may be vehicles queued and stopped at the traffic lights.

P-4 Roundabout.

Vehicles can only turn in the direction of the arrows. P-5 Opening bridge. There is a bridge nearby that can be raised or turned. When it is raised or turned, traffic is stopped and vehicles must wait. P-6 Tram crossing.

Danger because nearby there is a junction with a tram line that has priority to go through. P-7 Level crossing with barriers nearby. A level crossing is the place where train tracks cross a path or road. On the road, before you reach the train track, you will find a barrier. P-8 Level crossing without barriers nearby. There will be no barrier separating the road you are travelling on and the train track. P-9a and P-10a There is a level crossing, opening bridge, or quay about 300 metres away. P-9b and P-10b There is a level crossing, opening bridge, or quay about 200 metres away. P-9c and P-10c There is a level crossing, opening bridge, or quay about 100 metres away.

P-11 Level crossing without barriers

In that same place.

P-11a Level crossing without barriers where there is more than one train track. In that same place.

P-12 Airport

Danger from unexpected noises that aircraft can cause. P-13a Dangerous bend to the right. P-13b Dangerous bend to the left. P-14a Several dangerous bends nearby. The first goes to the right. P-14b Several dangerous bends nearby. The first goes to the left. P-15 Uneven surface. There are humps, dips, or the road is in poor condition. Humps are parts of the road that slope upwards and stick out to make vehicles reduce speed. Dips are also parts of the road, they are potholes or trenches in the road. P-15a There are humps in the road. P-15b There are dips in the road.

P-16a Very steep downhill slope. P-16b Very steep uphill slope. P-17 The carriageway becomes narrower.

P-17a The carriageway becomes narrower on the right.

P-17b The carriageway becomes narrower on the left. P-18 Roadworks. P-19 Slippery carriageway. P-20 Pedestrians.

Careful, you are coming to a place where there are often pedestrians. P-21 There are children near that place. For example, the exit of a school.

P-22 Place where cyclists often go or cross. P-23 Place where domestic animals may often go. For example, cows, sheep...

P-24 Place where animals in the wild may cross.

P-25 Vehicles can travel in both directions.

P-26 There may be stones or other obstacles on the road because it is an area where objects fall. For example, a road below a mountain.

P-27 The road ends at the quay of a port or at a watercourse. P-28 Small stones jump up from the road surface, called gravel.

P-29 There is strong side wind. P-30 The carriageway is at a different height on one side because there is a step. P-31 There is a lot of traffic on a section of the road.

P-32 There are vehicles that make traffic difficult because of a breakdown, an accident, or other causes. P-33 There is fog, snow, or smoke on the road and visibility is worse.

P-34 An area of the carriageway is very slippery because of ice or snow.

P-35 Warning of a danger different from all the previous ones.

Regulatory signs

They are signs that tell drivers and pedestrians about their duties, limits, and bans on the road.

Regulatory signs placed next to or above the sign that shows the name of the town or city mean that you must follow that rule in the whole town or city. For example, a circular sign with the number 40 above the name of a town means that it is forbidden to drive at more than 40 kilometres per hour in the whole town. If the circular sign is before or after, it means that you cannot drive at more than 40 kilometres per hour on that section, until you find another sign.

40
JACA
40
JACA

The names of regulatory signs start with the letter R and there are different types:

Priority signs

R-1 Give way. You must give way to all vehicles already travelling on the road you want to enter or that are in the lane you want to go into. R-2 Stop. You must stop the vehicle and let vehicles already travelling on the road you want to enter go through. R-3 Priority road. While you are on that road you have priority to go through at junctions before drivers who reach the junction from another road. R-4 End of priority road. The road you are travelling on stops having more priority than other roads.

R-5 Do not enter a narrow passage if, when you go through it, you obstruct vehicles coming towards you.

R-6 You have priority to go through a narrow passage before vehicles coming in the opposite direction.

Signs that forbid entry

R-100 No traffic. It forbids all vehicles in any direction. R-101 No entry. It forbids entry to all types of vehicles.

Vehicles may be travelling in the opposite direction to you because this ban may only be in one direction. R-102 No entry for motor vehicles. Mopeds can enter. R-103 No entry for motor vehicles. Two-wheeled motorcycles, without a sidecar, can enter. R-104 No entry for motorcycles.

R-105 No entry for mopeds and vehicles for people with reduced mobility.

R-106 No entry for vehicles that carry goods, even if they do not carry much load. For example, vans and lorries.

R-107 No entry for vehicles that carry goods and can carry a load in tonnes greater than the number shown on the sign. These vehicles are forbidden to enter, even if they are empty at that time.

R-108 No entry for vehicles carrying dangerous goods.

R-109 No entry for vehicles with goods that can explode or catch fire.

R-110 No entry for vehicles carrying more than 1,000 litres of products that can pollute water.

R-111 No entry for tractors and other motor agricultural machines.

R-112 No entry for motor vehicles with a trailer.

Articulated vehicles and single-axle trailers can enter.

R-113 No entry for vehicles pulled by animals. For example, horse-drawn carriages. R-114 No entry for bicycles.

R-115 No entry for handcarts pushed by people. R-116 No entry for pedestrians.

R-117 No entry for animals ridden by people.

Signs that limit passage

R-200 You must stop the vehicle at the place where it is placed to comply with the signs in each case. For example, a toll booth on the motorway or a police checkpoint.

R-201 No entry for vehicles carrying a load that weighs more than the number of tonnes shown on the sign.

R-202 No entry for vehicles with an axle load greater than the number of tonnes shown on the sign. In this case, vehicles cannot pass if they have an axle load greater than 2.4 tonnes.

R-203 No entry for vehicles that are longer than the size shown on the sign.

R-204 No entry for vehicles that are wider than the size shown on the sign.

R-205 No entry for vehicles that are higher than the size shown on the sign.

Other prohibition or restriction signs

R-300 Minimum distance from the vehicle in front. You must keep from the vehicle in front the metres of distance shown on the sign. R-301 Maximum permitted speed. You must not drive faster, in kilometres per hour, than the speed shown on the sign. This applies until you see another sign for end of speed limit, end of prohibitions, or another sign for maximum speed. R-302 No right turn.

R-303 No left turn and no U-turn.

R-304 No U-turn. You cannot turn round and go the other way. R-305 No overtaking other vehicles. You can only overtake two-wheeled motorcycles if you do not enter the opposite lane. You must obey this sign until another one appears that shows that you can overtake in that area.

R-306 No overtaking for lorries that can carry loads of more than 3,500 kilos. They can only overtake two-wheeled motorcycles if they do not enter the opposite lane. R-307 No stopping and no parking in that place.

R-308 No parking on that side of the carriageway. You can stop, but not park.

R-308a No parking on that side of the carriageway on odd days of the month. You can stop, but not park.

R-308b No parking on that side of the carriageway on even days of the month. For example, days 2, 4, 6... You can stop, but not park.

R-308c No parking on that side of the carriageway for the first 15 days of each month. You can stop, but not park.

R-308d No parking on that side of the carriageway between day 16 and the last day of each month. You can stop, but not park.

R-308e No parking in front of a dropped kerb. Dropped kerbs are spaces in the street reserved so that some vehicles can enter their garages, homes, or shops. R-309 Limited-time parking area. The driver must show the time when they parked the vehicle.

R-310 Remember that you should only make sounds and horn beeps with the vehicle when they are needed to avoid accidents.

Mandatory signs

R-400a Must go to the right.

R-400b Must go to the left. R-400c Must go straight on. R-400d Must turn right. R-400e Must turn left.

R-401a Must pass on the side shown by the sign.

R-401b Must pass on the side shown by the sign.

R-401c Must pass on the sides shown by the sign.

R-402 Must turn in the direction shown by the arrows.

R-403a Must turn right or go straight on. You cannot make a U-turn.

R-403b Must turn left or go straight on. You cannot make a U-turn.

R-403c Must turn right or left. You cannot make a U-turn.

R-404 Cars must use that carriageway.

Motorcycles without a sidecar are not required to.

R-405 Motorcycles without a sidecar must use that carriageway.

R-406 Lorries, vans and light vans must use that carriageway. R-407a Road reserved only for bicycles. No one else may use this road. R-407b Road reserved only for mopeds. No one else may use this road.

R-408 Vehicles pulled by animals must use that path. Others can also use that path.

R-409 People riding an animal must use that path. No one else may use that path.

R-410 Pedestrians must use that path. No one else may use that path. R-411 Minimum speed.

Vehicles must travel at least at the speed shown on the sign. You cannot drive more slowly. You must follow this rule until you find another sign that allows you to drive at a lower speed.

R-412 You must fit snow chains on the vehicle to continue driving. R-413 You must switch on dipped headlights. You must keep them on until another sign tells you that you can switch them off.

R-414 Vehicles carrying dangerous goods must use that road.

R-415 Vehicles carrying more than 1,000 litres of products that can pollute water must use that road.

R-416 Vehicles carrying explosive material or material that can catch fire must use that road.

R-418 Lane only for vehicles that have an electronic toll device. The electronic toll is an electronic device that is placed on the windscreen and that recognises your vehicle number plate so that the motorway barrier opens automatically.

Signs that end the prohibition or the obligation R-500 End of all prohibitions. It shows the place from which all prohibitions end. R-501 End of speed limit. You can go faster. R-502 End of no overtaking. You can overtake other vehicles.

R-503 End of no overtaking for lorries.

Lorries can overtake other vehicles.

R-504 End of limited parking area. Vehicles can park in that place. R-505 End of road reserved for bicycles. R-506 End of minimum speed. You can drive more slowly than the speed shown on the sign.

Signs that give information of interest

They give information of interest to drivers and pedestrians. They are called information signs. There are different types of information signs.

General information signs

S-1 Place where a motorway begins. S-1a Place where a dual carriageway begins. S-2 Place where a motorway ends. S-2a Place where a dual carriageway ends.

S-3 Start of a road reserved for cars.

S-4 End of a road reserved for cars. S-5 A tunnel begins. S-6 End of the tunnel. S-7 Recommended maximum speed. It recommends the maximum speed you should always drive at on that road. S-8 End of recommended maximum speed. The stretch of road ends where it is recommended to drive at that speed or more slowly. S-9 Recommended speeds. It recommends driving between those two speeds shown on the sign. S-10 End of recommended speeds. It shows the place from which it is not recommended to keep driving at those speeds.

S-11 One-way carriageway. They show the direction in which vehicles must travel on that road. They also ban U-turns (turning round). The arrows show the number of lanes S-12 One-way section of carriageway. It shows that on that section of street or road you must travel in the direction the arrow points. It is forbidden to do it in the other direction. S-13 Pedestrian crossing. S-15a No through road.

Vehicles can only leave that carriageway by the same place they entered. S-16 Emergency stopping area. It shows an area where vehicles can pull off and stop when the brakes fail. S-17 Place where vehicles can park.

S-18 Taxi rank. S-19 Bus stop. Mountain pass. Road that crosses a mountain. S-21 Driving over a mountain pass. It shows the conditions of the mountain pass and whether you can drive over it or not. The sign has three panels and each one shows one thing: S-21.1 a, b, c, d and e. Panel number 1. It can be different colours White, with the word OPEN. All vehicles can drive.

Green. All vehicles can drive. But lorries that can carry more than 3,500 kilos cannot overtake other vehicles.

Yellow. Cars and buses must drive at a maximum speed of 60 kilometres per hour.

Lorries that can carry more than 3,500 kilos and articulated lorries cannot pass.

Red. Vehicles must use chains and drive at a maximum speed of 30 kilometres per hour. Lorries and buses cannot pass. Black, with the word CLOSED.

No vehicle can pass on that road.

S-21.2 a, b, c and d. Panel number 2. It can have the following signs: R-306 when panel 1 is green.

R-106 and R-301 you must drive at less than 60 kilometres per hour when panel 1 is yellow. R-107 with the text 3,5 t and R-412 when panel 1 is red. S_21.3 a and b. Panel number 3. It can show the name of the place from which you must follow the instructions of panels 1 and 2. S-22 You can make a U-turn. S-23 There is a hospital nearby. It warns drivers to make little noise with their vehicles when passing in front. S-24 You can stop using dipped headlights.

S-25 You can make a U-turn using a road that is higher or lower.

S-26 a There is a motorway, dual carriageway, or road for cars exit in 300 metres. S-26 b There is a motorway, dual carriageway, or road for cars exit in 200 metres. S-26 c There is a motorway, dual carriageway, or road for cars exit in 100 metres.

S-27 There is a roadside emergency post to ask for help in case of an accident or breakdown. S-28 Area with priority for pedestrians. You must not drive faster than 20 kilometres per hour. S-30 Area where pedestrians can walk. You must not drive faster than 30 kilometres per hour.

S-31 End of the area with priority for pedestrians. You can drive at more than 30 kilometres per hour. S-32 Electronic toll.

Vehicles can pay the toll through electronic toll as long as the vehicle has the necessary technical equipment installed.

S-33 There is a path reserved for pedestrians and bicycles, separated from motor traffic, in parks, gardens, and woods. S-34 There is a place where you can park a vehicle inside a tunnel in case of emergency or breakdown. S-34a It shows that this place has an emergency telephone.

Signs that give information about lanes
S-50a, S-50b, S-50c, S-50d and S-50e

Only vehicles that travel at that speed or faster may use the lanes marked with a number.

S-51 Lane reserved for buses.

S-52, S-52a and S-52b

It warns that a lane can no longer be used and shows the lane you can continue in.

S-53c Road that changes from two lanes to three lanes. It also shows the maximum speed you can drive in each one.

S-60a. On a two-lane road it shows that the left lane is going to move to the left.

S-60b. On a two-lane road it shows that the right lane is going to move to the right.

S-61a. On a three-lane road it shows that the left lane is going to move to the left.

S-61b. On a three-lane road it shows that the right lane is going to move to the right. S-64 Cycle lane or cycle track on the road. Only cycles may use that lane.

Signs that give information about services

S-100 First aid post where you can get emergency treatment.

S-101 Place where there are ambulances to help and take people injured in traffic accidents.

S-102 Place where you can have the vehicle checked. S-103 There is a vehicle repair workshop. S-104 There is a telephone.

S-105 Place where you can put fuel in the vehicle.

S-106 Place where you can repair the vehicle and put in fuel.

S-107 Place where you can camp. S-108 There is a water fountain. S-109 There is a nice place to see. S-110 There is a hotel or a motel. S-111 There is a restaurant. S-112 There is a bar or café. S-113 Caravans can park.

S-114 You can stop to eat.

S-115 From that place you can start a walking trip.

S-116 You can camp in that place with a tent and with a caravan. S-117 There is a youth hostel. S-118 There is a tourist office. S-119 Area of a river where you need a special permit to fish. S-120 There is a national park.

S-121 There are monuments to see and visit. S-122 Sign to show other services.

S-123 Rest area to stop to rest.

S-124 Area to park vehicles that connects with a train station.

S-125 Area to park vehicles that connects with a lower train station.

S-126 Area to park vehicles that connects with a bus station.

S-127 Service area inside a motorway or dual carriageway. In service areas there are usually petrol stations, restaurants and places to rest for a while.

Direction signs

S-200 Shows all the exits there are at the next roundabout. The exits painted blue mean that they lead to a motorway or dual carriageway.

S-222a Shows the direction of a road and the exits that lead to a motorway or dual carriageway.

S-232 Shows where the entrance is to the motorway and at what distance.

S-235a Shows which is the next exit on a motorway or dual carriageway and how far away it is.

S-300 Shows the names of towns and cities on a road and how far away that town or city is.

S-301 Shows the names of towns and cities on a motorway or dual carriageway and how far away that town or city is. S-322 Shows in which direction there is a cycle lane or cycle path and the distance to it.

S-360 Shows what the number is of the road, the place that road goes to, and the name of the town or city at the next exit towards another road.

S-368 Shows the number of the motorway or dual carriageway, where they go to, and the name of the town or city at the next exit towards another road.

Signs to classify roads

S-400 European road.

Name given to that road in Europe. S-410 Motorway or dual carriageway. S-410a Toll motorway. S-420 National road, of the Spanish State.

S-430 More important road of an Autonomous Community. They are called regional roads of first level.

S-440 Roads that link towns within an Autonomous Community or are used to reach the first-level roads. They are called regional roads of second level.

C-241 Roads that link small towns. They are called regional roads of third level.

Signs that show where you are

S-500 Entry to a town or city. S-510 Exit from a town or city.

S-574 Shows the kilometre where you are on a motorway or dual carriageway, and you will see them every 10 kilometres. That is, from the start of the motorway or dual carriageway it marks kilometre 10, kilometre 20, kilometre 30, until the end.

S-574a Shows the kilometre where you are on a road, and you will see them every 10 kilometres, the same as on the motorway and dual carriageway.

S-574b Shows the kilometre where you are on a toll motorway, and you will see them every 10 kilometres, the same as the rest.

S-575 Shows the kilometre where you are on a regional road, and you will also see them every 10 kilometres. The colour matches the level of importance of the road.

Supplementary panels

S-800 Shows the distance there is from the sign to a danger or warning.

S-810 Shows how long the dangerous section is. For how many metres or kilometres there is danger.

S-820 and S-821 These signs are placed under a prohibition sign. They show how many metres that prohibition lasts following the arrow.

S-850 to S-853 Shows in which direction and way you can drive with priority. It is placed next to sign R-3, which means road with priority.

S-840 It is placed under the

Give way sign. It shows the distance at which you must stop. S-870 It is placed under another sign. It shows that the prohibition or warning on the other sign only applies to the lane or slip road it points to. S-880 It is placed under another sign. It shows that the prohibition or warning on the other sign only has to be followed by the vehicles shown on this sign. S-890 It is placed under another sign. It shows that the prohibition or warning on the other sign must be followed when there is snow, rain or fog. S-960 There is an emergency telephone.

S-980 There is an emergency exit.

S-990 Shows where the emergency exit is in tunnels.

Lines and markings on roads
White lines and markings
Continuous line

This line forbids drivers to:

- Cross the line.
- Drive on top of it.
- Drive to the left of the line when the road has two-way traffic. When the line is a bit wider than normal it shows that there is a special lane. For example, a lane that only some vehicles can use.

Broken line

This line forbids drivers to drive on top of it. You can only drive on top of it when the lane is less than three metres wide and it is necessary. When the gap between the broken lines is shorter than normal it means a continuous line is near or a dangerous situation. For example, a bend with poor visibility. When the gap between the lines is wider than normal it means there is a special lane nearby.

See video

Broken line. Line made up of many smaller lines in a row, with empty spaces between them.

Double broken lines

Double broken lines on both sides of a lane mean that in this lane sometimes you can drive in one direction and other times in the other. There will be traffic lights in that lane or other means to show which direction you can drive and avoid accidents.

Continuous and broken lines together

In this case, each driver must only take into account the line that is closer to the side they are driving on.

Edge and parking lines

Lines that mark where the road ends or places where you can park. You can cross them or drive on them.

Continuous transverse line

All vehicles must stop at this type of line when the following signs show it: The signs that show that stopping is compulsory.

- Pedestrian crossings.
- Traffic lights.
- Instructions from traffic officers.
- Signs at a level crossing or movable bridge.

STOP

In any case, the marking does not by itself make it compulsory.

Transverse line. Line that crosses the road across its width, instead of along it.

Broken transverse line

Vehicles must not cross it if they must give way to other vehicles following the priority rules and following the instructions shown by these signs:

- Give way sign.
- Green turning arrow of a traffic light.

Pedestrian crossing

Wide lines on the road that show that drivers must give way to pedestrians.

Cycle crossing

Broken and parallel lines on the road that show that cyclists have priority to ride through that place.

Stop sign written on the road

The driver must stop their vehicle before the stop line to give way to drivers travelling on the road they want to enter.

STOP

Speed limit sign written on the road

No vehicle must go at more kilometres per hour than the number shown by the sign in that lane. You must follow that instruction until another sign says you can drive at another speed.

Arrow to choose lane

All drivers must follow the direction or one of the directions shown by the arrows in the lane they are using. If the signs allow it you can also change lane.

End of lane arrow

It warns that the lane you are in ends and shows you where you must go.

Return to lane arrow

It warns that a continuous line is coming and tells drivers that they must move as soon as possible to the right of the lane.

Level crossing marking

The letters P and N written on the ground with two lines in the shape of a cross show that there is a level crossing nearby.

Reserved lane or area

It shows that that area or lane is reserved for certain vehicles to drive or stop. For example, buses and taxis.

BUS

Marking for the start of a reserved lane It shows where the reserved lane starts for some vehicles.

Cycle lane or cycle path marking

It shows that that lane is reserved for bicycles to use.

Zebra markings

It forbids vehicles to drive in that area.

Only vehicles that must use the hard shoulder can drive in that space.

STOP

Coloured lines and markings
Yellow line with zigzag markings

It shows that parking is forbidden in that area.

Continuous yellow line

It shows that stopping and parking are forbidden in that area.

Broken yellow line

It shows that parking is forbidden in that area.

Marking made up of yellow squares

It tells drivers that it is forbidden to enter that area if they could end up stopped in it blocking traffic.

White and red squares

Area to do an emergency stop. It is the only reason why a vehicle can enter that area.

Blue parking markings

At some times of the day you can only park for a limited time and you must pay.

Topic 9. The road

Contents

Driving on motorways and dual carriageways

- Emergencies on the motorway and dual carriageway

The road for vehicles and pedestrians What is the road? A road is each street, road and track, public and private, where vehicles and pedestrians can travel.

Parts of the road for vehicles

The parts of the road that matter to the driver are:

Lane
Carriageway platform
Carriageway
Hard shoulder
Carriageway platform

It is the whole area of the road that vehicles can use to drive and to stop. Within the carriageway platform are the carriageway and the hard shoulder.

Carriageway

Part of the road where vehicles travel. The same road can have several carriageways. A narrow carriageway is one that is less than 6.5 metres.

Hard shoulder

Paved area that is on both sides of the carriageway where most vehicles do not travel. Some roads do not have a hard shoulder. A road can have several carriageway platforms.

Lane

It is one of the parts into which the carriageway can be divided.

Its width must allow a line of cars to travel. On the same carriageway there can be several lanes, depending on how wide it is. They are usually separated by a white line. When the lane is less than 3 metres wide, it is called a narrow lane.

Other areas of the road
Straight

Section of road that does not change direction.

Bend

Section of the road that changes direction to the right or to the left. In some bends you cannot see the full width of the road. In that case it is called a bend with reduced visibility.

Crest

It is the slope of the road. A change of crest means that a section of the road changes its slope. There are changes of crest with reduced visibility where the road is very steep and you cannot see the vehicles in front of you or coming from the other direction.

Central reservation

Space between two separate carriageways where vehicles do not drive.

Lay-by

Part of the carriageway that widens so that vehicles can stop if they need to without affecting the other vehicles driving on that carriageway.

Emergency braking area

Part of the road prepared so that vehicles can stop when their brakes fail. When entering that area, the vehicle stops even if the brakes do not work.

Junction

Place where several roads, streets, or tracks cross.

Roundabout

Junction where there is a circular structure in the centre and vehicles must drive around that structure. You cannot go through the centre.

Level crossing

Place where a road or track and the railway tracks cross.

Traffic island

Area of the carriageway that is slightly raised or painted, used to guide traffic. They are usually placed at junctions. The parallel white lines that form the traffic island are called hatched markings.

Cycle route

Route prepared for cycles to use.

There are several types of cycle route:

Cycle lane

It is next to the carriageway. It can work in two directions or in one direction only.

Protected cycle lane It is separated from the carriageway or from the pavement by physical elements to give more protection to the people who use it.

Cycle pavement It is marked within the pavement.

Cycle track

There is a space separating the cycle route and the road.

Cycle path Route located in parks, gardens, and woods where only pedestrians and cycles can travel.

Pedestrian area

Part of the road reserved for pedestrians. The pavement, the walkway, and the refuge are considered pedestrian areas. The refuge is a pedestrian area located on the carriageway that vehicles cannot enter. For example, an area to wait between two pedestrian crossings.

Built-up area

Places where there are buildings. That is, towns and cities. On the roads entering and leaving built-up areas there are signs with their name written on them.

Watch video
Direction and sense
Direction

Straight or curved line that links two places.

Sense

The two options, outward and return, that can be used to go in the same direction. For example, two cars that go on the same road, one in the outward lane and the other in the return lane, go in the same direction, but in a different sense. There are one-way roads and others where you can drive in both senses, in different lanes.

Exit
Entry

A
B
TWO-WAY ROAD
ONE-WAY ROAD
A
B
Types of road
Depending on where it is
Urban road

Streets and tracks within the built-up area.

Interurban road

Roads and tracks outside the built-up area.

Through road

Road that goes through a built-up area.

Depending on its characteristics

Motorway

It is a road where only motor vehicles can drive and it must meet the following characteristics:

- You cannot enter it from any land or property next to the motorway.
- You can only enter through prepared and authorised access points.
- No railway, tramway, or path crosses it, and no other road crosses it.
- It has separate carriageways so that vehicles drive in one sense or the other. Motorways can be toll roads. That is, you have to pay to drive on them.

Dual carriageway

Road that has the same characteristics as the motorway with two main differences:

- You can enter them from some properties next to the dual carriageway.

- It has more access roads.
- You do not have to pay to drive.

Road for motor vehicles

Road where only motor vehicles can drive. It has a single carriageway. As on motorways, you cannot enter it from any land or property next to it. It is signposted with signs S-3 and S-4.

Conventional road

These are all the roads that do not meet the characteristics of motorways, dual carriageways, and roads for motor vehicles.

Service road or service lane

It is a road that runs parallel to the main road and links this road with houses, petrol stations, and other properties.

Colours that show the state of the road

White

You can drive normally.

Green

You cannot reach the maximum speed allowed on that road.

Yellow

There are stops and queues of vehicles or traffic is slow in some sections.

Red

There are too many vehicles and there are many stops and queues.

Black

The road is closed and you cannot drive.

Lanes

Where should vehicles drive? They must drive on the right and as close as possible to the edge of the carriageway, especially on bends and changes of crest with reduced visibility.

Watch video

You must always keep free the part of the carriageway that belongs to the opposite sense. On roads with two carriageways, vehicles must use the carriageway on their right. On roads with three carriageways, the middle one can have one or two senses and the side ones one sense only.

When you find on the road refuges or traffic islands, you must drive on the right-hand side of the refuge or traffic island. If the road has only one sense you can drive on either side of the refuge or traffic island.

Motor vehicles normally must not drive on the hard shoulder, except for a breakdown or some other unexpected event.

Watch video
Rules for using lanes
General rules
Type of road

Where should you drive?

Two-way and with two lanes On the right.

Two-way and with three lanes On the right. You must never use the left lane. The centre lane is only used to overtake and to change direction to the left.

Within towns or cities

Motor vehicles and special vehicles can drive in the lane they prefer on roads with two lanes going in the same sense and separated by a white line.

Watch video

In these cases you must keep in mind that:

- You should only leave that lane to change direction, park, or overtake another vehicle.
- You must not block the way or be an obstacle for other vehicles. When the lanes are not separated by a white line, all drivers must drive on the right. Other vehicles, such as cycles, mopeds, or vehicles for people with reduced mobility, must always drive on the right, even if there is a separation line between lanes.

Outside towns or cities

On roads with two lanes going in the same sense, motor vehicles and special vehicles that with load can weigh more than 3,500 kilos will drive in the lane on their right. They may use the lane on their left when the road or traffic conditions make it advisable and as long as they do not hinder other drivers.

Watch video

On roads with three or more lanes going in the same sense, some vehicles have the obligation to always go in the right lane and they are forbidden to use other lanes. These vehicles that must drive in the right lane may use the lane next to it, but cannot use any more, and under the same conditions

as the previous point. These vehicles are:

- Lorries, vans, and special vehicles that with load can weigh more than 3,500 kilos.
- Vehicle combinations that are more than seven metres long.

Reserved lanes

High occupancy vehicles (HOV)

Bus and taxi For manoeuvres

Lanes reserved for certain vehicles

Some lanes are reserved for a particular type of vehicle and the other vehicles cannot use them. These lanes are:

Lane reserved for the bus

Driving in that lane is forbidden for the other vehicles.

Taxis can drive there when the word TAXI is written in the lane. This lane will be separated from the other lanes by a wide white line.

Lane for high occupancy vehicles (HOV)

For a motor vehicle to be a high occupancy vehicle it must meet three requirements:

1. It can only carry people.
2. It cannot weigh more than 3,500 kilos.
3. The number of people travelling in the vehicle is the one shown in the HOV lane signs. The following vehicles can also drive in this lane:

- Motorcycles, cars, or adaptable mixed vehicles when they meet the three requirements needed for HOV motor vehicles (see previous paragraph).
- Vehicles for people with disabilities even if only the driver is travelling.
- Buses (that weigh more than 3500 kilos).
- Vehicles providing emergency services.
- Vehicles doing road works.

Lanes reserved for some manoeuvres

Manoeuvre. Movement or action done when driving a vehicle.

Watch video

Entry lane (acceleration) For vehicles that want to enter a road.

Exit lane (deceleration) For vehicles that want to leave a road. You must take a series of precautions when entering this lane:

- Move into the lane closest to the exit.
- Enter the exit lane as soon as possible.
- Drive more slowly within this lane.
- Pay attention to the conditions of the new road you are joining. For example, when changing from a dual carriageway to another road where you must drive more slowly.

Special cases for using lanes

Reversible

Opposite to the usual

Additional
Reversible lanes

They are lanes where you can drive in one sense or the other, depending on the circumstances. They are marked with broken white lines on both sides of the lane.

All drivers must use dipped headlights when driving in this lane, day and night.

Lanes that go in the opposite sense to the usual one They can be set up when there are many vehicles that want to go in the same sense of the road. This means there are fewer traffic jams and vehicles can drive better.

Only motorcycles and cars can use these lanes. How should you drive in these lanes?

- With dipped headlights on, both day and night.
- At a speed between 60 and 80 kilometres per hour.
- Without changing into other lanes. Not even to overtake.

Vehicles that are in the lane next to it, in the normal sense of the road, must take the same precautions.

Lanes can also be set up in the opposite sense to the normal one when road works are being done. In that case, all vehicles authorised to drive on a road with road works can drive in that lane.

Additional lanes

On carriageways with two senses of traffic, using the hard shoulders, they are sometimes set up when there are many vehicles on the same road so that there is one more lane to drive in. They are marked with bollard signs. How should you drive in these lanes?

- Carefully, so you do not move or drive over the bollard signs.
- At a speed between 60 and 80 kilometres per hour. Or more slowly if the signs say so.
- With dipped headlights on, both day and night.
- Without changing into other lanes. Not even to overtake.

A
R
C
E
N
A
R
C
E
N

The hard shoulder If there is no road, or part of it, that is meant for them to travel on, the following vehicles must travel on the hard shoulder:

- Cycles.
- Vehicles following cyclists.
- Mopeds.
- Vehicles pulled by animals and riding animals.
- Special vehicles that, when loaded, do not weigh more than 3,500 kilos.
- Vehicles for people with difficulty moving. When the road has no hard shoulder or it is not possible to travel on it, these vehicles may use only the essential part of the lane.

Bicycle riders can leave the hard shoulder and travel on the right-hand part of the lane on downhill slopes with bends.

See video

All vehicles travelling on the hard shoulder must do so in single file, one behind the other. Except bicycle riders, who may ride two abreast in the same line. But on stretches with no visibility, and when there is heavy traffic, they must also ride in single file, one behind the other.

Vehicles that must travel on the hard shoulder may not overtake other vehicles when the overtaking:

- Lasts more than 15 seconds.
- Needs a space of more than 200 metres to overtake.

Bicycle riders may overtake other vehicles without following these two rules.

Travelling on motorways and dual carriageways

The following may not travel on motorways or dual carriageways:

- Pedestrians.
- Animals.
- Bicycles.
- Mopeds.
- Vehicles for people with reduced mobility.
- Personal mobility vehicles.

Bicycle riders who are over 14 years old may travel on the hard shoulder of the dual carriageway, unless there is a sign that forbids it.

On toll motorways you must go through the booths prepared to collect the entry ticket to the motorway or to pay when leaving. These booths have a traffic light or green arrow when they are open and a traffic light or red cross when they are closed.

See video
Emergencies on the motorway and dual carriageway

Vehicles that have to travel more slowly on the motorway or dual carriageway due to a breakdown must leave this road at the first exit they come to. A vehicle that has to stop because of an emergency must do so on the hard shoulder or in the central reservation. To ask for help you can use the nearest emergency phone.

People travelling in the broken-down or crashed vehicle must not walk on the road.

When a vehicle has an accident or a breakdown, another vehicle that is authorised must remove it from the road to do so. For example, a tow truck.

Topic 10. Speed and distances

Contents

- Maximum speed
- Minimum speed
- Inappropriate speed
- Appropriate speed
- Situations in which you must drive at low speed

General speed

- Speed limits outside towns and cities
- Maximum speed limits inside towns and cities
- Minimum speed limits inside and outside towns and cities
- Why are speed limits necessary?

Types of distance

- Reaction distance
- Braking distance
- Stopping distance
- Lateral separation
- Safety distance

Types of speed

It is very important to drive at the right speed to avoid dangers and road traffic accidents. 2.2 seconds 1.2 seconds 30 metres

50
Km/h
90
Km/h

There are different types of speed when driving:

Maximum speed

Minimum speed

Inappropriate speed

Appropriate speed

Maximum speed

Maximum permitted speed that a vehicle can reach on the road it is travelling on. When the vehicle goes faster than the maximum permitted speed, it is travelling with excess speed. For example, it is excess speed to travel at a speed of 100 kilometres per hour on a road where the maximum permitted speed is 90 kilometres per hour.

Minimum speed

Minimum speed at which a vehicle must travel on the road it is travelling on. When a vehicle travels at a speed lower than the permitted one, it has an abnormally reduced speed. For example, it is an abnormally reduced speed to drive at 40 kilometres per hour on a motorway where the minimum permitted speed is 60 kilometres per hour.

Inappropriate speed

Speed that is within the permitted limits but is not suitable because of the driver's condition, the weather, the state of the road, or the circumstances of the vehicle. For example, driving at a speed of 70 kilometres per hour on a road where there are ice patches and the vehicle can skid. That speed would be appropriate in normal circumstances. But in that case it is an excessive speed because there is ice on the road.

Driving at an inappropriate or excessive speed is one of the main causes of road traffic accidents.

Appropriate speed

Speed that is within the permitted limits and that is suitable for the weather conditions, the road, the circumstances of the vehicle and the driver's condition.

Appropriate speed makes it possible to control the vehicle better in unexpected situations.

People are more likely to die in a road traffic accident when the vehicle is travelling at excessive speed than when it is travelling at an appropriate speed. What must you take into account to travel at an appropriate speed? You must always respect the speed limits. But you must also take into account the circumstances at each moment. These circumstances are:

- Your physical and psychological condition. For example, if you are very tired.
- The characteristics of the vehicle and the load you are carrying. It is better to drive more slowly when you are carrying a lot of weight in the vehicle.
- The state of the road. You must go more slowly if the road has speed bumps or is badly surfaced.
- The traffic situation. You must drive more slowly when there is heavy traffic.
- Weather conditions. You must be more careful when it rains, snows, there is fog or ice on the road.

Situations in which you must drive at low speed You must drive slowly and even stop the vehicle, in the following cases:

- When there are pedestrians on the road or they may cross it. Especially when there are children, older people, people who cannot see or with another type of physical disability.
- Near lanes reserved for cyclists and in all places where cyclists travel.
- At pedestrian crossings where there is no traffic light or traffic officer.
- Near markets, schools and other places where there may be children.

See video

- On roads or tracks where there are animals or they may appear.
- On stretches where there are buildings right next to the road. People or animals may come out of those buildings.
- When approaching a bus that is stopped. Especially if it is a bus that carries children.

- When approaching any vehicle stopped on the road, emergency service vehicles, and bicycles travelling in your lane or on the hard shoulder.
- When the road is slippery or there is water on it, gravel or other materials.
- When approaching level crossings, roundabouts or junctions where you do not have priority. In these cases, the maximum speed must be 50 kilometres per hour.

See video

- When approaching narrow places or places where visibility is poor.
- When meeting another vehicle when the circumstances do not allow it to be done very safely. For example, when you meet a lorry on a bend on a narrow road.
- When you are dazzled by the lights of another vehicle.
- When there is thick fog, rain, snow or smoke.

General speed

What is it?

Speed limit that each type of vehicle must follow on each road. There are general maximum and minimum speeds.

Speed limits outside towns and cities
Type of vehicle

- Cars.
- Motorcycles.
- Three-wheeled vehicles.
- Motor caravans that can carry loads under 3,500 kilos.
- Pick up.

Road type	Max speed (km/h)	Min speed (km/h)
Motorway and dual carriageway	120	60
Roads	90	45
Tracks	30	15

On some roads the limit may be 100.

Type of vehicle

- Buses.
- Car-derived vehicles.
- Adaptable mixed vehicles.

Road type	Max speed (km/h)	Min speed (km/h)
Motorway and dual carriageway	100	60
Roads	90	45
Tracks	30	15

Type of vehicle

- Lorries.
- Articulated tractor units.
- Vans.
- Motor caravans that can carry loads over 3,500 kilos.
- Articulated vehicles.
- Vehicles with a trailer.

Road type	Max speed (km/h)	Min speed (km/h)
Motorway and dual carriageway	90	60
Roads	80	40
Tracks	30	15

Type of vehicle

- Three-wheeled vehicles that are not motorcycles.
- Heavy quadricycles.
- Special vehicles that can reach 60 kilometres per hour on flat ground.

Road type	Max speed (km/h)	Min speed (km/h)
Motorway and dual carriageway	70	60
Roads	70	35
Tracks	30	15

Type of vehicle

- Special vehicles without brake lights.
- Special vehicles that tow a trailer.
- Motor cultivators.

Motorway and dual carriageway

Roads
Tracks
Maximum and minimum speed

May not travel

25
Types of vehicle

- Other special vehicles

Road type	Speed limit
Motorway and dual carriageway	May not travel
Roads	40
Tracks	30

Types of vehicle

- Cycles

Road type	Speed limit
Motorway and dual carriageway	45 on dual carriageways. On motorways they may not travel.
Roads	45
Tracks	30

Bicycles may exceed these speeds if circumstances allow.

Type of vehicle

- Two- and three-wheeled mopeds.
- Light quadricycles.
- Vehicles for people with reduced mobility.

Road type	Speed limit
Motorway and dual carriageway	May not travel
Roads	45
Tracks	30

For vehicles that do school transport with children or that carry dangerous goods, the maximum speed limit is 10 kilometres less than for other vehicles. For example, the speed limit on a road for a bus is 90 kilometres per hour. But for a bus that takes children to school the speed limit is 80 kilometres per hour.

Maximum speed limit inside towns and cities

Type of vehicle

- All vehicles in general

Street type	**Max speed (km/h)**
Streets without kerbs, where the carriageway and the pavement are at the same height	20
Streets with one lane in each direction	30
Streets with several lanes in each direction	50

Type of vehicle

- Vehicles that carry dangerous goods.
- Special vehicles that cannot reach a speed of more than 60 kilometres per hour.

Street type	**Max speed (km/h)**
Streets without kerbs, where the carriageway and the pavement are at the same height	20
Streets with one lane in each direction	30
Streets with several lanes in each direction	40

Type of vehicle

- Two- or three-wheel mopeds.
- Light quadricycles.
- Vehicles for people with reduced mobility.
- Cycles.

Street type	**Max speed (km/h)**
Streets without kerbs, where the carriageway and the pavement are at the same height	20
Streets with one lane in each direction	30
Streets with several lanes in each direction	45

Type of vehicle

- Personal mobility vehicles.

- Special vehicles with a trailer or semi-trailer.
- Vehicles without brake lights.

Streets without kerbs, where the carriageway and the pavement are at the same height

Streets with one lane in each direction

Streets with several lanes in each direction

20
25

The maximum speed for all vehicles on motorways and dual carriageways that are inside a town or city is 80 kilometres per hour, if it is not signed in another way.

Minimum speed limits inside and outside towns and cities

- On motorways and dual carriageways the minimum speed allowed is 60 kilometres per hour.
- On the rest of the roads the minimum speed allowed depends on each vehicle. For example, on a road where the maximum speed allowed for a car is 90 kilometres per hour, the minimum speed allowed will be 45 kilometres per hour. It is forbidden to drive below that speed, even if there are no other vehicles on the road. The only cases in which you can drive slower than the minimum speed allowed are:
- When there is heavy traffic, the vehicle has a breakdown or the road is in poor condition.
- When the vehicle is a cycle, is pulled by animals or is a special vehicle.
- In the case of vehicles that accompany other vehicles. For example, vehicles that accompany cyclists in a competition.

SPECIAL
CYCLIST

If your vehicle cannot reach the minimum speed and there is a risk of an accident, you must switch on the hazard lights. For example, if your vehicle has a breakdown and you have to drive slower than allowed.

Why are speed limits necessary?

Speed limits are set so that all vehicles can travel safely and in an easy and comfortable way. One of the conditions that is taken into account to set the speed is the type of road and its conditions. That is, the maximum speed at which you can drive safely on that road.

Inside towns and cities you must drive at less than 50 kilometres per hour. If you run over a pedestrian they will have a better chance of surviving if the vehicle is going at a low speed.

On roads that are not a motorway or dual carriageway you must not drive at more than 90 kilometres per hour so that accidents are less serious. On a motorway and dual carriageway there is a better chance of surviving an accident when the vehicle travels at less than 120 kilometres per hour. Things that can happen if you drive very fast

- It is harder to react to unexpected events. This means that many mistakes are made that can cost people their lives.
- You have less ability to see what happens at the sides. This effect is called tunnel vision because you

only see what is in front of you, as if you were going through a tunnel. 0 km/h 65 km/h 130 km/h 180o 70o 30o

- Driving fast for a long time can make you feel tired and aggressive because you drive with more tension.
- Driving very fast on a road in good condition can make you think you are going slower than you really are.

It is forbidden to hold speed competitions on any road, unless the authorities authorise it and take the necessary measures. A person who takes part in races that are not authorised will lose points from their driving licence.

Types of distance

Reaction distance

Braking distance

Stopping distance

Lateral separation

Safety distance

Reaction distance

Space you travel with the vehicle from when you notice an unexpected event, such as an obstacle, a sign or a noise, until you can react to the unexpected event. For example, the time that passes from when you see a red traffic light until you press the brake. A person's normal reaction time to an unexpected event is usually 0.75 seconds, almost one second. Although this depends on their reflexes, their physical and psychological state and the environment they are in.

The faster a vehicle goes the less reaction distance it will have because it will travel more distance in less time. Therefore, it will reach the danger sooner. The driver sees the traffic light The driver presses the brake

Reaction distance
Braking distance

Space the vehicle travels from when you press the brake until it stops completely. This distance can change for the following reasons:

- Speed. The faster the vehicle goes the more space it will need to brake.
- The load. A vehicle with a heavy load takes longer to stop.
- Technical condition of the vehicle. Condition of its brakes and other parts of the vehicle, such as the tyres.
- The weather. On a wet surface the vehicle may need more braking distance.
- The condition of the road.
- Characteristics of the driver. For example, age, physical or psychological state.

Stopping distance

Space you travel with the vehicle from when you notice an obstacle until you stop the vehicle completely. For example, the time that passes from when you see a red traffic light until you stop the vehicle completely.

Stopping distance is the sum of reaction distance and braking distance.

Reaction distance
Braking distance
Stopping distance

Lateral separation

Necessary separation distance between two vehicles that pass each other or that drive side by side. There must always be enough separation between them to avoid any danger. You must keep more lateral separation distance from other vehicles when:

- You drive fast.
- The road is in poor condition.
- Many vehicles pass yours.
- There is rain, snow, fog, wind or smoke.

Watch video
Safety distance

Minimum space you must leave from the vehicle in front so you do not crash in case of sudden and unexpected braking.

You must leave more safety distance from the driver in front when:

- You increase speed.
- The road is hard to see.
- The vehicle does not grip the road well.
- You are not in good physical or psychological condition. You must keep a distance from other drivers that allows you to stop in case of sudden braking in the following cases:
- Inside a town or city.
- In areas where overtaking is forbidden.
- Where there is more than one lane in the same direction.
- When you cannot overtake because there are many vehicles.
- When another vehicle signals that it is going to overtake. In places where you can overtake other vehicles you must leave enough distance to overtake or to allow another vehicle to overtake you comfortably.

Vehicles that can carry more than 3,500 kilos and combinations of vehicles that measure more than 10 metres must follow these rules and also leave a minimum separation of 50 metres from the rest of the vehicles.

Bicycles are the only vehicles that can travel without keeping the safety distance between cyclist and cyclist.

Topic 11. Manoeuvres

Contents

Safety rule R.S.M. What is it? The set of rules and steps you must follow when you do any manoeuvre with the vehicle:

Rear-view mirror
1
Signal
2
Manoeuvre
3

This rule is divided into three steps. Steps of the safety rule R.S.M.

Observe the road

Rear-view mirror

Warn about the manoeuvre

Signal

Do the manoeuvre

Observe the road

It means looking carefully at the traffic and the road directly, and by using the rear-view mirrors, before doing the manoeuvre. This way you will make sure you can do the movement at that moment.

Warn about the manoeuvre

After checking that there is no danger to do that manoeuvre, you must warn other drivers what you are going to do.

You can warn them with the vehicle lights or with your arm. The signals you make with your arm are more valid than the ones you make with the lights, as long as you do them well and other drivers can see them. When you make signals with the vehicle lights, these lights must be on during the whole time the manoeuvre lasts and you must switch them off the moment you finish it. When you make signals with your arm, you must make the signal just before starting the manoeuvre, with enough time for other drivers to see it.

How should you indicate the manoeuvre with your arm?

Manoeuvre
Arm movement

Lane change, change of direction or turning around To move the vehicle to the left: arm horizontal with the hand extended downwards. To move the vehicle to the right:

Arm bent upwards with the hand extended also upwards.

Reversing

Arm extended horizontally with the palm of the hand facing backwards.

Braking

Arm movements up and down with short and quick movements. How should you indicate the manoeuvre with the lights?

Manoeuvre
Lights

Lane change, change of direction or turning around

Switch on the indicator on the side you are going to move to.

Reversing

Switch on the reversing light.

Braking

Switch on the brake lights several times.

Do the manoeuvre

After observing and signalling you can now do the manoeuvre. You must do it correctly, at the right speed and without putting other vehicles or pedestrians in danger.

Safety rule when entering a road
Watch video

It is very important to follow the steps set out by the safety rule R.S.M. when entering a road with the vehicle. You must observe the road, always make sure that your entry onto that road does not create a danger for other drivers and warn about your manoeuvre with your arm or the lights.

When doing the manoeuvre you have to:

- Give way to other vehicles without causing them danger.
- Not be an obstacle for vehicles that are coming towards you.
- Go at the right speed. When you have to give way to another vehicle while you are in the acceleration lane to enter a road, stop at the start of the lane until the vehicle that is already on the road goes past.

Watch video

Then enter the lane that applies to you adapting to the speed of the other vehicles.

All drivers who travel on a road have the duty to allow other vehicles to enter that road and, if they can, they must make the entry manoeuvre easier for them.

Above all, for large vehicles that carry many people and are at a marked stop. For example, a bus that is at its stop picking up passengers and must re-enter the road. To make it easier for these vehicles to enter, vehicles that are already driving must:

- Move to one side to leave them more space, when possible.
- Reduce speed and even stop to let the vehicle leave the marked stop, when this happens inside a town or city.

Precautions when changing lane
Watch video

To change lane safely you must take the following precautions:

- Start the lane change far enough away from other vehicles or any other obstacle.
- Do the manoeuvre little by little, without sudden movements.
- Do not get in the way of vehicles driving in that lane and do not cut in front of them. They have priority because they were already in that lane.

Watch video
Overtaking other vehicles

Overtaking is the manoeuvre of passing other vehicles that are travelling more slowly. Which side do vehicles overtake on? You overtake on the left side of the vehicle you want to overtake in prohibited areas. There are some exceptions where you can overtake on the right side.

- To overtake a tram that is travelling in the central part of a road with two-way traffic.
- When the driver you want to overtake

is clearly indicating that they want to change direction to the left or stop on that side.

Watch video

You can also overtake on the right inside a town or city when the carriageway has at least two lanes going in the same direction and they are separated by a white line painted on the road. There are some situations that are not considered overtaking, even if you pass the other vehicle. These situations are allowed and are:

- Roads with traffic jams, full of vehicles.

Drivers in one lane move faster than those in another lane and their speed depends on the movement of the vehicles in front. In these situations it is forbidden to change lane to overtake.

- Driving faster in the lanes for entering and leaving the road than the vehicles that are already on it.
- Vehicles that travel faster in lanes that are only reserved for them. For example, a bus that goes in the lane reserved for that vehicle can travel faster than the vehicles in the lane to its left.
- Cyclists riding in a group and overtaking each other.

Watch video

- Overtaking a vehicle that is stopped on the road due to a breakdown or accident.

Watch video

- Overtaking an obstacle on the road, even if you have to move into the opposite lane to overtake it.

Special cases of overtaking
Situation

How to act?

Two-way roads and three lanes separated by broken white lines

Overtaking is done in the centre lane as long as it is not occupied by another driver travelling in the opposite direction. Roads with two, or more, lanes in the same direction You can use the lane on the left to overtake. The driver can continue their journey in the lane on the left if they keep overtaking. But they must return to the right one if another vehicle comes that is travelling faster than them.

Steps to follow to overtake another vehicle
Preparation
Checks
Ask to pass

(if necessary)

Overtake

End of overtaking

Preparation

- Keep a suitable distance from the vehicle you want to overtake.
- Judge the distance of the vehicles coming in the opposite lane and the speed at which they are coming towards you.
- Take into account the speed limits allowed on that road.

Checks before overtaking

You must check that:

- The driver you are going to overtake has or has not signalled with the lights or their arm that they are going to move towards the side you are going to overtake on. If they have signalled, you must wait for them to do their manoeuvre before overtaking.
- You can return to your lane after overtaking without risk to other vehicles.
- The lane you are going to use to overtake is not occupied by another vehicle that also wants to overtake. Ask to pass to overtake (if it is considered necessary)
- Signal with the vehicle lights.
- Use the vehicle horn. This can only be done outside towns and cities.

Overtake

To overtake you must increase the vehicle's speed. But you must not go over the maximum permitted speed limit. To overtake a moped or a cycle, inside or outside built-up areas, you must:

- Keep a separation of at least one and a half metres from the moped or the cycle.
- Move into the lane to your left to overtake. Do not overtake in the same lane as the moped or cycle. On roads outside built-up areas, you must also leave a separation of at least one and a half metres in these cases:
- When you drive a two-wheeled vehicle and you want to overtake any other vehicle.
- When you drive any vehicle and you want to overtake people, animals, motorcycles, vehicles stopped on the road and vehicles that are doing assistance work. For example, ambulances or recovery vehicles at least, one and a half metres, on inter-urban roads.

In the rest of the cases, you must leave enough space between your vehicle and the one you want to overtake. The space will depend on the speed and the road conditions. You must stop overtaking if you see that another vehicle is coming in the opposite direction or that the vehicle you want to overtake suddenly speeds up. In those cases you have to slow down and return to the lane you were driving in.

End of overtaking

When you have finished overtaking, you must switch on the correct lights to show that you are going to return to the lane you were driving in. Then you have to return to your lane as soon as possible and without forcing other drivers to change their speed.

Duties of the driver who is being overtaken by another vehicle

- Allow the other driver to overtake you and make the manoeuvre easier for them. It is forbidden to increase speed at that moment.

Watch video

- Place your vehicle at the right edge of the lane without moving onto the hard shoulder.
- Reduce speed if any dangerous situation arises while the other vehicle is overtaking you.
- Allow the vehicle that has overtaken to return to its lane.
- Drivers of very large or heavy vehicles who cannot move to the right of the lane must reduce speed

and indicate to the driver behind that they can overtake. This signal can be made with the arm extended and moving the palm of the hand backwards and forwards. It can also be done by switching on the right indicator of the vehicle.

When is overtaking forbidden?

- When you cannot see the whole road well

ahead. Except if the lanes are marked and you do not have to move into the opposite lane to overtake, for example:

- On bends where you cannot see well.
- When the road has crests and there is not good visibility.
- When there is fog, heavy rain or the sun dazzles and does not let you see well.
- Behind a vehicle that is also overtaking and does not let you see the road ahead.
- Several vehicles at the same time. This can only be done when you can return to the right at any time without bothering the vehicles you have overtaken.
- A vehicle that is already overtaking another, if you have to move into the opposite lane to overtake.
- When you could put cyclists travelling in the opposite lane in danger.
- At pedestrian crossings, at junctions with cycle tracks, and at level crossings and near them.

At level crossings you can overtake two-wheeled vehicles that let you see well because they are small, as long as we warn with sound or light signals. At pedestrian crossings you can overtake:

- When you do it very slowly, so that if a pedestrian appears we can stop the vehicle.
- Two-wheeled vehicles.
- In tunnels with traffic in both directions when the vehicle you are going to overtake only has one lane going in its direction.
- At junctions and near them. You can overtake at a junction when:
- it is a roundabout,
- you overtake on the right,
- you overtake a two-wheeled vehicle,
- there is a sign at the junction that gives you priority. You can overtake on any path or road:
- Pedestrians.
- Bicycles.
- Cycles.
- Mopeds.
- Animals.
- Vehicles pulled by animals. Before overtaking them, you must make sure there is no danger for them or for other drivers.

Changing direction and turning round
Changing direction

To change direction with the vehicle, you must make sure that vehicles coming in the opposite direction are far away and are travelling at a speed that lets you change direction safely.

When they are very close or coming very fast, you cannot change direction. If the change of direction is to the left, but you cannot see well, you cannot do it either. If there is a suitable place on the road such as a turning bay or something similar, the change must be done there, but if there is a sign, you must follow what the sign says. How should you change direction?

Watch video

When the change of direction is to the right you must move as close as possible to the edge of the carriageway. When the change of direction is to the left you must position yourself:

- As close as possible to the left edge of the carriageway when this carriageway is one-way.
- Next to the white line separating lanes when the carriageway is two-way. When there is no white separation line you must position yourself as close as possible to the opposite lane, but without crossing into it.
- In the centre lane when the road is two-way and has three lanes separated by white lines. When turning to change direction, you must go through the centre of the junction and leave it on your left.

You must not cross it diagonally.

Well done Badly done
Watch video

On inter-urban roads, two-wheeled cycles and mopeds that want to turn to change direction to the left must position themselves on the right of the lane and off the carriageway if possible, and wait until it is safe to turn. They must also go through the centre of the junction.

Turning round
Watch video

Turning round means making a U-turn to keep travelling on the same road or path, but in the other direction. You can turn round at a roundabout or a turning bay on the road.

Turning bay. A half-circle shaped diversion in the road to change direction or turn round.

The turning round manoeuvre must be done with a single turn of the vehicle and without using reverse gear. You can only use reverse gear to turn round in circumstances where you cannot do anything else. For example, to get out of a street closed by roadworks. You must start the U-turn next to the white line on the road that separates the lanes going in different directions. When it is not possible to do it from there, you must position yourself close to the right edge to start the turn from there.

You cannot make a U-turn if you make drivers who come behind you or from the opposite direction slow down, or you can be a danger to them. In those cases, you must stop on the hard shoulder on your right and wait until it is safe to turn to make a U-turn. If there is not enough hard shoulder or it is not a safe place, you must keep going until you find a place where you can make a U-turn.

Situations where it is forbidden to make a U-turn

- Bends and crests where you cannot see well.
- When there is fog or heavy rain.
- At level crossings.
- In tunnels.
- On motorways and dual carriageways. You can do it in places prepared for it.
- On one-way carriageways and roads.
- In all places where it is forbidden to overtake, except where there is a sign that says you can make a U-turn.

Reversing
Watch video

It is forbidden to drive in reverse on any road or track. You can only drive in reverse in the following cases:

- When doing manoeuvres to park the vehicle.
- When it is not possible to drive forwards or to change direction or make a U-turn. For example, to get out of a street closed because of roadworks. In these two cases you must not reverse more than 15 metres and you must not reverse into a junction of roads or streets. You must drive at a very low speed while you are reversing.

Before you start reversing you must make sure there is no danger for other drivers and pedestrians. You have to stop the vehicle and stop reversing in any dangerous situation. For example:

- Someone makes sound signals with their vehicle horn.
- You notice another vehicle coming closer, a person, or an animal.
- On motorways and dual carriageways reversing is always forbidden.

Stopping the vehicle
Watch video

It may be necessary to stop the vehicle while you drive for the following reasons.

- An emergency. For example, an accident, a breakdown, or one of the passengers feels unwell.
- There are many vehicles on the road and there are traffic jams.
- Instructions from signs or traffic officers who ask you to stop.

On motorways, dual carriageways, and enclosed places or places where you cannot see well, you must switch on the hazard warning lights and sidelights whenever you have to stop the vehicle.

Stopping the vehicle because of an emergency

Vehicles that have an accident or breakdown can be in the following situations.

Situation

What to do? The vehicle keeps working, but it cannot reach the speed needed on that road and it blocks traffic. If the vehicle weighs less than 3,500 kilos, drive on the hard shoulder on the right. If there is no hard shoulder, put the vehicle as far as possible to the right of the road. If you are on a motorway or dual carriageway, take the first exit. The vehicle works, but you need a recovery service

Leave the road at the first exit using the right hard shoulder. The vehicle cannot keep driving Park the vehicle on the right hard shoulder of the road or in the place where it gets in the way least for other drivers. (For example, the left hard shoulder or the central reservation).

People travelling in a vehicle that cannot keep driving must get out of the vehicle and go to a safe place. Before getting out of the vehicle, they must put on the yellow high-visibility vests that all drivers must carry in their vehicles. If there is no safe place, they must stay inside the vehicle with the seat belt fastened. You must switch on the hazard warning lights to show that there is a vehicle stopped on the road. It is also compulsory to place on top of the car the emergency warning light (orange light). If you want, you can place the warning triangles. They are placed 50 metres from the vehicle. One at the front and one at the back, on the right side of the road, and they must be seen from 100 metres away.

Vehicles that pick up and tow broken-down vehicles must be prepared for roadside assistance (for example, recovery trucks). Not just any vehicle can do it.

Stopping and parking
Difference between stopping and parking
Stopping

Leaving the vehicle stopped for less than two minutes. When stopping, the driver does not leave the vehicle. And if they leave, they stay close to it.

Parking

The vehicle stays without moving for more than two minutes. The driver can leave the vehicle and go somewhere else. Where can you stop and park?

Watch video

On roads outside towns and cities, you must stop and park on the right side of the road, leaving the hard shoulder free, as much as possible.

On roads and streets inside towns and cities you can park on the carriageway and on the hard shoulder. When the road is two-way, the vehicle must stop or park on the right. When the road is one-way you can do it on the right or on the left. How do you position the vehicle? As a general rule, you must stop and park the vehicle in a line parallel to the edge of the carriageway. You must always check that your vehicle lets other vehicles use the rest of the space. What must you do before leaving the vehicle?

Drivers of a motor vehicle or moped must:

- Stop the engine and switch off the system that allows the vehicle to move.
- Apply the parking brake.
- Leave first gear engaged if you park on an uphill slope.
- Leave reverse gear engaged if you park on a downhill slope.
- Put it in park if the vehicle is automatic.

Drivers of vehicle combinations, also, have to leave their vehicle well secured when they park on a slope.

Watch video

They can do it in 2 ways:

1. Putting the correct wheel chocks. You cannot use stones or other items.

Chock. Wedge put between the vehicle and the road so the vehicle does not move.

2. Resting one of the front wheels against the kerb.

On uphill slopes, the wheel turns towards the centre of the carriageway. On downhill slopes, the wheel turns outwards.

Dangerous stops and parking

There are different situations where it is dangerous to stop the vehicle or park it. What are these situations?

- When a parked vehicle does not allow other vehicles to pass. When a vehicle is less than three metres from the opposite side of the road or from the continuous line that separates directions. Some vehicles will not be able to pass through that space.
- When a parked vehicle does not allow another vehicle that is stopped or parked to rejoin the road.
- When people or animals cannot enter or leave their homes or any other place because a vehicle blocks them.
- When vehicles cannot use a dropped kerb that is signposted.
- When a vehicle is an obstacle so that people with physical disabilities cannot enter or leave the road using the areas prepared for this.

When a vehicle stops or parks in places used to control traffic. For example, traffic islands. When a vehicle stops another vehicle from turning where the sign says. Parking is considered dangerous when:

- A vehicle is parked in a loading and unloading area at times when it is not allowed.
- A vehicle is parked double-parked and the driver has left.
- A vehicle parks at a public transport stop. For example, a bus stop.
- A vehicle parks in places where there is a sign that forbids parking.
- A vehicle parks in a place reserved for emergency and security services. For example, ambulances.
- A vehicle parks in the middle of the carriageway.

Where is it forbidden to stop and park? It is forbidden to stop and park in:

- Bends and crests where you cannot see well.
- Tunnels.
- Level crossings, pedestrian crossings, and cyclist crossings.
- Lanes or parts of roads reserved for other vehicles.
- Junctions or near them when you do not let other vehicles turn.
- On tram tracks or very close to them.
- Places where the parked vehicle does not let you see the signs.
- Motorways or dual carriageways. You can only stop or park in rest areas or service areas.
- Lanes and areas reserved for buses, taxis, or bicycles.
- Areas reserved for people with disabilities. You can stop, but not park:

- In loading and unloading areas.
- On pavements, walkways, and other areas where pedestrians walk. For two-wheeled vehicles, as long as it is regulated by the local council and they do not get in the way of pedestrians.
- In front of dropped kerbs where there is a dropped kerb sign.
- Double-parked.
- In places where you need a permit to park at certain times or for a set time. You can park if you have the permit that allows it.

Topic 12. Priority rules for driving

Contents

Giving way to other vehicles
General rules

To give way to another vehicle that has priority to go, you must:

- Reduce speed little by little so that the driver of the other vehicle realises you are giving way.
- Stay stopped or drive slowly until the other driver has passed and you can drive without danger.

Rules for giving way in specific places

Junctions without signs

Junctions with signs

Roundabout
Junctions without signs

At junctions without signs you must give way to vehicles coming from your right, even if they come from a narrow road or a road in poor condition.

At junctions without signs, there are some exceptions that you must keep in mind. They have priority:

- Vehicles that travel on rails before those that travel on the road. For example, a tram has priority before a car.
- Vehicles that travel on a road that is paved before vehicles that go on an unpaved road.

Pave. Cover the ground of roads and streets with asphalt and other materials that allow vehicles and people to pass more safely.

- Vehicles that are already inside a roundabout before those that want to enter the roundabout.
- Vehicles that are already on a motorway before those that want to enter that motorway.

Junctions with signs

At junctions with signs you must always follow the instructions shown by that sign. When there are no lines that show which is the entry to that junction, you must wait and give way from the place where you can best see the junction and the vehicles coming to it. You must never enter a junction, a pedestrian crossing or a cycle crossing if you think your vehicle will get stuck in the middle of the junction or the crossing, blocking traffic. For example, you must not enter a pedestrian crossing when there are vehicles right in front that will not let you pass. In that case, you must wait before the pedestrian crossing. If you are stopped at a junction with traffic lights and your vehicle is an obstacle for other drivers, you must leave the junction in any way possible, to continue on the road you wanted to take, as long as no vehicles are coming towards you.

Roundabouts

You must drive in the same way as on the rest of the roads, depending on whether they are urban with or without boundaries, interurban, etc.

Watch video

Right of way for cyclists, pedestrians and animals

Cyclists

Cyclists have priority to go in the following situations:

- When they are riding in a cycle lane, a cycle crossing or a well-signed hard shoulder.
- When another vehicle wants to turn right or left and there are cyclists nearby.
- When cyclists are riding in a group and one of them has already entered a junction. The other vehicles must wait until the whole group of cyclists has passed.

Pedestrians
Watch video

Pedestrians have priority to go in the following situations.

- At pedestrian crossings, on pavements and in other pedestrian areas.
- When a vehicle is going to turn to enter another road where pedestrians are crossing.
- When a vehicle crosses a hard shoulder where pedestrians are walking and they have no other place to go because there are no pavements.
- When a vehicle needs to cross

a pedestrian area. For example, the pavement that a car must cross when leaving a garage.

Drivers must also give way to:

- An organised group of people who all go together for a purpose. For example, a group of children on a trip with their school.
- People getting on or off public transport at the stop. For example, people getting off the bus, up to the nearest pedestrian area.

Animals

Animals have priority to go in the following situations.

- On drovers' roads where there is a sign for "Domestic animals crossing". And below it a sign that says drovers' road.

Drovers' road. A path made so that animals can go along it.

- When a vehicle is going to turn to enter another road where animals are crossing.
- When a vehicle crosses a hard shoulder where animals are moving and they have no other place to go because there is no drovers' road.

Right of way for vehicles in emergency service

Which vehicles are prepared to do emergency services?

- Police cars.
- Fire engines.
- Civil protection and rescue vehicles.
- Ambulances and other vehicles for medical assistance. When these vehicles are on an emergency service, they must warn that they are coming by switching on the siren and the lights made for this. In these cases, these vehicles have priority on all roads (streets and roads).

All other vehicles and pedestrians must give way to them.

The following vehicles also have priority:

- Maintenance teams that repair the road.
- Roadside assistance vehicles when they are going to help vehicles that have broken down or had an accident.

Drivers of vehicles prepared to do emergency services can drive faster than the speed limit and do not have the duty to obey signs. The only signs they must obey are those given by traffic officers. They must make sure they do not put pedestrians or vehicles in danger when going through a junction or going through a red traffic light. On the motorway and dual carriageway they can change direction, change the direction of travel, reverse and enter the central reservation when they can make sure that there is no danger for other vehicles.

How should you act when there are vehicles doing emergency services? All vehicles must make it easier for them to pass as soon as they hear the siren or see the lights. They will move to the right and stop if necessary.

Pedestrians will keep the road clear and wait on the pavement.

Any vehicle doing an emergency service

Watch video

In some situations drivers of vehicles that are not prepared to do emergency services must ask for priority from other vehicles because of some situation. For example, they are taking a person who is very ill to hospital or a woman about to give birth. In these cases, the driver of the vehicle can warn about the emergency situation in the following ways:

- Sounding the vehicle horn intermittently.
- Switching on the vehicle's hazard lights.
- Waving a handkerchief through the window. The driver must obey the traffic rules. Especially at junctions. Other drivers must let them pass.

Right of way in narrow sections
Narrow sections with signs

You must always follow the rules shown by vertical signs, traffic lights or traffic officers when going along a road or path that is very narrow.

Narrow sections without signs

If there are no signs to pass through a narrow section, the vehicle that entered first has priority in the narrow section.

If there is doubt about which vehicle entered first, vehicles with more difficulty doing manoeuvres will have priority.

Which vehicles have priority to pass through a narrow section? The order of priority of vehicles is as follows:

1. Special vehicles that exceed the weight and size set by the rules that regulate vehicles.
2. Vehicle combinations.
3. Vehicles pulled by animals.
4. Motorhomes and cars pulling a trailer that weighs less than 750 kilos.
5. Buses.
6. Lorries, articulated lorries and vans.
7. Cars and car-derived vehicles.
8. Heavy quadricycles, light quadricycles and special vehicles that have the weight and size set by the rules that regulate vehicles.
9. Three-wheeled mopeds, motorcycles with sidecar and other three-wheeled vehicles.
10. Motorcycles, two-wheeled mopeds and bicycles.

When they are vehicles of the same type or there are doubts about which one should go, the following has priority:

- The vehicle that has to reverse for a longer distance.

- When the distance to reverse is the same, the vehicle that is wider, longer or can carry more load will have priority to go. The lorry can go first. The car has to reverse. The bus can go first. The lorry has to reverse.

Narrow sections on a slope

On steep slopes, the vehicle going uphill has priority to go, unless it has a safe place to stop closer than the other vehicle. If there is doubt, the same rule will be followed as in narrow sections that are not on a slope. The car has priority. The lorry must reverse.

Crossing level crossings and movable bridges

You must reduce speed and drive more carefully when you approach a level crossing or a movable bridge.

You cannot cross the level crossing or movable bridge when:

- It is closed.
- Its barriers are moving to open or close the crossing.
- There is a traffic light that tells you to stop. In those cases all vehicles must wait in their lane, one behind another, until the crossing is clear. Once the crossing is clear, you must cross quickly. Make sure first that there is no risk of being trapped inside the level crossing. If there are no barriers or traffic lights, you must check that no train or tram is coming before crossing. When a vehicle is trapped inside a level crossing or a movable bridge all the people travelling in it, except the driver, must get out of the vehicle. The driver will try to start the vehicle once the rest of the passengers have got out. If they cannot, they will also get out of the vehicle and try to warn about the situation to any trains or trams that may pass and to the drivers of other vehicles approaching the area .

Crossing tunnels and underpasses

No vehicle can enter a tunnel when at the tunnel entrance there is a traffic light that forbids entry.

Watch video

Inside the tunnel you must keep a distance of at least 100 metres from the vehicle in front (or 4 seconds).

Vehicles that can carry a weight greater than 3,500 kilos must keep a distance of 150 metres from the vehicle in front (or 6 seconds).

Rules for stopping the vehicle inside a tunnel When there is heavy traffic and vehicles cannot move forward inside a tunnel, all passengers must stay inside the vehicle. The driver must switch off the engine and keep the sidelights on. When braking, they will switch on for a moment the hazard lights so that other drivers can see them.

Emergencies inside a tunnel

When a driver has to stop the vehicle inside a tunnel because of an emergency, the steps they must follow are:

- Switch off the engine, keep the sidelights and hazard lights on so that other drivers can see the vehicle.
- Take the vehicle to the nearest area reserved for emergencies. If there is no emergency area, they must move it as close as possible to the right edge of the road.

- Switch on and place the warning light, if you have one, or if not place the warning triangles on the carriageway.
- Ask for help through the nearest emergency post inside the tunnel and follow the instructions you are given.
- All the people travelling in the vehicle must leave it and go to the nearest refuge or exit.
- When the vehicle can be moved, despite the breakdown, drive until you leave the tunnel or reach the nearest emergency area.

If there is a fire inside a tunnel, drivers of vehicles must follow these steps:

- Move the vehicle to the right to let emergency vehicles pass.
- Switch off the engine.
- Leave the key in and the vehicle doors

open. That is, not locked.

- All the people travelling in the vehicle must leave it and go to the nearest refuge or exit. Always in the opposite direction to the fire.

Topic 13. Carrying people and loads

Contents

Rules for carrying people, animals and loads

General rules
Weight limit

All vehicles have marked a maximum number of kilos they can carry depending on their features. It is forbidden for vehicles to carry more load than they can, counting the weight of passengers, luggage and other materials or loads. When a vehicle carries more weight than it should, some of its parts get damaged, such as: the tyres, the acceleration systems and the braking systems.

Carrying animals

Animals must travel in the back of the vehicle and be secured to the seat. When possible, a separating mesh or another device will be placed between the part where the animals travel and the part where the driver travels.

Animals can never travel loose inside the vehicle.

Carrying people

General rules

Carrying people on a bicycle

Carrying people on a moped and motorcycle

General rules

The number of people who can travel in a vehicle depends on the number of seats that vehicle has. For example, in a car with five seats six people cannot travel.

Watch video

Each person must travel in their seat during the whole journey without moving to other spaces in the vehicle.

However, people can travel in the parts of vehicles meant to carry loads when the vehicle has the necessary authorisation. The driver must spread the passengers in the vehicle and place the load so that they have enough space to drive and see the road well from all sides of the vehicle. The driver must pay special attention to:

- Being comfortable in their seat.
- Making sure the other passengers use their seats and do not move from them.
- Placing the objects they carry well so they do not get in the way. All travellers will get in and out of the vehicle when it is stopped and on the side closest to the pavement or the hard shoulder.

Carrying people on a bicycle

On bicycles made for one person only the rider may travel. However, when the rider is an adult, they can carry as a passenger a child under seven years old. The child will travel in an allowed extra seat. This seat will be placed behind the seat of the adult who is riding.

Carrying people on mopeds and motorcycles

Watch video

On mopeds and motorcycles one person can travel with the rider when the person meets the following requirements:

- They are over 12 years old.
- They sit behind the rider.
- They sit correctly and place their feet on the footrests that are on the sides of the vehicle.
- They wear the helmet correctly fitted.

Watch video

As an exception, children who are over seven years old can travel on a motorcycle or moped when they meet two requirements.

- They travel with one of their parents or with an adult who takes responsibility for them.
- They follow all the safety rules that other passengers must follow.

Carrying loads

A load is any object that is carried in the vehicle. It can be luggage, goods, or anything else.

In vehicles made to carry people, you can carry luggage and other types of loads as long as they meet these requirements:

- The load is suitable for the vehicle's features. For example, it does not exceed the weight that vehicle can carry.
- The load is well distributed and secured so that the vehicle does not lose its stability.
- The driver of the vehicle shows that they are carrying that load, when necessary.
- The load does not cover the lights or the warning signals of the vehicle. It also lets other vehicles clearly see the signals the driver makes with their hand.

Objects carried in a vehicle must be placed in the boot.

When it is necessary to place an object in another place in the vehicle, you must make sure that the object will not move in case of a crash or when braking. The driver must also make sure that the object lets them see the whole road well. It is forbidden for the loads carried by a vehicle to:

- Move around inside the vehicle.
- Fall off.
- Drag along the road.
- Make noise, give off dust or smoke.

Transport of loads that can fall or that give off dust will always be done in special vehicles prepared to carry these materials. Motorcycles, three-wheeled vehicles, mopeds, cycles and bicycles can carry a trailer or semi-trailer when it meets the following features:

- The trailer weighs at most half of what the vehicle weighs.
- It travels during the day and is clearly visible.
- The vehicle travels more slowly than the maximum speed allowed for that vehicle.
- No people travel in the trailer.

Allowed sizes for vehicles and loads
Allowed sizes for vehicles
Width

The maximum width allowed for a vehicle is 2.55 metres, including its load if it carries one. 2.55 m

Height

Vehicle	**Maximum height allowed (Including the load)**
Vehicles in general	4 metres 4 m
Buses	4.20 metres
Recovery trucks that remove vehicles	4.50 metres

Length

Vehicle	**Maximum length allowed (Including the load)**
Motor vehicles, except buses	12 metres 12 m
Trailer only	12 metres 12 m
Vehicles with trailer	18.75 metres 18.75 m
Articulated vehicles, that carry a semi-trailer	16.50 metres 16.50 m
Combination of vehicles in euromodular configuration	25.25 metres 25.25 m

Allowed sizes for loads

As a general rule, the load cannot be wider, higher, or longer than the vehicle carrying it. That is, it cannot stick out from the vehicle. But the load placed on the roof rack of the vehicle may stick out.

Roof rack. The top part of the vehicle where luggage and other objects can be placed. In cases where the load has to stick out because there is no other way to carry the object, vehicles must meet different conditions depending on the type of vehicle.

Vehicles that carry loads

Vehicles that carry passengers

Vehicles that only carry loads

In vehicles used to carry goods, loads that cannot be bent or divided can stick out when they meet some requirements. Some examples of these loads can be: pipes, beams, or posts.

Type of vehicle	How much can the load stick out?
Vehicles that are 5 metres or less in length	One third of the metres the vehicle measures. For example, if the vehicle measures 3 metres, the load can stick out 1 metre. The load can stick out at the front and at the back 1/3 Maximum
Vehicles that are more than 5 metres in length	The load can stick out 2 metres at the front and 3 metres at the back. Maximum 2m Maximum 3m
Vehicles that measure at most 2.55 metres in width when they carry the load	The load can stick out 0.40 metres on each side of the vehicle. It is not allowed to place the load like this. The panels cannot go crosswise because they take up more. It is allowed to place the load like this. Larger dimension NOT ALLOWED Smaller dimension Larger dimension ALLOWED

Combinations of vehicles in euromodular configuration The load cannot be longer than the vehicle.

Vehicles that also carry passengers

The load cannot stick out at the front or on the sides of the vehicle. The load can stick out at the back of the vehicle when it meets these features.

- Loads that can be divided into smaller loads can stick out at most, 10 per cent of the vehicle's length at the back of the vehicle.

For example, in a vehicle that is three metres long, a bicycle placed at the back can stick out 30 centimetres at most.

- Loads that cannot be divided into smaller loads can stick out, at most, 15 per cent at the back of the vehicle.

For example, in a vehicle that is 3 metres long, a board placed on top can stick out at the back 45 centimetres at most.

- In vehicles that are less than one metre wide, such as, for example, motorcycles, the load:
- Cannot stick out at the front.
- Can stick out at the back 0.25 metres (25 centimetres).
- Can stick out on each side 0.50 metres (50 centimetres) from the centre of the vehicle. Whenever the load sticks out from the vehicle you must take all necessary measures to avoid accidents or damage to other people and vehicles.

Marking the load that sticks out
Loads that stick out at the front

- During the day they are not marked.
- At night or on days with low light they are marked with a white light.

Loads that stick out at the back

- They are marked during the day and at night.

- To mark them, a plate is placed with red and white diagonal lines at the back of the load.
- Two plates with red and white diagonal lines are placed at the back of the vehicle when the load is as wide as the vehicle.

Each plate is placed at one end of the load that sticks out.

Loads that stick out at the back must be lit with a red light when the vehicle travels at night or on a day with low light.

Loads that stick out on the sides

Loads that stick out more than 40 centimetres on the sides of the vehicle must have a light on at the ends of the load when it is night or a day with low light. The lights that mark the load at the front of the vehicle will be white and those that mark the load at the back will be red.

Loading and unloading operations of objects are always done off the road. When circumstances force it to be done on the road, the following precautions will be taken:

- Not cause damage to other vehicles or pedestrians travelling in that area.
- Respect the rules about the places where you can stop and park and the times when it is allowed.
- Stop the vehicle on the side that is closest to the pavement or the hard shoulder.
- Take as little time as possible to load or unload.
- Avoid noise and disturbance that are not necessary. It is forbidden to leave loads on the road, the hard shoulder, or in pedestrian areas.

Plates and signs on vehicles
Number plate

All motor vehicles, except motorcycles, must carry two number plates. One at the front and one at the back.

Motorcycles and mopeds must carry only one plate, at the back.

Trailers and semi-trailers that can carry more than 750 kilos must carry their number plate at the back and, in addition, the number plate of the vehicle that tows it.

Trailers and semi-trailers that can carry less than 750 kilos will only carry the number plate of the vehicle that tows them at the back.

Priority vehicle sign

Light signal made up of one or more blue lights. It is carried by vehicles that provide emergency services. These vehicles are: police cars, fire engines, civil protection vehicles, rescue vehicles, and ambulances. These vehicles can use the light signal at the same time as the siren to make sounds.

Speed limit plate

It shows that the vehicle that carries it cannot travel at more kilometres per hour than the number shown on the plate. It is placed on the back of the vehicle.

Slow vehicle plate

It shows that the vehicle that carries it cannot travel faster than 40 kilometres per hour. It is placed on the back of the vehicle.

Long vehicle plate

It shows that the vehicle that carries it is more than 12 metres long. This plate is placed on the back of the vehicle. It is rectangular, yellow, and with red edges.

Plate for a vehicle that carries dangerous goods These plates can have numbers. The numbers at the top inform about the type of danger that this material can cause. For example, if it is toxic material or if it can burn. The numbers at the bottom show the material that the vehicle is carrying.

New driver plate

It shows that the driving licence of the person who is driving the vehicle is less than one year old. This sign is placed on the rear window of the vehicle, in a place where it can be seen clearly.

Sign to show danger

It shows that the vehicle is stopped on the road because of any emergency, because of a breakdown, or that the load it carries has fallen onto the road surface. It is a yellow sign that is placed on the highest part of the vehicle so that it can be seen well.

Vehicle roadworthiness test sign

Sticker that shows that the vehicle has passed the roadworthiness test and is in good condition to drive. It also shows the date when it must have the next check. This sticker is placed on the right side of the windscreen at the front of motor vehicles. On trailers and semi-trailers it is placed in a place where it can be seen well.

Warning sign for escorting special transport Sign that a vehicle carries on the top to show that a special transport vehicle is travelling nearby. For example, a combine harvester or an excavator.

SPECIAL
Warning sign for escorting cyclists

Sign that a vehicle carries on the top to warn that there are cyclists travelling nearby.

CYCLIST

Topic 14. Driving safely

Contents

- Driving in clouds of dust or smoke
- Driving in strong wind
- Driving in heat

The road On some stretches of road you must be more careful than on others when driving because more difficulties can appear. These places are:

Bends

Road works area

Bends

Bends are dangerous parts of the road where you must drive more carefully. The reason is that, when you take a bend at speed, a force is created that can push the vehicle out of the bend and send it to the side it should not go to.

Bends are more dangerous when:

- The bend is very tight.
- The vehicle carries a lot of weight.
- The vehicle is travelling very fast. In those cases, the vehicle can go straight on, as if there were no bend.

So, it can leave the road or go into the opposite lane.

Steps to take a bend well

Before entering the bend you must:

- Reduce speed or brake if necessary.
- Move as close as possible to the right edge of the carriageway. In the bend you must:
- Turn the steering wheel smoothly.
- Do not accelerate or brake suddenly.
- Accelerate little by little. When leaving the bend you must:
- Turn the steering wheel smoothly to drive straight again.
- Increase speed little by little.

Skidding on bends

Skidding happens mainly on bends. The reason is that the force that pushes vehicles on bends can make their tyres not grip the road, slide, and lose control.

Skid. To slide or slip on the road. When a vehicle skids it moves away from the direction it was going and goes to another side. A skid can happen because of:

- Driving very fast.
- Making a sudden turn with the steering wheel.
- Pressing the brake pedal very hard.

- Having tyres or shock absorbers in bad condition.
- Sharing the load badly in the vehicle.

How can you make the vehicle stop skidding?

Type of vehicle	Which wheels skid?	What should you do?
Front-wheel drive. The engine power goes to the front wheels	Rear	Do not brake. Turn the steering wheel towards the side that the wheels are moving to. Accelerate smoothly.
Rear-wheel drive. The engine power goes to the rear wheels	Rear	Do not brake. Stop accelerating smoothly. Turn the steering wheel towards the side that the rear wheels are moving to.
Rear-wheel drive	Front	Stop accelerating. Keep the steering wheel straight until the wheels stop skidding.

Roadworks area

Roadworks can be a danger. Whenever you drive on a road with roadworks, you must follow the instructions of the roadworks staff. To show that a road has roadworks the following signs are used.

- Vertical warning signs and signs that inform you about the rules you must follow. These signs will have a yellow background.
- Signs painted on the road, in yellow.
- Other signs that are only placed when there are roadworks.

Driving at night

Driving at night is more dangerous than driving during the day.

Watch video

The reasons are:

- The road is less well lit.
- Distances, people, objects, and vehicles are harder to see.
- Other vehicles can dazzle you with their lights. You must pay special attention when you go from an area that is well lit to another that is poorly lit. Your eyes take a few seconds to get used to the change in light.

When driving at night you must take special care with:

Speed
Bends
Dazzle
Overtaking

Speed

You must respect the speed limits and drive more slowly if you cannot see the road well. This way it will be easier to stop the vehicle if there is a danger or something unexpected.

Dazzle

When a vehicle dazzles you you cannot see the road well because too much light enters your eyes.

Watch video

What can you do to avoid dazzling other drivers?

- Keep the lights well adjusted. The dipped beam may bother others because it is not adjusted properly.
- Share the weight you carry in the vehicle well. The dipped beam can point too high and dazzle because there is too much weight at the back of the vehicle.
- Switch off the main beam and switch on dipped beam when another vehicle comes towards you from any direction. When a vehicle dazzles you, reduce speed or even stop the car, taking the necessary safety precautions.

Do not move into other lanes when reducing speed or stopping. You can guide yourself by the line on the right edge. Do not use dark glasses or sunglasses at night. Other vehicles may dazzle you less, but you will see the road much worse.

Overtaking

When you are going to overtake another vehicle at night, you must switch off the main beam and switch on dipped beam.

Main beam can dazzle the other driver through the rear-view mirrors. When you are already overtaking, switch the main beam back on as soon as possible. That is, when you can no longer dazzle the other driver.

If another vehicle is overtaking you, keep the main beam on while they overtake you. These lights will light the road for both of you during the overtaking.

Bends

When two vehicles travelling in opposite directions meet at night on a bend, the vehicle that is travelling on the inside of the bend is the one that must switch off the main beam and switch on dipped beam. The reason is that its lights are the ones that light the road directly and can dazzle the other driver. The driver travelling on the outside of the bend can keep the main beam on because their lights shine off the road. They cannot dazzle other drivers. They should only change to dipped beam if they see that at some moment they are dazzling another driver.

The weather

Rain
Snow
Ice
Fog
Dust
Wind

Heat
Driving in rain

When it rains a lot it can be harder to drive for the following reasons:

- The road is harder to see.
- The tyres grip the road less. When this happens the vehicle needs more space to brake.

The precautions you must take to drive in rain are:

- Keep the tyres in good condition.
- Check that the windscreen wipers work well.
- Brake smoothly so that the wheels have more time to stop.
- Check that the brakes work well after going through a puddle.
- Keep more distance from the vehicle in front.
- Reduce speed.

Watch video

You must be especially careful when the first drops of rain start to fall. These drops mix with dust, grease, and other dirt on the road, making mud that can make the vehicle skid. When the wheels skid you must lift your foot off the accelerator, but without braking.

Motorcycle riders must pay special attention to the road markings when it rains because they can slip on them.

Aquaplaning

What is it?

Losing control of the vehicle because the tyre cannot clear all the water it picks up from the road and it slides. It is easier to aquaplane and lose control when the vehicle is travelling at high speed and when the tyres are very wide or very worn.

So, the best way to prevent aquaplaning is to travel slowly in areas where there is water. After passing through that area, you must check that the brakes work well.

Driving in snow

Driving in snow is harder for the following reasons:

- The road, the signs, and the vehicles are harder to see.
- The tyres grip the road less. When this happens, the vehicle needs more space to brake. The precautions you must take to drive in snow are:
- Start the vehicle with the wheels straight.
- Choose a gear that allows the vehicle to start at a higher speed, but with low engine power.

Gearbox. Vehicle mechanism that converts the engine power into movement of the wheels. With the gearbox the driver can choose to go faster or slower and with more or less engine power.

- Do not make sudden movements with the steering wheel and do not change gear suddenly. When going downhill you must do it more slowly than normal and choose low gears (first or second) so that they hold the vehicle back.

- Reduce speed little by little.
- Keep a greater safety distance from other vehicles.
- Use the brake as little as possible and do it smoothly.
- Drive over the tracks made by other vehicles with their wheels. When the sun comes out after it has snowed, it is a good idea to use sunglasses so that the light does not dazzle you. The R-412 sign shows that you must fit snow chains on the vehicle to keep driving in snow. The chains are fitted, at least, on the driven wheels. You must fit at least one chain on each side of the vehicle.

Driving with ice

Ice makes the wheels not able to grip the road and the vehicle skids a lot. The precautions you must take to drive with ice are the same as for driving with snow. Drive slowly and keep a large safety distance from other vehicles because you will need more space to brake. There may be ice on the road, during the night or early morning, in the following places:

- In damp areas.
- In areas where the sun does not reach.
- On bridges and flyovers.
- On speed humps.
- Where there are P-34 or P319 signs with a panel where the word Ice is written.

Ice can also stay on the vehicle windows. In these situations, before you start driving, you must:

- Scrape the ice off all the windows, without scratching them.
- Clean the rear-view mirrors.

Driving in fog

Fog is very dangerous for driving for the following reasons:

- You see the road, the signs, and the vehicles worse.
- The tyres grip the road less because the ground is wet. The precautions you must take to drive in fog are:
- Keep the vehicle well ventilated so that the windows do not mist up.
- Switch on dipped headlights and fog lights.
- Drive slowly.
- Keep a large safety distance from other vehicles.
- Pay close attention to the signs and road markings.
- Try not to overtake other vehicles if it is not necessary.
- Be more careful when you approach junctions.

Driving with clouds of dust or smoke

They are a danger because they do not let you see well the road, the signs, or other vehicles. Also, they can appear suddenly. To drive with clouds of dust and smoke, you must take the same precautions as for driving in fog.

Driving with strong wind

It is dangerous to drive with strong wind, especially when it blows from the side, because it can make the vehicle lose control, roll over, or leave the road.

Wind affects two-wheeled vehicles more and those that are towing a trailer.

Wind is more dangerous for driving in the following situations:

- When you pass another vehicle coming the other way.
- When you overtake a vehicle that takes up a lot of space.
- When you pass in front of buildings, trees, and other objects that can cause the wind to appear suddenly. The precautions you must take to drive with strong side wind are:
- Drive more slowly.
- Hold the steering wheel firmly and steer against the wind.
- Keep the windows closed.
- Keep in mind that trees, branches, or stones may fall.

Driving in heat

It is dangerous to drive in heat for the following reasons:

- Sleepiness and tiredness appear sooner, which can cause distractions and mistakes.
- You may need more time to react to unexpected events.
- Aggressiveness may increase towards other drivers. The precautions you must take to drive on a very hot day are:
- Use the vehicle’s air conditioning and keep it at a temperature between 20 and 23 degrees.
- Take more breaks during the journey.
- Drink plenty of water or juices.
- Pay more attention when you drive after eating because you may feel sleepy.
- Wear light-coloured, light, loose clothing.

Topic 15. Vehicle mechanics and maintenance

Contents

- Transmission system
- Steering system
- Suspension system
- Braking system

Checking wheels and tyres

Vehicle systems

What are they? All the mechanisms and parts of the vehicle that make it work and be safe. Vehicles have eight main systems:

- Fuel system
- Electrical system
- Lubrication system
- Cooling system
- Transmission system
- Steering system
- Suspension system
- Braking system

Fuel system

What is its function? To bring air and fuel to the engine so that the vehicle can work.

The air can carry dirt from the street. That is why the system has a filter that is responsible for cleaning that air so that it reaches the engine in good condition. You must clean this filter from time to time because, when it is dirty, fuel passes through it worse. This makes you use more fuel than when it is clean. Also, when the filter is very dirty or broken, black smoke may come out from the fuel through the exhaust pipe.

Exhaust pipe. A pipe that cars have at the back to expel the gases that are created in the engine. You must check the filter more often in summer than in winter and when you drive on very dusty tracks.

Electrical system

What is its function? To give energy to the car so that it switches on, the engine can start and work, and other parts work, such as the lights or the horn. This system has several parts:

Battery

Ignition circuit

Charging circuit

Lighting circuit

Battery

It provides the energy needed to start the engine. It also supplies energy to the rest of the vehicle when it is needed. The battery must be kept clean, dry, and well fixed in its place.

Ignition circuit

It makes sure that the spark of electricity needed so that the air and fuel turn into energy and the vehicle starts to work. This electric spark is made in an engine part called a spark plug. To keep the ignition circuit in good condition, you must check the condition of the spark plugs and change them when necessary. You must also check that the engine is in good condition.

Charging circuit

It is responsible for carrying the electric current to the battery and to all the systems that need it while the vehicle engine is running.

Watch video

The part that allows the vehicle to produce electricity and store it in the battery is called the alternator. The alternator is moved by a belt.

You must check from time to time that this belt is in good condition and change it when necessary.

Lighting circuit

It is made up of all the parts that allow the vehicle lights to be switched on.

Watch video

It is important to check the lighting system. You must change the bulbs that give light when they shine less.

Lubrication system

What is its function? To distribute oil to all the parts of the vehicle engine to create a layer that covers the engine parts to protect them and stop them rubbing against each other. The oils used for vehicles are special substances that prevent the vehicle parts from wearing out more. The instruments that control the oil in the lubrication system are:

- Dipstick. It shows how much oil there is in the engine.
- Pressure gauge or warning light. It controls the oil pressure in the engine. The oil must have the right pressure to be distributed well through the whole engine and cover all its parts.

Pressure gauge
Warning light
Watch video

When the warning light comes on it means there is little oil or that it does not have the pressure it needs. In that case you must stop the vehicle and not continue until you know what the problem is and fix it. To check the oil level you must look at the dipstick when the engine is stopped and cold. The oil level must be between the minimum and maximum marks.

Types of oils

Depending on where they come from

Depending on their thickness

Depending on where they come from
Type of oil

How is it made?

Mineral

From oil. Nowadays it is used very little.

Synthetic

It is made in a laboratory by mixing substances from oil and from other materials.

Semi-synthetic

A mix of mineral and synthetic oils.

Why are synthetic oils used more? Because they have the following advantages:

- They resist cold and heat better.
- They allow the vehicle to start better when it is very cold.
- They protect the engine better because they are thicker and less runny than mineral oils.
- Less oil is needed.
- It lasts longer.

Therefore, you need to change the vehicle oil less often.

Depending on their thickness

Oil is more runny when it heats up and thicker when it is cold. That is why it is better to use an oil that is more runny when the temperature is low. This way it will circulate better through the engine when you start the vehicle, even if it is very cold.

However, when the engine is hot it is better to use a thicker oil. If it stays too runny it will not cover well the engine parts. To avoid these problems there are oils that adapt well to cold and heat. They are called multigrade oils and they have this label.

10W/40

10W indicates that this oil spreads well through the engine, even if the temperature is low. 40 indicates that it can withstand very high temperatures without being damaged.

To top up the vehicle with oil you must lift the engine cover and pour the oil through the cap that is there. From time to time it is necessary to change the oil and the filter. To change it, the vehicle must be level and the engine must be stopped and warm. You must avoid the oil falling on the ground because it is very polluting. If oil spills while you top up the vehicle, you must collect it. When a bluish white smoke comes out through the exhaust pipe it means that the vehicle has too much oil.

Cooling system

What is its function? To make sure the engine stays at a good temperature so it can keep working and to stop its parts wearing out and breaking because of too much heat. Without this system, the engine would heat up very quickly when working and would break down.

A liquid called coolant passes through the engine parts and absorbs the heat so that it stays cool. Then this liquid goes to a part called the radiator. There it cools down again and returns to the engine pushed by a water pump. You must check that the belts that drive the water pump are in good condition and that the whole cooling circuit works well.

Watch video

You must also check that the vehicle has enough coolant. It must stay between the minimum and maximum marks when the engine is cold. White smoke may come out of the exhaust pipe. That smoke is water vapour. This is normal when starting, especially when it is cold.

However, it may indicate that there is a broken part or a fault if the white smoke comes out when the vehicle has been running for a while and the vehicle is hot. At the start of winter it is important to check that the coolant is in good condition and will not freeze. If the vehicle is in a very cold place during winter, the liquid can freeze and break the engine or the cooling system.

Transmission system

What is its function? To carry the engine power to the wheels so that the vehicle can move. A vehicle engine can drive some wheels or others. Depending on which wheels get the drive, the vehicle can be:

Which wheels does the drive from the engine reach?

- Front-wheel drive It reaches the front wheels.
- Rear-wheel drive It reaches the rear wheels.
- Four-wheel drive It reaches all the wheels.

First gear is the one that sends the most force to the wheels, but it is the slowest.

Steering system

What is its function? To transmit the movement of the steering wheel to the front wheels so that the driver can turn the vehicle without problems.

Signs that the steering system is failing:

- It takes a lot of effort to move the steering wheel.
- The tyres may have low pressure and need more air.
- The steering wheel is too loose.
- The vehicle pulls to one side when you let go of the steering wheel on a straight road.
- Some tyres may be more inflated than others.
- The tyres wear out very soon.
- The steering wheel vibrates while you drive. The wheels may not be well balanced.

Power steering

A system that helps the driver use less force to turn the steering wheel and control the vehicle. There is also a type of power steering called progressive.

Progressive power steering helps you turn the steering wheel very easily when the vehicle is going slowly and makes it a bit harder when the vehicle is travelling faster. This makes it easier to control the vehicle.

Suspension system

What is its function? To keep contact between the tyres and the road at all times.

Thanks to this system, the vehicle does not lose stability and allows passengers to travel more comfortably.

Without the suspension system the driver would have more difficulty getting over small obstacles on the road and driving without jolts.

What happens when the suspension system fails?

- The vehicle dips forwards or the back lifts a lot when braking.
- The vehicle sways from side to side or leans a lot when taking a bend.
- You notice bumps or wind too much.
- The tyres wear out very quickly.
- The vehicle lights move up and down when they are on.

What consequences does a suspension system in poor condition have?

- The vehicle loses stability.

Especially on bends and when it is windy.

- There is more risk of aquaplaning.
- The vehicle needs more space to brake. Especially when the road is wet.
- The system that stops the brakes from locking works worse.
- Other parts of the vehicle wear out and break more easily.
- The lights move up and down and can dazzle other drivers.
- The driver can feel tired more easily.

Braking system

What is its function? To reduce speed or stop the vehicle and stop it from moving again when it should not. When you press the brake pedal, a force is sent to the vehicle's wheels so that they stop and stop working. This force goes through a liquid called brake fluid and reaches the parts that brake the wheels, which are the pads or the shoes. The pads and the shoes rub against other parts of the wheels and make them stop. What should you check?

- That there is brake fluid in the vehicle's reservoir. And that it is between the minimum and maximum marks and that it is in good condition.
- That the pads or the shoes are not worn and are well fixed.
- That the tyres are well inflated, with the pressure they need.

Watch video
Wheel and tyre check

To keep the vehicle's wheels and tyres in good condition, you must:

- Check that the tyres are well inflated, at least once a month. Especially if you are going to make a long trip, or load the vehicle a lot.

You must do this check before you start driving.

Watch video

- Look at the tyre tread to make sure it is not too worn.

You must always carry a spare wheel in the vehicle and other systems. This wheel will be a bit more inflated than the others.

Topic 16. Road traffic accidents

Contents

Avoid road traffic accidents
Risk factors

Most road traffic accidents can be avoided. Some circumstances of the driver, the vehicle, the road and the surroundings mean that there is more risk of having an accident. The circumstances that make a person have more risk of having an accident are called risk factors.

What are the main risk factors?

Risk factor Some causes

Number of accidents

Human

Speeding. Drinking alcohol. Distractions.

Between 70 and 90 out of every 100.

Road condition The road is wet.

Losing control on a bend.

Between 10 and 35 out of every 100.

Vehicle fault The brakes fail. A tyre gets a puncture.

Between 4 and 13 out of every 100.

Watch video

Many of these risk factors can be avoided if the driver behaves responsibly. For example, by checking the tyres before starting a trip to check that they are in good condition and will not blow out.

Accidents can also be avoided by reducing speed and leaving more safe distance from the vehicle in front on a road with rain. Where and when are there more road traffic accidents?

- Most fatal accidents happen on roads outside towns or cities.
- Most accidents happen on straight roads and not on bends.
- There are fewer accidents on motorways and dual carriageways than on the rest of the roads.
- The times of year when there are more accidents are: Easter, December and summer.
- There are more accidents at weekends and on public holidays. Especially in the early hours of the morning.
- During the day, most accidents happen at the times people go to and leave work. Especially at the end of the working day.
- The accidents that happen most inside cities are pedestrians being run over.

Consequences of road traffic accidents for society Road traffic accidents are one of the main causes of injury and death. Especially in young people.

Also, each road traffic accident has very negative consequences for the whole of society.

These consequences are:

Human
Material

Medical and healthcare costs

Spending on security resources

Human

The death of a person means physical and psychological suffering for the people around them.

Material

Road traffic accidents cause damage to vehicles, the road and the environment.

Spending on medical and healthcare resources

A lot of staff and money must be invested to give first aid to the injured, treatment, rehabilitation and the changes that each injured person needs.

Spending on security resources

A lot of staff and money must also be invested in the work that police officers and firefighters do at accidents.

Also, each accident means extra costs that insurance companies and organisations that provide services to the injured have to pay. For all these reasons, the World Health

Organization considers that road traffic accidents are a health problem that affects the whole of society. And it asks all of us to work together so that there are fewer deaths and injuries in road traffic accidents.

People with more risk of having road traffic accidents There are studies that show that some groups of people are more likely to have an accident. These groups of people are called risk groups. Within these risk groups, we must tell the difference between the risk that pedestrians face and the risk that drivers face.

Pedestrians

When a vehicle runs over a pedestrian, the pedestrian usually suffers very serious injuries. They are more likely to die when the accident is on a road outside the city because vehicles travel faster.

Pedestrians are run over more in areas where drivers cannot see them well because there is some kind of obstacle. To avoid being run over, pedestrians must take the following precautions:

- Cross at pedestrian crossings.
- Cross when the traffic light is green.
- Walk in the allowed spaces on roads and hard shoulders.
- Wait on the pavement to cross. Not on the road.
- Always look before crossing.
- Check that no other vehicle is coming before you get out of yours.

Drivers must also take measures not to run over pedestrians.

- Drive at a moderate speed. Especially inside the city.
- Drive more carefully when you reach places where there are pedestrians and other vehicles parked with people inside. Some of these people may get out of the vehicle without looking.
- Do not go through when the traffic light is red or amber.
- Let pedestrians cross when they have priority.
- Take special care when a pedestrian crosses while talking on a mobile phone. They may be distracted.
- Pay attention to pedestrians who walk near places for parties and entertainment.
- Take special care at the exit of garages.
- Do not park the vehicle on the pavement.

Vehicles on the pavement force pedestrians to walk on the road.

BEEEP

- Do not park on pedestrian crossings.
- Do not make changes to the outside of vehicles. The outside of today’s vehicles is designed to cause less harm in collisions with pedestrians.
- Drive more slowly when passing near a bus that is stopped.

Especially if it is a bus that takes children to school.

- Drive more carefully on rainy days.
- Take many precautions when driving in reverse. There is a system called a pedestrian detector that can detect when there is a pedestrian in front of a vehicle. It warns the driver and, if they do not respond, the system activates the brakes and stops the vehicle.

Children and older people

The pedestrians who have the most risk of having a road traffic accident are children and older people.

Children

- They are shorter than adults. This means they see less of the road and drivers see them worse too.
- They pay less attention to the road because they have less sense of danger.
- They do not know the traffic rules.

Most collisions with children happen when they leave school.

Older people

The main problems that many older people have as pedestrians are:

- The noise in the surroundings does not let them hear a vehicle when it comes closer.
- They may have difficulty knowing how fast a vehicle is coming.
- They may have difficulty telling the colour of traffic lights.
- There are few pedestrian crossings in some areas.
- Some streets are poorly lit and it is harder to cross them.
- Some kerbs are too high.
- There are obstacles on pavements such as plant pots or badly parked cars.
- They may have orientation problems when they do not know the streets well.

Older people have more accidents when they go alone than when they go with children. When they go with children they are more careful to set a good example and to protect them.

Drivers

The drivers with the highest risk of having accidents are:

Young people
People over

65 years old

Two-wheeled vehicles

Bicycles

Motorcycles and mopeds

Young people

In Spain, road traffic accidents are the main cause of death for young people between 15 and 29 years old. These accidents happen, above all, in big cities, on the way to and from work, and in places where people go out. The times when more young people die in road traffic accidents are: the summer months, Christmas, and weekends. The hours when there are most accidents of this type are at night and in the early hours.

Main causes of road traffic accidents among young people:

- Driving too fast.
- Using alcohol and drugs.
- Not following traffic rules.
- Taking many risks when driving.
- Little experience.
- Competing with other drivers.

Km/h
20
40
60
80

100 120 140 160

180
200
220
240
STOP 80

- Not wearing a seat belt or a helmet because they think they do not need it.
- Following advice from adverts that encourage drivers to take risks.

People over 65 years old

These people are usually very careful when driving and they follow traffic rules. But they can have more accidents for the following reasons:

- They may have less ability to pay attention and need more time to react to something unexpected.
- Many people over 65 years old see and hear worse.
- Some older people take medicines that affect driving. In an accident, older people have more serious injuries and less chance of surviving.

Drivers of two-wheeled vehicles

Two-wheeled vehicles are more fragile and are harder to see than other vehicles.

So, their drivers must drive with special care.

Bicycles

Cyclist accidents happen more at weekends. Especially when the weather gets good. There are more accidents in cities, but more cyclists die on roads outside cities. Most of these accidents happen on straight roads or at junctions, due to a crash with another vehicle. The causes of these accidents can be because of mistakes made by cyclists or because of mistakes made by drivers of other vehicles.

Mistakes made by cyclists

Mistakes made by drivers of other vehicles Distractions.

Riding in the wrong direction or where it is forbidden. Making forbidden turns.

Not obeying the Stop sign or give way to other vehicles.

Entering the road without care. Distractions. Driving very fast.

Overtaking when they should not. Making forbidden turns. To avoid running over cyclists, drivers of other vehicles must:

- Leave space in front until you have overtaken them, and leave enough side distance when you are overtaking them.
- Do not overtake other vehicles if cyclists are coming in the opposite direction.
- Be more careful when there is rain, snow, fog, or poor visibility.
- Do not use the horn near cyclists so you do not frighten them.
- Be careful when there are parked vehicles.

Cyclists can appear between these vehicles.

- Drive more carefully near buildings and leisure areas.

Motorcycles and mopeds

The most common accidents in this type of vehicle when overtaking are head-on and side crashes.

Most of these accidents happen inside the city. They have more serious consequences when the people travelling on the motorcycle or moped are not wearing a helmet.

The causes of these accidents can be because of mistakes made by the moped or motorcycle drivers or because of mistakes made by drivers of other vehicles.

Mistakes made by motorcycle drivers

Mistakes made by drivers of other vehicles

Using the lane of the road that goes in the opposite direction. Distractions.

Not respecting the motorcycles' right of way.

Using the lane of the road that goes in the opposite direction. Distractions.

To avoid accidents with motorcycles and mopeds, drivers of other vehicles must:

- Check that there are no two-wheeled vehicles coming before changing lanes.
- Do not stay next to a two-wheeled vehicle. Drive in front of it or behind it.
- Keep a safe distance from these vehicles.

- Respect situations where they have right of way.
- Be more careful when there is rain, snow, fog, or poor visibility.

What to do in case of an accident

Steps to follow

All people who witness a road traffic accident must:

- Help the victims or ask for help so someone can help them.
- Work together so there is no more damage or more injured people.
- Help traffic become safe again when possible.
- Explain what happened to the authorities and emergency services.
- Give your name and contact details to the authorities and to other people who are part of the accident.

Drivers must also give their vehicle details. When you witness an accident, but you cannot do anything to help or the traffic officers have already arrived, it is best to keep driving so you do not get in the way at the accident scene.

Failure to help

You must always try to help the person who has had an accident. Anyone who does not do so is committing a crime called failure to help, that is, not helping, and they can go to prison. This crime is committed by a person who:

- Does not help the victims of an accident when they can do so without putting themselves in danger.
- Does not report the accident so that someone comes to help.
- Causes an accident and runs away instead of helping.

Important moments after an accident

After an accident there are three moments when people are at risk of dying:

First moment

First seconds after the accident. It is called immediate death.

Second moment One or two hours after the accident. In this period, resuscitation work is very important to try to save life. It is called the golden hour.

Last moment

Days or weeks after the accident. It is called late mortality.

PAS rule

What is it? These are the guidelines that must be followed to help a person who has had an accident. These guidelines are:

Watch video

It is very important to follow the PAS rule in the first 10 minutes after the accident. Acting quickly helps to:

- Save lives.
- Prevent a more serious accident from happening.
- Prevent the person from suffering more harm.
- Prevent material damage. For example, more vehicles and road elements being damaged.

Protect
Alert
Help
Protect

What must you protect?

- Yourself. Do not put yourself in danger or take risks that could cause you to have an accident.
- The victims.
- The accident scene. Mark the place so it can be seen well, avoid more accidents, and help the emergency services find the place more easily.

What steps must you take to protect?

- Stay calm.
- Stop and park the vehicle in a safe place, leaving access for the emergency services.
- Stop the engine and switch on the hazard lights.
- Put on the reflective vest before getting out of the vehicle. This vest must be approved to make sure it meets all safety rules. It can be yellow, orange, or red.

Approved. It is officially registered and meets the features set by law.

- Observe the situation to assess how serious the accident is and decide how to help.
- Mark the accident area to warn other vehicles.
- Apply the handbrake of all vehicles in the accident to make sure none of them moves.
- Do not smoke or light a fire at the accident scene to avoid a fire. If the fire has already started, try to put it out with a fire extinguisher or by throwing earth, sand, or blankets over the fire. Do not throw water on the fire.
- Help the area become safe again so traffic can move again.
- Try not to touch or move people or things if there are dead people or people with very serious injuries.

Alert

Before alerting the emergency services, try to collect as much information as possible to make it easier for the right services to come and do their work well. For example, say if there is a fire so the fire brigade comes. Or explain how many injured people there are so they know how many ambulances to send. To ask for help you must call the emergency number 112 from your mobile. If you do not have a mobile, you can go to the nearest SOS post to alert them.

You must stay at the accident scene or nearby until the authorities and emergency services arrive. Some vehicles have a system called eCall. This system sends a message automatically to 112 in case of an accident so the emergency services go to the place, even if nobody alerts them.

Help

You must help the injured quickly but stay calm. You must stay with the injured and not leave them alone. General rules for helping.

- Do not move the injured unless it is necessary to keep them safe.
- Keep the neck protected from sudden movements. You can place clothes, sand, or your hands at the sides of the injured person's neck. Never under the neck.
- Do not remove the injured person's helmet.
- Loosen any clothing they are wearing that is tight on the body.
- Do not give medicines, food, or drinks to the injured.
- Do not put ointments or alcohol on their wounds.
- Cover the injured with clothes or thin blankets so they keep their body temperature.
- If you are not sure how to act, talk to the injured and give them emotional support. It is better not to touch or move them if you are not sure how to do it because you could make their injuries worse. How should you move an injured person? When it is necessary to move an injured person because their life is in danger in that place, it must be done by three people to move them as if they were one single block. The three people must be coordinated so they do not move the neck and so the head, neck, and body stay aligned.

To take an injured person out of their vehicle, you must follow these steps:

- Put your arms under their armpits, carefully.
- Hold their arm with one hand and support the chin with the other.
- Take the person out of the vehicle slowly and keep their head, neck, and trunk in the same line, as if it were one block.
- Place them in a safe place.

First aid
Vital signs

In road traffic accidents, the parts of the body that are usually most damaged are the head and the legs. To know how serious the injured people are, the first step is to check their vital signs, always following the same order.

Consciousness
Breathing
Consciousness

To know if a person is conscious, you can ask: Are you OK? If they do not answer, you can give a small pinch to see if they react with any movement or sound.

How to act depending on their reaction? The injured person responds by answering or moving The injured person does not respond and does not move Do not move the injured person.

Check if they have bleeding, burns or fractures.

Watch them every few minutes to see if there are changes. They are unconscious. Check if they are breathing.

Breathing

Steps to follow to check if a person is breathing:

- Look at their chest to see if it moves.
- Bring your face close to their mouth to hear their breathing and feel on your cheek the air that comes out of their mouth. When the person is face down, turn them to put them on their back and lift their chin to open their airways.
- Check if there is normal breathing for 10 seconds.

Normal breathing for an adult is between 15 and 20 breaths per minute.

Normal breathing for a baby is between 30 and 40 breaths per minute.

Normal breathing for a child is between 26 and 30 breaths per minute. How to act depending on their reaction?

Breathes normally

Does not breathe normally It is a sign that the heart is working.

Check if they have bleeding, burns or fractures.

Stay with them until the emergency services arrive.

Put them on their side if you have to leave them alone to help other injured people.

Start cardiopulmonary resuscitation, until the emergency services arrive, as follows:

- Place both hands with fingers interlocked on top of the chest of the person who is not breathing.
- With your elbows straight, push your hands up and down about 100 times per minute.
- Give 2 mouth-to-mouth breaths every 30 chest compressions so that air goes into the injured person's mouth.

Press

30 times

Blow

2 times Continue like this until medical help arrives.

Knowing the condition of the injured people

When the injured person is conscious and breathing you must check if they have other types of injuries or wounds. To assess their condition you must take into account if there are:

Bleeding
Burns
Fractures
External
Internal
Bleeding

What is it?

Losing a lot of blood because blood vessels have broken. A person can go into shock or die if they lose a lot of blood.

Shock. Danger of death because the body does not receive the blood it needs. When the bleeding happens because an object has become stuck in the person's body, you must not try to remove the object.

Bleeding can be external or internal.

External bleeding

Blood comes out of the person's body. To control external bleeding three methods can be used:

Direct pressure

- Press on the wound with your hand or with your fist using a gauze pad or a cloth that is as clean as possible.
- Do not remove the gauze pad or the cloth, even if they soak with blood.
- If the wound is on an arm or a leg, keep the arm or leg raised.

Arterial pressure

Artery. Tube that carries blood from the heart to all parts of the body.

- It is used when direct pressure fails.
- Press the main artery of the arm or the leg to stop blood circulation.
- Keep the area pressed until the emergency services arrive.

Tourniquet

- It is only used for bleeding in arms and legs when direct pressure and arterial pressure have not worked.
- Tourniquets are usually made with a folded cloth. Do not use very thin objects.
- It is done above the place

where the wound is. Never below.

- You must write on a piece of paper the time when the tourniquet was applied and the place where it has been put.
- Only medical staff can remove the tourniquet.

Internal bleeding

It happens inside the body and cannot be seen. But you can know there is this type of bleeding because the injured person goes into shock. What symptoms does a person in shock have?

- They breathe very fast and shallow.
- The pulse is fast and weak.
- Their skin is pale and their sweat is cold and sticky.
- They say things that do not make sense. In these cases you must place the injured person lying down with their feet higher than their head. If they vomit, place them on their side and keep their feet higher than their head.

There is a type of internal bleeding where the wound cannot be seen, but the blood does come out of the body through an opening such as the nose, the ear or the mouth. This type of bleeding is called externalised. In these cases you must:

- Treat the injured person as very serious.
- Do not plug or stop this bleeding.
- Put the injured person on their side if there is a risk of choking.
- When blood from the bleeding comes out through the ear you must cover it with gauze, without pressing.

Burns

What to do in case of burns?

- Do not touch the burnt area.
- Do not remove clothing that has got stuck to that area.
- Pour cold water over the burnt area.
- Do not cut or prick the blisters.
- Put on a disinfected and damp dressing.
- Do not bandage two burnt areas together.
- When the burnt area is on an arm or a leg, keep that arm or leg raised so it swells less.
- You can give the injured person water to drink as long as they do not have other injuries, are conscious and do not vomit.

Fracture

A fracture is a broken bone. What are the symptoms of a fracture?

- Severe pain in the area.
- The area swells or becomes deformed.
- The person cannot move that area or has difficulty doing so.
- When the fracture is in the spine the person cannot move and loses feeling. What to do in case of a fracture?
- Keep the fracture area still.

- Prevent the injured person from moving that area.
- Cover the wound if there is one.

Transporting injured people

You must always wait for the ambulance to arrive to take injured people to hospital. You should only take injured people to hospital without waiting for the ambulance in these cases:

- The accident has happened in an isolated place where you cannot ask for help from the emergency services.
- The emergency services take more than 30 minutes to arrive and the injured person is unconscious or has bleeding that cannot be controlled. Transport should be done, if possible, in a van or lorry so you can lay the injured person down. You must drive slowly and smoothly.

Topic 17. Preventive and efficient driving

Contents

Prepare the trip

Safety checks

Before starting a trip in a car or in a vehicle that carries goods, you must check that you have:

- The warning light or the triangles that you must place on the road in case of breakdown or accident.
- A reflective vest.
- A spare wheel. It is also advisable to carry a first aid kit.

People who wear glasses are advised to carry a spare pair.

Choose the route

Before starting the trip, you must plan the route you are going to follow by looking at a map. It is important to choose the safest and most comfortable way taking into account which roads have roadworks, the weather and the number of vehicles that use each road that day. You must also plan the breaks you are going to take during the trip. When you travel in winter, you must take the following precautions:

- Carry snow chains in the vehicle.
- Avoid driving at night.
- Travel with a full fuel tank.
- Carry warm clothes, water, food and a torch in the vehicle.

Preventive driving

What is it? A way of driving that allows you to get ahead of unexpected events that may happen during the trip. In preventive driving, the driver:

- Collects the necessary information before starting the trip to drive safely.
- Gets ahead of unexpected events that may happen and reacts well to them.
- Adapts to the circumstances at each moment of the trip. To drive in a preventive way, you must follow a technique that is based on three principles:

What does it consist of?

Vision

Look in all directions to gather information and do not focus on things that could be a distraction.

Anticipation Predict the movements and reactions of other drivers so you can react in time.

Space

Keep a safe distance from other vehicles at all times.

Vision

Look far ahead

Look to the sides

Look far ahead

You must look far ahead to control what happens in the space your vehicle will travel in the next 20 seconds.

This way you can predict dangerous situations and avoid harsh braking and sudden acceleration. The faster you drive, the further ahead you must look.

Look to the sides

We take longer to see what happens to the sides than what happens in front. So, it is necessary to check all the time what is happening to the left and to the right during the trip. This is done through the rear-view mirrors because they allow you to see what happens behind and to the sides of the vehicle. You must look at them during the trip, even if you are not going to do any manoeuvre.

How often you look at them will depend on the type of road you are on. You look quickly, briefly so you keep paying attention to the rest of the road.

You must keep in mind that there is always a space on both sides of the vehicle that you cannot see, even if your mirrors are well adjusted. These spaces are called blind spots and they are dangerous because you cannot control what is happening in that place.

Blind spot

Blind spot

Watch video

To make sure there is no other vehicle in the blind spot area, you can turn your head and look through the window. It is important to check that there is no other vehicle in the blind spot before turning the vehicle towards that side to change lane.

Blind spot detector (BSM)

There is a system that warns you when there is another vehicle in the blind spot area moving towards your vehicle. In those cases a light comes on that is placed in the rear-view mirror. Some blind spot detectors light up whenever another vehicle is in the blind spot area. Other detectors only light up if you switch on the indicator to change lane and there is another vehicle in the blind spot.

Anticipation
Watch video

To avoid risks and react quickly to unexpected events on the road, you must:

- Keep a suitable speed for your circumstances, the vehicle's and the road's at all times.
- Switch on the lights that are necessary at each moment so other vehicles can see you and know what movement you are going to make.
- Do not stay for a long time in the blind spot of another vehicle's mirrors. Especially if you are driving next to or behind a very large vehicle.
- Always signal the manoeuvres you are going to do for as long as necessary.

STOP

?

Control the space

Front space

Rear space

Side space

Space when stopping the vehicle

Front space

It is advisable that between your vehicle and the one in front there are two or three seconds of separation. That is, that it takes you two or three seconds to reach it at the speed you are travelling. However, there are circumstances in which you must leave at least one more second of space. These circumstances are:

- There is rain, snow, fog or it is night.
- The vehicle behind gets too close.

Rear space

Sometimes it is the vehicle behind that comes too close. What you must do in these situations is:

- Leave more safety distance from the vehicle in front so you do not brake sharply if something unexpected happens.
- Signal earlier the manoeuvres you are going to do.
- Brake smoothly and with enough time so that the driver of the vehicle behind can react.

Space at the sides

You must drive at a suitable speed and keep the necessary separation distance when overtaking other vehicles or passing them. For example, when changing lane or opening the vehicle door to get out. You should try to stay out of the blind spot area of other vehicles.

Space when stopping the vehicle

When you stop the vehicle for some reason, such as a traffic light or a traffic jam, you must keep a distance of at least two or three metres from the vehicle in front. This way:

- You will have time to warn the driver in front if they roll their vehicle backwards.
- You will be able to overtake the vehicle in front if it stays parked for some reason.
- You will not hit the vehicle in front if another vehicle hits yours from behind. Also, the driver of the last vehicle in the queue, to avoid rear-end collisions, can:
- Keep a distance of five or six metres from the vehicle in front.
- Keep the brake pressed and switch on the hazard lights in a traffic jam so they can be seen better.
- Look in the rear-view mirror and watch the vehicles coming from behind.

Efficient driving

What is it? A way of driving in which less fuel is used, it helps to pollute less and makes driving safer. What do you achieve with efficient driving?

- Improve air quality because fewer polluting gases are released into the atmosphere.
- Save money because less is spent on fuel and on vehicle maintenance.
- Increase safety because techniques similar to preventive driving are used.
- Make passengers travel more comfortably because sharp braking is avoided and there are no big speed changes.
- Reduce the noise that vehicles make when driving.

Pollution

Vehicles pollute because they produce toxic gases that reach the atmosphere. The measures you can take to pollute less with your vehicle are:

- Check the engine often so it does not give off clouds of smoke.

- Do not accelerate sharply when the vehicle is stopped and starts moving again.
- Use the horn only when necessary.
- Secure the load well so it does not move and make noise when it hits the vehicle.
- Prevent oil or other vehicle liquids from falling onto the road.
- Do not wash the vehicle on the road. It should be washed in places prepared for this.
- Do not throw objects onto the road that can dirty it, start fires or cause accidents.
- Do not make unnecessary noise or release more gases or smoke than allowed. In vehicles there is a device in the exhaust pipe called a catalytic converter. The catalytic converter reduces the pollution produced by the gases that come out of the exhaust pipe. When the vehicle battery runs out you must not try to start the engine by pushing the vehicle.

Unburnt fuel can reach the catalytic converter and destroy it.

Techniques for driving efficiently

When starting and beginning to drive When accelerating and changing gear When choosing the speed When reducing speed and stopping When starting and beginning to drive

- Start the engine without pressing the accelerator.
- Start driving with the vehicle right after starting the engine.
- Start driving at low speed and increase it when the engine warms up.

Driving at high speed with a cold engine makes it use more fuel and makes the engine wear out sooner.

Watch video

When accelerating and changing gear

- Accelerate little by little pressing the pedal smoothly.
- Start driving in first gear. In this gear the engine has a lot of power, but it goes at very low speed.
- Change to second gear after a few seconds. In this gear the vehicle has a little less power and goes a bit faster.
- After that you can increase speed smoothly and little by little. The vehicle uses less fuel in fourth and fifth gear, which is when the engine runs faster and has less power. In the city you should try to use the highest gear possible, always respecting the speed limits.

When choosing the speed you drive at From 80 or 90 kilometres per hour, the vehicle uses much more fuel and pollutes more.

Changing speed many times also makes more fuel be used. It also makes the driver get more tired. You should travel at a constant speed, that does not change all the time, and avoid sharp braking and sudden acceleration. When reducing speed and stopping the vehicle To reduce the vehicle’s speed you have to:

- Lift your foot off the accelerator and let the vehicle roll.
- Use the brake smoothly to reduce speed little by little.
- Switch off the engine when the vehicle is going to be stopped for more than one minute.

What makes more fuel be used?

- The air outside
- Not doing vehicle checks
- Driving in towns or cities
- Using air conditioning
- Carrying a lot of weight The air outside

Most of the fuel that is used is used to fight against the air that does not let the vehicle move forward. Some items fitted to the vehicle make it harder to fight against the air and, therefore, it uses more fuel. For example, a roof rack or deflectors.

Deflectors. An accessory fitted to the vehicle to divert the air and stop it entering the inside.

So that the vehicle does not use so much fuel because of the air, the following measures can be taken:

- Do not fit a roof rack on the vehicle. It is better for luggage and other loads to go in the boot.
- Do not lower the vehicle windows while it is moving.
- Do not tow a trailer or caravan.

Especially if the trailer or caravan is wider or taller than the vehicle.

Not doing vehicle checks

Doing the vehicle checks and checking that it is in good condition helps to use less fuel. To save fuel you must check:

- The ignition system. You must check the spark plugs following the instructions given by the manufacturer.
- The cooling and lubrication systems.
- Tyre pressure. They must be properly inflated.
- The engine fuel system. You must change the air filter when necessary and keep the idle speed at the correct value.

Idle speed. The ability of the engine to run without the driver pressing the accelerator. An idle speed that is too high will make more fuel be used.

Driving in towns or cities

In towns and cities more fuel is used even though vehicles drive more slowly than on other roads.

Watch video

The reasons are that there are more traffic jams, the vehicle has to stop more times, you have to change gear more, and the engine is worked harder. To save fuel in towns and cities and not suffer traffic jams, it is advisable to:

- Not use the vehicle for short distances.
- Use public transport.
- Avoid driving at the busiest traffic times.

- Choose the route where there are fewer vehicles, even if it is the longer route.

Watch video
Using air conditioning

When using air conditioning in the vehicle more fuel is used. It is advisable to have a temperature inside the vehicle of 23 or 24 degrees.

Carrying a lot of weight

A vehicle uses more fuel when it carries a lot of weight. So you should try to carry only the amount of luggage and loads that are necessary. It is also important to distribute the load well in the vehicle to use less fuel.

Measures to save fuel that are false Some measures used to save fuel are false and must not be done because they damage the vehicle. These measures are:

- Put fuel in the vehicle different from what the manufacturer states.

Unsuitable fuel will break the vehicle's engine.

- Go down hills in neutral. The vehicle can lose stability and you will have to use the brakes too much.
- Not spend money on the necessary vehicle maintenance.

Worn parts and faults will make more fuel be used.

Neutral. Position of the gearbox in which the engine's movement is not transmitted to the wheels.

New techniques to use less fuel

Start-Stop function

This system starts and stops the engine automatically when the vehicle stops for a moment at a traffic light, in a traffic jam or for other reasons. This way, when the vehicle is stopped it does not use fuel and does not release gases that pollute the environment.

Eco mode

This system gives the option to set the engine to less power and speed to save fuel. It takes into account all the elements that make a vehicle use fuel to know how it can work without using full power and speed.

By limiting the vehicle's speed and temperature less fuel is used. Eco mode is switched on and off manually in each vehicle.

Environmental labels

What are they? A way to classify vehicles depending on the pollution caused by their gases and the damage they do to the environment. There are four different types of environmental labels.

Zero emissions
Eco
B
C

The information shown on each label is:

The aim of this classification is to give preference to vehicles that harm the environment less. City authorities can:

- Create low-emission zones in the city. These are areas that only vehicles that harm the environment less can enter. This is done to improve air quality.

Low-emission zones are shown with a vertical sign.

Only vehicles with the environmental label shown on the lower part of the sign will be able to enter that zone.

- Decide which vehicles can enter the city centre on days when there is a lot of air pollution.
- Give financial or traffic benefits to vehicles that support caring for the environment. For example, they can allow vehicles with the “zero emissions” label to use the VAO lane, even if only the driver is travelling in the vehicle.

Environmental labels must be placed in front of the windscreen, on the right side of the vehicle’s front glass. In vehicles that do not have a windscreen, it will be placed in a visible place.

Shared vehicles

Another measure to care for the environment is the use of shared vehicles. This initiative means that a person can hire a vehicle through an application (App) for hours or minutes and share it with other people.

Shared-use vehicles carry this label.

Annex points. The driving licence with points

Contents

The aim of the points driving licence is to reduce the number of road accidents. How many points does each driver have?

Drivers who start driving have 8 points during the first two years. If in that time they do not lose any points, they will move to 12 points.

Drivers who keep the 12 points for three more years will get another two points. So, they will have 14.

Drivers who keep the 14 points for another three more years will get another point. The maximum number of points a driver can get is 15.

Type of driver	Points
New drivers	8
Drivers with more than two years of driving licence	12
Drivers with more than five years of driving licence	14
Drivers with more than eight years of driving licence	15

Losing points

A driver loses points when they commit some offences or penalties while driving. A person who loses all the points will not be able to drive again until six months have passed. After those six months, they will have to do a 24-hour course and an exam at the Traffic Headquarters to get the points back and the driving licence. If, in the next three years, they lose all the points again, they will have to wait 12 months more to get them back again.

Drivers who only lose some points can get them back again if they do not lose any more in two years. Also, they can do a road safety course lasting 12 hours to get back up to 6 points.

Which offences take points away from the driving licence? Offences that take away from 2 to 6 points

- Driving faster than the allowed speed. The number of points lost for this offence will depend on the speed the vehicle is going in each case.

Km/h
20
40
60
80

100 120 140 160

180
200
220
240
Offences that take away 3 points

- Making a U-turn where it is not allowed.
- Driving with helmets or headphones connected to the mobile phone or another electronic device.
- Having devices in the vehicle to detect speed cameras.

Radar. A system used to detect an object, know how far away it is and how fast it is moving.

Offences that take away 4 points

- Driving with an alcohol level between 0.25 and 0.50 milligrams of alcohol per litre of air.

Professional drivers and those who have had the driving licence for less than two years will lose the 4 points with an alcohol level between 0.15 and 0.30 milligrams of alcohol per litre of air.

- Driving a vehicle with a licence that does not authorise you to drive it. For example, driving a car when the driver's licence only authorises them to drive motorcycles.
- Not following the right of way rules for other drivers or pedestrians and not stopping at Stop signs, Give Way signs and at red traffic lights.
- Overtaking other drivers in places where it is forbidden.
- Reversing on motorways or dual carriageways.
- Not respecting or following the signs and orders of traffic officers.
- Not keeping a safe distance from the vehicle in front.
- Driving a vehicle when the driver has lost the driving licence for committing offences.
- Not wearing the seat belt, the helmet, child restraint systems for children and other safety systems that are required.

Offences that take away 6 points

- Driving with an alcohol level higher than 0.50 milligrams of alcohol per litre of air.

Professional drivers and those who have had the driving licence since less than two years ago will lose the 6 points with an alcohol level higher than 0.30 milligrams of alcohol per litre of air.

- Driving after taking drugs or when some of these drugs are still in the body.
- Refusing to take the tests to detect alcohol or drugs in the body.
- Driving in the opposite direction to the one allowed, taking part in illegal races or putting other people's lives in danger while driving.
- Driving vehicles that have installed mechanisms so that speed cameras do not detect them and do not know what speed they are going.
- Driving for more than half of the time that is allowed or taking breaks shorter than allowed (for professional drivers). For example, driving for 6 hours in a row if the time limit is 4 hours or resting for 10 minutes when you must take breaks of at least 20 minutes.
- Putting in the vehicle elements that change how the devices that limit speed work, or the ones that count the time the professional driver has been driving.
- Driving with the mobile phone in your hand.
- Throwing objects onto the road that can cause fires or accidents.
- Overtaking while putting cyclists in danger or without leaving a gap from them of at least 1.5 metres.

Collaborates: